The Homelessness Trap: Labor Market Trajectories of Individuals Experiencing Homelessness

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Abstract

Job loss may push workers into homelessness, and homelessness may keep them out of work. I provide the first causal evidence on both directions, quantifying the risk of homelessness upon job loss and how much shelter access mitigates the negative impact of homelessness on labor market trajectories. To do so, I combine the universe of shelter requests with administrative employment and unemployment insurance records in France. To identify each direction, I use a matched difference-in-differences design around economic layoffs and an instrumental variables design based on the leniency of call center agents allocating shelter. Losing a job raises the risk of homelessness by 50 percent. Getting a shelter bed at the first request cuts job loss by 26 percent for those still working and raises employment a year later by 58 percent for the unemployed. This evidence suggests that homelessness is an uninsured risk of unemployment for low-skilled workers. In turn, rapid shelter access can partly mitigate its negative consequences for employment.

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1 Introduction

> *"Can't get no job without a home, and you can't get a home without a job. They take one and you lose both."*
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> — George, 28, former construction worker, Philadelphia, 1987[1]

Homelessness is a persistent social problem in many advanced economies, and reducing it has moved to the forefront of social policy. Over the last decade, politicians have vowed to end homelessness in their electoral campaigns, and many policies have been put in place at the local, national, and European levels. Yet, despite these efforts, at least 2.6 million individuals are homeless across OECD countries today. This number reflects a rising trend: in many OECD countries, homelessness has increased by around 30 percent over the last decade (OECD, 2026).[2] Designing policies that fight homelessness requires understanding both what leads individuals into it and what keeps them there.

Unhoused individuals are often thought to be far from the labor market. They are not. Linked administrative records show that about half of unhoused individuals in the United States worked in the year of their homelessness spell (Meyer, Wyse, Grunwaldt, Meyer and Wu, 2023), and close to a third in Canada worked in the year before (Beauregard, Couture, Green, Hicks and Warburton, 2026). For many, homelessness is thus not the end point of a long exclusion but an episode in a working life, and possibly a consequence of losing a job. This raises two questions. Does job loss push workers into homelessness? And does homelessness, in turn, keep them out of work? If both are true, unemployment and homelessness can reinforce each other and trap at-risk workers in both states. Whether homelessness is an uninsured risk of unemployment, and whether rapid shelter access protects employment, bear directly on the design of unemployment insurance and of emergency shelter, the latter being the most common response to homelessness across advanced economies.[3]

Establishing causal evidence on these questions has been hard for two reasons. First, measuring homelessness is notoriously difficult. Most statistics rely on point-in-time counts that combine scarce street counts with shelter records and surveys, which fail to characterize flows into and out of homelessness (O'Flaherty, 2019). Even when flows can be measured, unhoused

[1] Quoted in Jon D. Hull, "Slow Descent into Hell," *Time*, 2 February 1987.

[2] For example, the United States reports a 35 percent increase in homelessness between 2018 and 2025. Over the same period, homelessness rose by 32 percent in England, 60 percent in Canada, and 72 percent in Ireland. For France, no comparable estimates exist for this period; however, the number of unhoused individuals is estimated to have more than doubled between 2012 and 2025. The aggregate estimate across OECD countries is driven by the United States, which has the largest number of unhoused individuals.

[3] In France, the state program financing shelter and pathways to housing (*programme 177*) was allocated 2.9 billion euros in 2025, most of it for the roughly 205,000 emergency and transitional housing shelter places (Sénat, Commission des affaires sociales, 2024). In the United States, shelter is funded mainly at the local level. For example, New York City alone budgets about 4 billion dollars a year for its Department of Homeless Services, over 98 percent of which goes to shelter and services for the unsheltered (New York City Council, Finance Division, 2025; Park, Molly Wasow, 2024).

individuals remain hidden in employment records, making it difficult to track their labor market trajectories over time. Second, the relationship between homelessness and unemployment runs in both directions, and health shocks, substance use, or family breakdown affect both housing and employment. Addressing this requires a setting with exogenous variation in each direction, which is hard to find.

In this paper, I provide the first causal evidence in both directions by assembling a novel dataset that allows me to exploit quasi-random variation on each side. I link nationwide records of the French shelter system to administrative employment and unemployment records, which lets me follow unhoused individuals' labor market trajectories before, during, and after homelessness. To disentangle the bidirectional relationship between homelessness and unemployment, I rely on two quasi-experiments. First, to isolate the effect of job loss from confounding factors that also drive homelessness, I use layoffs for economic reasons, that is, dismissals motivated by the firm's situation rather than the worker's, and compare laid-off workers to matched workers in firms without such layoffs. Second, in the spirit of a *judge IV* design, I exploit quasi-random variation in the leniency of emergency hotline call-center agents to identify the effect of shelter placement on labor market outcomes. Finally, I combine both causal estimates in a four-state Markov chain to quantify how unemployment and homelessness may reinforce each other.

My empirical setting is France, which combines one of the highest homelessness rates in the OECD (about 1 in 200 residents according to point-in-time estimates; Dihal, 2025) with nationwide administrative data on its shelter system. France has responded with Housing First policies that expanded shelter capacity to 205,000 beds, a 38 percent increase over the last decade. Unlike most countries, it manages this system through a single nationwide database that records the universe of shelter stays as well as all shelter requests, regardless of their outcome. I am the first to exploit these data to construct daily, nationwide flow measures of homelessness. Adding unmet requests raises the number of individuals observed as unhoused by 25 percent, and in total I observe 1.9 million unhoused individuals between 2018 and 2025, which shows that point-in-time estimates severely understate the number of people experiencing homelessness. I then link these shelter records, on names and dates of birth in a secure environment, to matched employer–employee data and unemployment insurance records, which allows me to recover the labor market trajectories, covering earnings, employment spells, and unemployment benefits, of over one million individuals who experienced homelessness between 2018 and 2025.

I first provide new descriptive evidence on the labor market trajectories of unhoused individuals. At the time of the first shelter request, 11 percent are working, 24 percent worked within the previous two years, and 55 percent worked within the previous five years. In their last contracts before the onset of homelessness, most unhoused individuals had very low earnings and were often employed on precarious short-term contracts. In line with this, I find a strong negative correlation between earnings and the risk of becoming homeless in the following year. Among those who worked for at least six months in the two years before the first request, em-

ployment and earnings drop sharply around the time of the request. These descriptive patterns are consistent with job loss causing homelessness, with homelessness causing job loss, or with a common shock causing both. My causal analysis therefore relies on two quasi-experiments to disentangle these effects.

The first quasi-experiment estimates the causal effect of job loss on homelessness. To isolate the effect of job loss from confounding factors such as health shocks or divorce, I focus on economic layoffs, that is, layoffs due to the firm's poor economic performance rather than the worker's individual performance.[4] Using one-to-one matching, I compare workers laid off for economic reasons with similar workers employed in firms that did not undergo economic layoffs. I find that job loss raises the risk of becoming homeless by 50 percent over the three years following the layoff. The increase in first shelter requests is sharpest in the quarters after unemployment benefits expire, pointing to a lack of resources. Heterogeneity analysis reinforces this interpretation: the effect is concentrated among single workers with low earnings in their previous job. Among low-income earners, the effect is smaller for tenants of social housing in high-rent areas, as social housing lowers their rent burden.

I then turn to the reverse direction and estimate the causal effect of rapid access to emergency shelter on the labor market attachment of individuals experiencing homelessness. I exploit a unique feature of the French shelter system: all requests go through a single nationwide emergency hotline, the *115*, where each call is handled by an agent assigned quasi-randomly within a *département*,[5] year-month, and shift.[6] As the system is highly saturated, with 60 percent of weekly requests going unmet (Dihal, 2025), agents differ systematically in how often they meet requests. This generates quasi-random variation in whether an individual is sheltered immediately or exposed to unsheltered homelessness. I use agent leniency as an instrument for whether an individual's first request is met. I first provide evidence supporting the validity of the instrument: leniency strongly predicts placement on the day of the call as well as shelter use in the following months, while an individual's labor market history does not predict the leniency of the agent handling their call. I then estimate the effect of rapid shelter access on labor market attachment. Among callers employed at the time of their request, a met request reduces the probability that their contract ends within one year by 26 percent, notably through fewer dismissals for misconduct and at the end of the probationary period. Among unemployed callers, it raises the probability of being employed one year later by 58 percent, from a baseline of only 8 percent, with gains concentrated in stable jobs that would pass landlord checks rather than in one-day

[4]Economic dismissals are the most regulated way to end an open-ended contract in France. The firm must document an economic cause, show that redeployment was impossible, follow legal criteria in choosing whom to dismiss, and consult employee representatives when two or more workers are dismissed within 30 days. A firm separating from a worker for individual reasons therefore has little incentive to declare an economic dismissal. A personal dismissal or, since 2008, a separation by mutual agreement is cheaper and less exposed to litigation (Carry and Schoefer, 2024).

[5]There are 101 *départements* in total: 96 in metropolitan France and 5 overseas. Each *département* has exactly one call center handling all requests for emergency shelter.

[6]There are three shifts: morning, afternoon, and night.

or one-week contracts. Effects are largest for callers who worked within the three years before the request. As individuals whose first request is denied spend 50 fewer nights in shelters over the following year, reduced exposure to street homelessness appears to be a central mechanism.[7] Even emergency shelter, which provides only a short-term roof over one's head, can thus protect labor market attachment, and its effect likely provides a lower bound on the potential effect of rehousing.

Finally, I combine the causal estimates from the two quasi-experiments in a four-state Markov chain to quantify how unemployment and homelessness may reinforce each other. To keep the estimates comparable, I focus on low-skill workers, defined as those with a high school degree or less, who are also more exposed to homelessness. I compute raw transition rates for the housed and, to address selection, use the causal ratios from the two quasi-experiments to pin down the transitions for the unhoused. The effect of emergency shelter serves as a lower bound for the effect of actual rehousing, since individuals in shelters are still counted as unhoused in the Markov chain. Shutting down each channel in turn shows its importance in explaining steady-state homelessness. For instance, setting the inflow into homelessness for the unemployed to the rate of the employed reduces steady-state homelessness by half, highlighting the importance of the higher risk of homelessness when unemployed. Following individual trajectories illustrates this interaction: an unhoused worker takes 29 percent longer to find work than a housed one, and this longer unemployment spell in turn prolongs their time in homelessness.

Contributions. This paper connects two literatures that have largely been studied in isolation. A large literature studies the consequences of unemployment and the policies that can mitigate them (see Le Barbanchon, Schmieder and Weber, 2024, for a recent review), while a smaller but growing literature does the same for homelessness (see Evans, Phillips and Ruffini, 2021, for a review). I show that for individuals at risk of homelessness, the two are closely intertwined, as unemployment and homelessness interact and reinforce each other. Beyond bringing these strands together, the paper contributes to each of them.

First, I add to the literature on the causes and consequences of homelessness. Existing work shows that local housing and labor market conditions and individual shocks, including health and mental health episodes, shape entry into homelessness (Beauregard et al., 2026; Bruneel-Zupanc, Gaillard, Hellwig, Wangner and Werquin, 2026; Curtis, Corman, Noonan and Reichman, 2013; Quigley and Raphael, 2001; van Ours and Moschion, 2021), while studies of eviction identify job loss as a main risk factor for rent default without separately identifying its effect on homelessness (Abramson, 2024; Collinson, Humphries, Mader, Reed, Tannenbaum and Van Dijk, 2024). Evidence on the consequences of homelessness, such as markedly elevated mortality, is largely correlational (Cobb-Clark and Zhu, 2017; Hérault and Ribar, 2017; Meyer, Wyse and

[7] I find consistent results when exploiting variation in quick access to transitional housing shelters, which are often seen as a more stable housing solution than emergency shelters. Using an instrumental variable approach based on average wait times due to capacity constraints at the time of the request, I find that quick access to transitional housing increases job finding among those unemployed at the time of the request.

Logani, 2023; Moschion and Johnson, 2019), and eviction studies cannot isolate the role of homelessness from the other channels through which an eviction order affects households. Exploiting two quasi-experiments, I provide the first causal evidence in both directions: the effect of job loss on homelessness and the effect of rapid shelter access on the labor market outcomes of individuals experiencing homelessness.

Second, I contribute to the literature evaluating housing interventions for unhoused individuals (Brounstein and Wieselthier, 2024; Christopher, Duggan and Martin, 2025; Cohen, 2024; Costantini, 2026; Faraji, Ridgeway and Wu, 2018). Most of this work evaluates Housing First programs, which provide permanent housing, and finds improvements in housing stability and reductions in service use. Closest to this paper, Christopher et al. (2025) study the effects of temporary shelter on crime and health. I estimate the causal effect of emergency shelter placement on labor market outcomes, which has so far only been studied in the context of permanent housing.

Third, I contribute to the measurement of homelessness. Identifying unhoused individuals is a central obstacle in this literature (O'Flaherty, 2019), and recent work has made progress by exploiting other data sources, including food and nutrition programs, welfare records, and hospital records (Beauregard et al., 2026; Bruneel-Zupanc et al., 2026; Christopher et al., 2025; Costantini, 2026; Von Wachter, Schnorr and Riesch, 2020). I propose a novel approach that combines the universe of nationwide shelter stays with all requests made to the national emergency hotline, including denied requests, which allows me to measure homelessness flows for the whole country at a daily frequency. Linking these records to employment and unemployment registers, I trace labor market trajectories before, during, and after homelessness.

Finally, I contribute to the literature on the consequences of job loss. A large body of work documents its lasting effects on earnings, health, and mortality (Bennett and Ouazad, 2020; Bhalotra, Britto, Pinotti and Sampaio, 2025; Black, Devereux and Salvanes, 2015; Britto, Pinotti and Sampaio, 2022; Browning, Moller Dano and Heinesen, 2006; Jacobson, LaLonde and Sullivan, 1993; Roulet, 2020). I identify homelessness as one of its most severe and least documented consequences. A related literature examines the links between housing and labor markets for low-income households (Dauth, Mense and Wrede, 2024; Dustmann, Fitzenberger and Zimmermann, 2022). I extend this work to the most extreme form of housing insecurity and combine both causal effects in a four-state Markov chain to quantify how unemployment and homelessness may reinforce each other.

The remainder of the paper is organized as follows. Section 2 characterizes homelessness in France and the French homeless shelter system. Section 3 describes the shelter records and the linked administrative data. Section 4 provides new descriptive facts on the labor market trajectories of unhoused individuals. Section 5 details the matched difference-in-differences design used to quantify the risk of homelessness upon job loss and presents its results. Section 6 develops the instrumental variables research design and estimates how much shelter access mitigates the

negative impact of homelessness on labor market trajectories. Section 7 combines the estimates from the two quasi-experiments to quantify how unemployment and homelessness may reinforce each other. Section 8 concludes.

2 Institutional Background

France offers an informative setting to study homelessness: it has one of the highest homelessness rates among advanced economies, a large nationwide shelter system that has grown under Housing First policies, and shelter records that track entries into and exits from homelessness.

Measuring homelessness. A central challenge in estimating the size of the homeless population is how homelessness is defined and measured. A common framework, adopted by many countries including France, is the European Typology of Homelessness and Housing Exclusion (ETHOS) (Feantsa, 2005). This framework disaggregates homelessness into several distinct categories. The first category consists of individuals *living in the streets or public spaces without a shelter that can be defined as living quarters*. The second category includes those in emergency accommodation, *people with no place of usual residence who move frequently between various types of accommodation*. The third category encompasses individuals in accommodation for the homeless, where the *period of stay is time-limited, and no long-term housing is provided*. Finally, the framework covers three additional categories that narrower definitions of homelessness do not always include: individuals living in institutions, those in non-conventional dwellings, and people living temporarily with family or friends due to a lack of housing.

Homelessness estimates in France. In France, an individual is defined as homeless if they fall within the first three ETHOS categories or reside in a non-conventional dwelling due to a lack of housing. In 2025, France reported one of the highest homelessness rates among OECD countries: 1 in 200 residents were unhoused. Point-in-time estimates show rapid growth over the last decade, with the total number of unhoused individuals more than doubling from 143,000 in 2012 to approximately 350,000 in 2025. Of these 350,000 individuals, roughly 334,000 reside in shelters.[8] This figure excludes an estimated 590,000 individuals who are staying with friends or family due to housing difficulties (Fondation pour le logement (2025)).

Policies for unhoused individuals. France grants its residents a legal right to housing. In addition, anyone in France has a right to shelter, which is fully unconditional, regardless of administrative status. Despite these rights, the number of unhoused individuals has risen. In response, France launched a nationwide Housing First plan in 2018. The plan seeks to shorten the path from the street to ordinary housing by expanding social housing allocation, rental interme-

[8]Unlike many other countries, France also counts asylum seekers staying in dedicated shelters as unhoused. Excluding them, the unhoused population is roughly 240,000, that is, 0.35 percent rather than 0.5 percent of the population, a share still above the OECD average.

diation, and support services for unhoused households, alongside continued growth in shelter capacity.

Most unhoused individuals nonetheless pass through the shelter system. Figure 1 summarizes the pathways available to an individual following a housing shock. These pathways form a ladder with four rungs: the street, emergency accommodation, transitional housing, and ordinary housing. From the street, the main entry point into the shelter system is the nationwide emergency hotline *115*, the fourth most used emergency hotline in France after the medical, fire, and police emergency numbers. In most cases, individuals call the hotline themselves, but street outreach workers or other social workers can also contact it directly. The hotline distributes beds in emergency shelters and hotels, which form the second rung. Emergency accommodation provides a bed, often for a few nights at a time, with limited support. During this period, however, a social worker should be in contact with the individual to discuss longer-term solutions, such as applying for a place in transitional housing. Transitional housing, the third rung, provides a place for several months, together with comprehensive social support, and is often seen as the first stable housing solution. Ordinary housing, mainly social housing through priority allocations, sits at the top of the ladder. Individuals on the street, in emergency accommodation, and in transitional housing are considered a priority group for social housing allocation. However, the French social housing stock has few units accessible to the poorest households (Madec, Parodi, Timbeau, Joutard, Portefaix and Aubisse, 2023). This is another reason why many unhoused individuals pass through the shelter system, and why the shelter system is the main focus of this paper's analysis.

Shelter capacity has grown by 38 percent over the last decade, to about 205,000 places in 2025, of which roughly 147,000 are emergency beds and 54,000 transitional places. Despite this increase, shelters cannot meet the underlying demand. On average, 60 percent of weekly requests for emergency shelter go unmet, and the median wait for a place is six and a half months.[9]

Both the shelter system and the emergency hotline *115* are managed by the SIAO (*Service Intégré d'Accueil et d'Orientation*; in English: *Integrated Service for Reception and Orientation*), which comprises 96 local units organized at the level of the *département*, the primary administrative unit in France. These units maintain a comprehensive inventory of all available shelter capacity, spanning both short-term emergency facilities and time-limited transitional housing. Based on individual requests, the SIAO then coordinates the placement and referral of individuals to available accommodation. Every request, met or not, is recorded in the SIAO's shelter management system, which is one of the main sources of variation that this paper exploits, both to measure homelessness and to measure the impact of the shelter system on the labor market trajectories of individuals experiencing homelessness.

[9]These figures represent the author's calculations based on the SIAO shelter management records. Yet, it likely represents a lower bound, as many individuals experiencing homelessness report that they have either never called or have stopped calling the *115* hotline after repeated failed attempts (Fondation pour le logement (2025)).

3 Data

I combine shelter records with administrative employment and unemployment data in France to recover labor market trajectories in and out of homelessness for over one million individuals from 2017 to 2025. The nationwide homeless shelter records cover the universe of shelter stays and every request for shelter made through France's nationwide emergency hotline, including unmet requests. This allows me to recover flows into and out of homelessness for a nationwide sample at the daily level. The matched unemployment insurance records and employer–employee data allow me to recover labor market trajectories, covering earnings, employment spells, and unemployment benefits, before, during, and after homelessness spells.

3.1 Nationwide Homeless Management Information System

The nationwide homeless management information system consists of two components: shelter stays and shelter requests. Thanks to unique pseudo-anonymized individual and household identifiers, I can link these components and track individuals' entries into and exits from homelessness over time.

Shelter stays. The nationwide shelter stay records include the near-universe of shelter stays in France. The data date back to 2010, but coverage is limited before 2017 and has improved significantly since 2021, which is why some of my analyses focus specifically on this period. Between 2018 and 2025, the records contain 2 million emergency shelter stays and half a million transitional housing stays. They provide detailed information on each stay. First, they record the placement type, that is, whether an individual was placed in a hotel, an emergency shelter, or transitional housing. Second, they record the entry and exit dates, with a note on whether the stay can be automatically renewed if no alternative solution is found. For transitional housing only, the situation at exit is known, that is, whether the individual found a more permanent housing solution, either by entering housing in the private market or social housing or, as for many young adults, by returning to their parents' home. In addition to stay-specific information, the records contain socio-demographic characteristics and information on the individual's situation at the time of the request, described in more detail in the section on shelter requests below. I use this data source to study the impact of the nights spent in shelter following a met shelter request.

Shelter requests. There are two primary channels for requests. First, the main entry path into the system is for individuals to signal their housing distress to the nationwide emergency shelter hotline *115*. Every call to this hotline is registered. Following this first request, social workers may make follow-up requests for the individual to enter emergency or transitional housing shelters. Second, if the individual is already followed by a social worker, for example in a day center or a food distribution service, the social worker can also make the request to the *115* on

the individual's behalf. Both the hotline and the request data records who is filing the request. The request files include one line per household member, while the hotline files only record the primary caller requesting shelter for the household.

Emergency shelter hotline. The vast majority of requests are requests for shelter, although some individuals also call with requests for related services. In total, I observe 11 million calls requesting shelter between 2018 and 2025. Each call record includes the date and time of the call, the type of request, and its status (met or not met). It also includes information relevant for placing the individual, such as their age, nationality, household type, and whether they are a victim of violence. Household type is recorded in great detail, ranging from single mothers with children to minors and single men, as it is one of the main criteria, together with violence, used to prioritize the allocation of places. The record further includes the caller's geographic area and the call-center agent who picked up the call. Because the details of each request are recorded regardless of its outcome, I can exploit variation in the placement of individuals into shelters. As the emergency shelter hotline is the entry point into the shelter system, I use this database to construct my measure of first entry into homelessness and to study labor market trajectories around this event.

Emergency and transitional housing shelter requests. The shelter request data provide a parallel set of records covering emergency shelter and transitional housing requests. These records contain the same fields as the emergency hotline data. In addition, they include a wealth of background information on the individual's situation, including the stated reason for the request, where the individual slept the night before, and their situation at the time of the request. They also indicate whether individuals are on the waiting list for specific types of accommodation and whether they have applied for social housing. Whenever a request is met, the records also provide all the shelter stay information described above. This database therefore provides the most comprehensive measure of unhoused individuals.

Construction of outcome variables. To study the impact of job loss on homelessness, I use both shelter stays and shelter requests, regardless of whether the requests are met. I consider two definitions of homelessness. First, I define an individual as homeless in a given quarter if I observe them staying in a shelter or making at least one shelter request during that quarter. While entry into homelessness is well observed through these channels, it is difficult to determine whether someone who is no longer observed in the shelter system, either through a stay or because they have stopped calling, has found a housing solution. I therefore define an individual as having exited homelessness if I do not observe them for six consecutive months, and I test the robustness of this measure using different time horizons. Second, to study the timing of entry into homelessness, I use the calls made to the emergency shelter hotline and construct an indicator equal to one in the quarter in which the very first call for shelter is made.

To measure shelter access, I construct from the call database an indicator for whether the

first request for shelter was met. I further construct measures of the number of nights spent in emergency and transitional housing shelters, as well as indicators equal to one if an individual spent at least one night in each type of shelter within one month of their first request. Finally, I distinguish the outcomes by shelter type, examining whether individuals entered transitional housing within different time horizons of their emergency shelter call.

3.2 Measuring Homelessness Flows

Existing homelessness statistics mostly recover point-in-time estimates. Despite recent advances in capturing homelessness through different type of records (e.g.food aid nutrition programs (Bruneel-Zupanc et al., 2026) or hospital and welfare records (Beauregard et al., 2026; Costantini, 2026)), two challenges remain: measuring homelessness flows at high frequency for a nationwide sample, and recovering what happens before and after homelessness.

The French point-in-time estimate combines the shelter stock with street and informal settlement counts. Yet, as highlighted above, the French shelter system is highly capacity constrained. The produced point-in-time estimate might thus severely underestimate the number of individuals experiencing homelessness. This subsection addresses the first challenge by combining shelter stays with shelter request records. The linkage to employment and unemployment records, described below, addresses the second.

Measuring flows. The ideal dataset for measuring homelessness flows would fulfill three requirements. First, it would be nationwide, as unhoused individuals may be more mobile than housed individuals. Sub-national datasets thus frequently face attrition, as they cannot distinguish individuals who have been housed from those who remain unhoused but have moved, for example, to warmer parts of the country. Second, it would be at a high frequency level, as many homelessness spells are estimated to be short (Kuhn and Culhane, 1998). Third, it would ensure high coverage of unhoused individuals according to a common definition, such as the ETHOS framework introduced above (Feantsa, 2005).

The shelter stay records fulfill the first two criteria: they are nationwide and high frequency, with records at the daily level. However, while they cover the universe of shelter stays, and thus ETHOS categories 2 and 3, they do not cover rough sleeping (ETHOS 1). The emergency shelter hotline records also fulfill the first two criteria, with an even higher, hourly frequency. By recording unmet requests, they additionally capture individuals who could not be sheltered, which extends coverage to rough sleeping (ETHOS 1). Combining the two data sources increases the number of individuals observed as homeless by 25 percent relative to shelter stays alone. Using this flow measure, I estimate that 1.9 million individuals experienced homelessness in France between 2018 and 2025.

Data coverage. I conduct several exercises to obtain a rough estimate of my data's coverage of the sheltered and unsheltered populations in France. Panel A of Figure A1 reports published figures

on homelessness in France based on point-in-time estimates and street counts, and compares them with the estimated number of unhoused individuals on an average night in 2025 in the shelter records. Comparing the published figures for the general shelter stock with the number of individuals I observe in shelter beds confirms that my records cover the near-universe of shelter stays. Remaining discrepancies can be explained by the total number of shelter beds varying over the year with additional winter shelter programs. For the unsheltered population, I compare the estimated number of individuals sleeping rough and in informal settlements with the number of individuals whose shelter request could not be met. These estimates indicate high coverage of both the sheltered and unsheltered populations in France.

Still, concerns remain about the coverage of the unsheltered population, as some individuals may never call the emergency shelter hotline. These concerns are partly mitigated because social workers in day centers and community social centers can also request a shelter place. Notably, these requests record the individual's situation at the time of the request, which is the street in roughly 50 percent of cases. This allows me to construct a point-in-time measure of the unsheltered population in my sample, using both pending requests where the individual is currently staying on the street and calls refused by the emergency shelter hotline. I can construct this measure for exactly the night on which volunteers conduct the yearly street count in Paris, the *Nuit de la Solidarité* (English: Night of Solidarity). Panel B of Figure A1 therefore compares my estimate directly with the street count, showing a coverage of roughly 80 percent.

Point-in-time vs. flow estimates. The same exercise allows me to compare point-in-time and flow measures. Figure A2 reports the number of distinct individuals in my shelter records on a given night, which recovers the point-in-time estimate. Considering the flow over one month already increases the estimate of unhoused individuals by 12 percent, and considering the entire year almost doubles it.

Geographic variation. Figure A3 turns to regional variation. Panel (a) reports all individuals observed as homeless in 2025 by geographic areas, at *département* level. Homelessness is heavily concentrated in Paris and the surrounding *départements*: 52,157 individuals in Paris and 28,016 in Seine-Saint-Denis, which borders Paris to the north-east. Panel (b) repeats the exercise for individuals who called the *115* at least once. The spatial pattern is almost identical, with a rank correlation of 0.96, but the level is not: only 230,573 of the 540,162 individuals in panel (a), or 43 percent, appear in panel (b). This channel misses individuals who enter through a request made by a social worker, notably for transitional housing whose spots cannot be allocated through the hotline, as well as those whose most recent call predates 2025. Panel (c) reports the number of individuals in a shelter bed on an average night, combining emergency and transitional housing places. The geography again reproduces panel (a), with a rank correlation of 0.97. However, the ratio of individuals observed over the year to beds occupied on a given night ranges from 1.9 in Paris and Seine-Saint-Denis to 3.5 in Nord and 3.7 in Bouches-du-Rhône, compared with 3.0

nationally: the Parisian system keeps individuals for far longer highlighting potential re-housing constraints due to tight rental markets. Panel (d) reports the share of *115* calls that end without a placement. This share is positively but less strongly correlated, with a rank correlation of 0.41 with panel (a). Among the largest *départements*, the rate of non-met requests in Paris is high at 77 percent, while Seine-Saint-Denis, at 40 percent, lies well below the national figure of 60 percent.

The homeless population in 2025. Table 1 characterizes the homeless population in France in 2025, capturing the full inflow and outflow of roughly half a million individuals. Importantly, homelessness spells are highly skewed: 26 percent of observed spells last less than one month, while 41 percent were still ongoing at the end of 2025. The table breaks down the population by household type and age. For each group, it reports the group's overall share, the share of children and women within it, and the share of bed nights it accounts for. While single adults, the majority of whom are men, are the largest group, they do not account for the most shelter bed nights, indicating that this group is more exposed to street homelessness. The largest contributors to bed nights are single parents with children, the vast majority of whom are single mothers, and couples with children. Children under 18 make up 30 percent of all individuals observed as unhoused. Overall, many unhoused individuals are young, with slightly more than 30 percent aged 18 to 34. In this age group, the gender split is fairly equal, whereas men are the majority among older unhoused individuals.

3.3 Administrative Employment and Unemployment Records

While the homeless management information system provides a detailed view of entries into and exits from the shelter system, it lacks information on individuals' labor market outcomes. To reconstruct these trajectories, I merge the shelter records with administrative employment and unemployment data based on names and dates of birth in a highly secure environment.

Administrative employment records. To analyze labor market outcomes, I use the *Déclaration Sociale Nominative* (DSN) for 2017–2025. The DSN consists of monthly declarations submitted by employers for every employee, enabling me to track the universe of work contracts for individuals registered with the French public employment service since 2017.[10] For each contract, it records the start and end dates as well as the reason for termination. These detailed termination reasons allow me to identify all economic layoffs, from which I construct the analysis sample for the job loss design. They also allow me to examine, for all unhoused individuals, why their contracts ended, for example due to a dismissal, health-related reasons, or a quit. In addition, I observe individuals' monthly earnings, occupation group, and contract type (permanent, fixed-term, or temporary agency contract), the postal codes of their residence and of their employer, and the

[10]Given the weak labor market attachment of the population of interest, most individuals are likely to have registered with the public employment service at some point, so this restriction should capture the near-universe of relevant employment contracts. Employment rates may nonetheless be underestimated, as these administrative records do not cover self-employment.

firm and establishment identifiers. I use this information, combined with socio-demographic information from the unemployment registers, to construct the matched control group for the economic layoff events.

Administrative unemployment records. I also have detailed information on individuals' unemployment spells from the French public employment service registry. Individuals register with the public employment service either to receive unemployment benefits, which requires registration, or to receive job search assistance. At registration, they fill out a detailed form that records socio-demographic information, such as age, marital status, education level, and geographic location, as well as job search characteristics, such as whether they are looking for a full-time or part-time job, their reservation wage, and the maximum distance they are willing to commute. I use these records in particular for detailed heterogeneity analyses in both quasi-experiments. Furthermore, I observe the unemployment insurance records, that is, all claims for unemployment insurance, whether they were granted, the length of the unemployment insurance spell, the benefit amount, and the potential benefit duration.

Construction of outcome variables. I use the administrative employment records to study the effect of job loss on subsequent labor market outcomes. For each quarter before and after the job loss event, I define an indicator equal to one if the individual is employed in that quarter, as well as the individual's average monthly earnings in that quarter. I also use these records to study the impact of shelter access on labor market trajectories. For individuals employed at the time of their first call to the emergency shelter hotline, the key outcome is whether they remain in their job. I therefore construct an indicator equal to one if they are still in the same contract they held at the time of the call, at different time horizons. I further exploit the detailed information on the reason for the end of the contract provided in the administrative employment records. I focus on reasons influenced by individuals' behavior, that is, reasons other than the expiry of a fixed-term contract. Specifically, I consider dismissals for misconduct, dismissals for poor performance during the probationary period, health-related terminations unrelated to workplace accidents, and quits. For individuals unemployed at the time of the call, I focus on subsequent job finding, measured both as whether the individual is employed in a given month after the request and as job finding hazard rates. I also construct the number of days worked, average monthly earnings, and indicators for different contract types. Finally, I define an individual as being in a stable employment situation, one that would likely pass a landlord check, if they hold a permanent contract or a fixed-term contract lasting at least six months.

3.4 Record-Linkage Procedure

While I only observe the pseudo-anonymized identifiers, the full shelter stay and request records include the names, dates of birth, and phone numbers of unhoused individuals. These identifiers are used to link the shelter records at the individual level to the administrative employment

and unemployment records, based on names and dates of birth, in a secure environment.[11] Individuals are linked if they held a work contract or were registered with the French public employment service or the French youth public employment service at any point since 2017. The linked data allow me to study detailed labor market trajectories around the first homelessness episode. Appendix A describes the matching procedure in detail.

The match rate is roughly 50 percent among unhoused adults. Before reconstructing individuals' labor market trajectories, I therefore assess whether the matching process introduces selection. Table A1 compares the characteristics of the matched and unmatched samples after excluding children and recent migrants, who are less likely to appear in French administrative labor market records. Both groups are balanced across key demographics, including household composition, age, and gender. The only significant difference is that non-French nationals are less likely to be matched. This discrepancy is likely due to the imperfect identification of all recent arrivals: the unmatched group may still contain individuals who lack the legal right to work, which precludes formal employment and registration with the unemployment agency. Among French citizens only, the match rate is roughly 70 percent among adults. These estimates are in line with overall estimates of adults being part of the labor force. Thus, as not everyone is part of the labor force, a match rate around 70 rather than 100 percent is expected.

4 Labor Market Trajectories of Unhoused Individuals

This section provides new descriptive facts on the labor market trajectories of unhoused individuals in France using the merged sample. I first estimate the labor market attachment of individuals becoming homeless for the first time, which I capture through their first call to the emergency shelter hotline.[12] I then describe the employment conditions of unhoused individuals, both for those employed at the time of their request and for those with a prior employment history. Next, I use the full set of employment registers to characterize the workers at risk of becoming homeless in the following year. Finally, I plot individuals' labor market trajectories around their first request for shelter through the emergency shelter hotline.

Fact 1. Unhoused individuals are not detached from the labor market. At the time of their first shelter request through the nationwide emergency hotline, 11 percent of unhoused individuals are working, 24 percent worked within the previous two years, and 55 percent worked within the previous five years. These figures, however, mask important heterogeneity, as highlighted in Figure 2. Recent labor market attachment is highest among younger unhoused individuals: 44 percent of young adults under the age of 25 worked in the two years before their shelter request.

[11] I also used phone numbers in the merge, as they were initially expected to be more reliable than names in some cases. In practice, however, merging on phone numbers added very little. Appendix A provides the exact details.

[12] I restrict the sample to individuals calling the hotline, as this provides the best approximation of the onset of homelessness. By contrast, individuals whose shelter request is made by a social worker may already have been homeless for a longer period.

Yet 44 percent of this group had not worked at all before their request. Older individuals, by contrast, are more likely to have any prior employment record, with only 26 percent having no record at all. Nationality also plays an important role. French citizens are more likely to have a prior employment record, with only 26 percent having no record at all and 61 percent having worked in the previous five years. Non-French citizens are the most likely to have no observed prior employment contract, which is linked to potential issues with the right to work, yet they are slightly more likely to be working at the time of their shelter request, at 12 percent. This finding confirms recent evidence from other countries showing that unhoused individuals are not detached from the labor market (Beauregard et al., 2026; Meyer, Wyse, Grunwaldt, Meyer and Wu, 2023; Von Wachter et al., 2020). It raises, however, questions about the labor market conditions of unhoused individuals, as employment and homelessness have long been thought to be mutually exclusive, on the assumption that income from work should cover the rent.

Fact 2. Unhoused individuals face precarious labor market conditions. Table 2 and Table 3 examine the labor market conditions of unhoused individuals. Table 2 focuses on workers employed at the time of their first request, and Table 3 on unemployed individuals with a prior labor market history. Workers employed at the time of the request have median monthly earnings of around 1,446 euros; half of them are on open-ended contracts, 60 percent work full-time, and most work in blue-collar occupations. However, their median number of days worked is very low, highlighting precarious labor market attachment. Conditions are better for the slightly more than half who earn at least 1,500 euros per month, who work more days and are more likely to work full-time and on permanent contracts, with median monthly earnings of 1,750 euros. Those who worked at least 185 days over the previous two years, roughly 30 percent of employed workers, have median monthly earnings of around 1,556 euros and a median of 378 days worked. Across these groups, socio-demographic characteristics are similar: women make up around 40 percent, the median age is between 36 and 38, and roughly 20 percent have children.

Table 3 examines the past employment histories of the unemployed. Among all those with a prior employment record, the median time since the last job is slightly less than two years. Their earnings in this last contract are very low, with a median of only 247 euros, which can be partly explained by a low share of permanent contracts (21 percent) and a high share of temporary agency contracts (31 percent). In addition, 54 percent are blue-collar workers. Those whose last job ended less than two years ago look similar to this overall sample. By contrast, those who worked at least half a year over the previous two years have significantly more days worked in that period and higher earnings, with a median of around 877 euros. In this group, the share of permanent contracts is also higher, at around 36 percent, and the share of blue-collar workers slightly lower, at around 48 percent. However, this group represents only 15 percent of those with a prior labor market attachment.[13]

[13] Table D1 and Table D2 repeat this exercise for individuals requesting transitional housing. These individuals are typically accompanied by social workers, with whom they may also work on their employment prospects, which is

Fact 3. Precarious workers face a higher risk of becoming homeless. The low earnings and high share of precarious short-term contracts among unhoused individuals raise the question of how labor market precariousness relates to homelessness risk. To investigate this, I take all employment contracts in a given year from the employment registers, regardless of whether the individual is ever observed as homeless, and plot them against the probability that the individual is in a shelter or makes a shelter request in the following year. Figure 3 shows a strong negative correlation between earnings and the risk of becoming homeless in the following year. This risk, however, varies with contract type and local rent conditions. It is substantially lower for workers with a permanent contract than for those with a more precarious fixed-term contract, and much higher for those living in high-rent areas.

Fact 4. Labor market trajectories of unhoused individuals drop sharply at the onset of homelessness for those attached to the labor market. Figure 4 examines the labor market trajectories of individuals around their first observed spell of homelessness. For this analysis, the onset of homelessness is proxied by the date of the individual's first recorded request for shelter. To identify potential shifts in labor market outcomes, I restrict the sample to individuals with at least a marginal labor market attachment, defined as having worked for at least 180 days within the preceding five years. The figures reveal a sharp and substantial decline in both employment and earnings around the time of the first shelter request. However, some of the decline starts already before. These outcomes remain depressed several years after the initial request, a pattern consistent with a vicious cycle between unemployment and homelessness. While baseline earnings are notably low, these figures represent unconditional earnings: in the two years before the first shelter request, only about 45 percent of this marginally attached group were employed.

These patterns show that labor markets and homelessness are not segmented and that their interaction may have important policy implications. They also point to both directions of this relationship. On the one hand, the strong gradient in homelessness risk by earnings and contract type (Fact 3) suggests that precarious employment and job loss may push individuals into homelessness. On the other hand, the persistent decline in employment and earnings after the first shelter request (Fact 4) suggests that homelessness itself may cause job loss and lower job finding. The descriptive evidence cannot separate these two channels: the sharp drop in employment occurs precisely around the onset of homelessness, where both directions overlap, and common shocks, such as health shocks, substance use, or divorce, may drive both. This motivates the two quasi-experiments developed in the next sections, each isolating one direction of the relationship.

why this population is slightly more attached to the labor market overall. Furthermore, only half of this population is staying on the street, with others being in emergency shelters or friends and family.

5 The Effect of Job Loss on Homelessness

To recover the casual effect of job loss on homelessness risk, I restrict the analysis to workers that were laid off for an economic reason, that is a reason independent of their individual performance, and compare them to similar workers working in firms not experiencing economic layoffs. I find that job loss increases the risk of becoming homeless by 50% in the three subsequent years after the layoff.

5.1 Empirical Framework

The ideal experiment to estimate the causal effect of job loss on homelessness would randomly displace workers. In practice, however, job loss is rarely random: confounding factors, such as health or family shocks, may simultaneously influence both the probability of displacement and the risk of homelessness. To address this endogeneity, I focus on workers displaced in economic layoffs at their firm, which isolates the impact of job loss from other idiosyncratic shocks. Using a one-to-one matching approach, I compare the homelessness risk of these displaced workers with that of workers who are similar on a wide range of socio-demographic and job characteristics at the time of the layoff.

5.1.1 Sample Construction and Matching

The primary specification uses the universe of collective economic layoffs between 2017 and 2023, defined as events in which at least two workers were dismissed for economic reasons from a given firm in a given month.[14] I restrict the sample to workers aged 20 to 60 at the time of displacement who held an open-ended contract and were French citizens. The restriction to French workers serves two purposes: it ensures better match quality with the administrative employment records, and it rules out differences in administrative constraints linked to work and residence permits. Each treated worker is matched one-to-one to an observably similar control worker who was not displaced in the same calendar year and was not employed in a firm engaged in economic layoffs during the analysis period. Matching is exact on earnings bins, four-digit occupation, geographic area (*département*), age bins, job-tenure bins, and high school completion. When there are multiple potential matches, one is selected at random. This procedure yields a matched

[14] Under French law, an economic dismissal becomes collective when it concerns at least two workers within 30 days. Collective dismissals require the employer to consult employee representatives and to inform the labor administration, and dismissals of at least 10 workers within 30 days in firms with at least 50 employees additionally require a job protection plan (*plan de sauvegarde de l'emploi*). Restricting the sample to collective dismissals ensures that the layoff reflects a firm-level shock subject to additional oversight, rather than an individual dismissal declared as economic. The full set of legal grounds for economic layoff that I include is: *licenciement suite à liquidation judiciaire ou à redressement judiciaire*; *licenciement suite à fermeture définitive de l'établissement*; *licenciement pour motif économique*; *licenciement pour fin de chantier ou d'opération*; *autre fin de contrat pour motif économique*; *rupture pour motif économique dans le cadre d'un contrat de sécurisation professionnelle (CSP)*; and *rupture anticipée d'un contrat de travail ou d'un contrat de mission suite à fermeture de l'établissement*.

sample of 53,619 treated workers, drawn from a pool of 240,370 displaced workers, and an equal number of controls.

5.1.2 Matched Difference-in-Differences

To estimate dynamic treatment effects, I run the following regression:

$$Y_{it} = \alpha + \gamma Treat_i + \sum_{t=-P, t\neq 0}^{T} \delta_t (Treat_i \times Time_t) + \phi_t + \epsilon_{it} \tag{1}$$

where Y_{it} denotes the outcome of worker i in quarter t relative to the quarter of the economic layoff. Each matched control worker is assigned a placebo displacement date corresponding to the actual layoff date of their treated counterpart. $Treat_i$ is an indicator equal to one for individuals displaced in an economic layoff and zero for matched controls. I further include year fixed effects, ϕ_t, to account for macroeconomic shocks specific to the layoff cohort. Standard errors are clustered at the individual level.

To summarize the overall magnitude of the effect, I also estimate the following pooled specification:

$$Y_{it} = \alpha + \gamma Treat_i + \beta (Treat_i \times Post_t) + \lambda Post_t + \epsilon_{it} \tag{2}$$

The only modification relative to Equation (1) is the aggregation of the post-event periods into a single estimate: I replace the individual time-to-event indicators with an indicator $Post_t$, equal to one for all quarters following the economic layoff.

5.1.3 Validity of the Design

This identification strategy relies on the assumption that workers displaced in an economic layoff are dismissed solely for economic reasons unrelated to their individual performance. Specifically, the counterfactual assumes that, in absence of the layoff, these individuals would have remained at their firms. Under this assumption, treated and control workers would have followed parallel trends, which can be assessed visually by comparing pre-event trajectories across the outcomes of interest. As noted by Britto et al. (2022), this design is robust to recent methodological concerns about staggered difference-in-differences: by relying on a group of never-treated workers, it avoids the biases associated with heterogeneous treatment effects in standard two-way fixed effects models (Callaway and Sant'Anna, 2021; De Chaisemartin and d'Haultfoeuille, 2020).

A main concern for the validity of the design is that workers selected for layoff in economic layoff events may not be a random subset of the firm's workforce. The literature therefore often focuses on plant closures. As in this case all workers are laid off, this selection is not an issue. However, the number of plant closures in my data is too small to be exploited. To test the robustness of the design, I therefore employ a second identification strategy that focuses only

on workers laid off in economic layoff events. As all workers in this sample were selected for layoff, the comparison relies only on the timing of displacement rather than on who was selected. Following Callaway and Sant'Anna (2021) and Sant'Anna and Zhao (2020), this strategy exploits the quasi-random timing of displacement events and compares not-yet-treated workers with those already treated. It uses multiple years of displacement events without imposing matching restrictions. A key advantage of this approach is that, unlike in the matched design, pre-trends are not mechanically aligned by the matching algorithm, which provides a more transparent test.

5.2 Main Results

I find a 50 percent increase in the probability of becoming homeless within the three years following the economic layoff. While the baseline probability of becoming homeless is very low, this finding shows that homelessness is an important dimension to consider when assessing the negative consequences of job loss.

Labor market outcomes. Columns (1) to (3) of Table 4 report the pooled estimates from Equation (2). In line with the literature, I find large, significant negative effects on labor market outcomes over the three years after the layoff: employment falls by 32 percent, employment in stable open-ended contracts by 41 percent, and average monthly earnings by 31 percent from a baseline mean of 2,109 euros. Figure B1 traces the dynamic path of labor market outcomes in quarters relative to the economic layoff. Consistent with the existing literature on job displacement, all three outcomes drop sharply in the first quarter after the layoff and recover only partially over the subsequent three years. For example, employment drops by 80 percent relative to baseline in the quarter after the layoff and is still 20 percent lower three years later. Pre-event trends are flat, supporting the parallel trends assumption.

Homelessness. Column (4) of Table 4 shows the effect of job loss on the probability of becoming homeless. The homelessness risk following job loss increases by approximately 50 percent relative to a baseline mean of 0.05 percent at the time of the layoff. Figure 5 presents the dynamic effect on homelessness risk. The effect emerges within the first year: by the third quarter after the layoff, the probability of becoming homeless is around 50 percent higher than at baseline. It remains at roughly this level during the second year and rises again in the third year, reaching close to twice the baseline three years after the layoff. The point estimates are noisy, reflecting the very low baseline incidence, and individual quarters are mostly significant only in the third year. Panel (b) focuses on male workers, who are more exposed to economic layoffs. For men, the increase is more gradual and steady and becomes statistically significant from around quarter 7 onwards. In the third year, their homelessness risk is roughly twice the baseline, which is itself slightly higher than for the full sample (0.07 percent vs. 0.05 percent).

Entry into homelessness. To understand the mechanisms behind the increased risk of home-

lessness, Figure 6 presents the effect of job loss on the first call for shelter. The timing of this first call allows me to assess, for example, links to the expiry of unemployment benefits. For male workers, first calls for shelter already rise somewhat in the first quarters after the layoff, but spike especially between quarters 6 and 9, that is, after around two years, when unemployment insurance benefits are exhausted for most workers in the sample.[15]

Income trajectories of displaced workers becoming unhoused. To shed further light on the timing of entry into homelessness, Figure B9 describes the income trajectories of treated workers who became unhoused after the economic layoff. As this exercise conditions on individuals who became unhoused and does not include a control group, it is purely descriptive. While I cannot calculate the exact date of benefit expiry, I can measure individuals' resources by combining earnings and unemployment benefits.[16] The figure tracks their monthly income alongside their cumulative rate of entry into homelessness. Three features stand out. First, income remains constant in the months immediately following the layoff, reflecting income replacement through unemployment benefits. Second, income then declines steadily, with sharper drops around months 12, 18, and 24, corresponding to standard potential benefit duration thresholds in French unemployment insurance, which depend on prior employment and age.[17] Third, entry into homelessness accelerates exactly as income falls. While descriptive, these patterns are consistent with the timing of first calls for shelter: workers can sustain housing payments while their income remains adequate, but as unemployment benefits are exhausted and savings are depleted, the rent burden becomes unsustainable.

5.3 Mechanisms

The aggregate effect on homelessness masks important heterogeneity: not all individuals in my sample face the same risk of becoming homeless after losing their job. The effects are driven by low-income and single job seekers, pointing to a lack of resources as the mechanism behind my main results.

To test formally for heterogeneity, I interact the $Treat_i \times Post_t$ indicator with different socio-demographic and job characteristics. Figure B2 and Figure B3 plot the interaction coefficients for socio-demographic and job characteristics, respectively.

[15] Several unemployment insurance regimes were in place during my analysis period. Until October 2017, the maximum benefit duration was 24 months for workers under 50 and 36 months for those aged 50 and above. From November 2017 to January 2023, it was 24 months for workers under 53, 30 months for those aged 53 and 54, and 36 months for those aged 55 and above. From October 2021, the calculation of benefit duration within these maximums changed, mainly shortening entitlements for workers with fragmented careers. From February 2023, maximum benefit durations fell by 25 percent, to 18 months for workers under 53, 22.5 months for those aged 53 and 54, and 27 months for those aged 55 and above.

[16] Unemployment benefits include the allowance paid under the *contrat de sécurisation professionnelle* (CSP) during the first year after an economic layoff.

[17] The drop around month 18 may reflect both workers with shorter prior employment histories and the 18-month maximum benefit duration that applies to workers under 53 whose entitlement opened from February 2023 onwards.

Household insurance. Figure B2 reports heterogeneity across socio-demographic characteristics, including marital status. The effects are larger for single individuals and smaller for married individuals. This points to the role of household insurance, for example through a working partner, in cushioning the income shock after job loss.

Unemployment insurance. Workers laid off for economic reasons can join the *contrat de sécurisation professionnelle* (CSP), under which they receive a specific allowance rather than standard unemployment benefits during the first year after the layoff. This makes it difficult to calculate the exact potential benefit duration. Instead, I proxy for benefit duration using two key inputs into its calculation: age and the number of days worked before registration. First, workers aged 53 and over are entitled to longer benefits. In line with this, I find smaller effects for this group (see Figure B2). SSecond, benefit duration depends on the number of days worked during the 24 months preceding the layoff. For most of my sample period, workers with one year of tenure were therefore entitled to 12 months of benefits plus any days worked in other jobs during the preceding year, whereas workers with more than two years of tenure reach the maximum duration. In line with these shorter entitlements, Figure B3 shows that the effect is largest among workers with less than one year of tenure in the job they lost. Conversely, workers with more than two years of tenure show smaller effects, consistent with longer entitlements but potentially also with higher savings accumulated on the job, which provide a larger income cushion.

Earnings. In line with these findings, workers with low earnings, below 2,000 euros gross per month, face a higher homelessness risk, while those earning more than 3,000 euros gross per month are largely protected against homelessness.

Welfare benefits. Once unemployment insurance runs out, welfare benefits provide another source of income. However, young adults are not entitled to welfare benefits before the age of 25, which leaves them in a particularly vulnerable position once their unemployment benefits expire. Consistent with this, I find larger effects for this group.

Housing. Finally, housing and the worker's overall rent burden may play a crucial role in the risk of becoming homeless. Figure B5 shows how living in social housing affects the estimated effects. Overall, individuals living in social housing are more at risk of becoming homeless, which likely reflects that they also have lower earnings. However, when I focus on low-income individuals in high-rent areas, where social housing may reduce the rent burden the most, I find that social housing protects individuals against becoming homeless.

Overall, the heterogeneity results point to a lack of resources as a driver of entry into homelessness. The effects are concentrated among workers with fewer income sources to fall back on, whether through a partner, longer unemployment insurance entitlements, higher earnings, or access to welfare benefits. Conversely, a lower rent burden through social housing protects low-income workers in high-rent areas.

5.4 Robustness

In this subsection, I assess the robustness of my results through a range of exercises. I first use an alternative research design to address potential selection into economic layoffs. I then show that the results are robust to excluding the Covid-19 period. Finally, I assess how mental health events and moving responses might affect my results.

Variation in the timing of job displacement. To address potential selection into economic layoffs, I employ the alternative identification strategy described in the research design above, which exploits the staggered timing of economic layoffs and uses not-yet-treated workers as controls. Figure B8 presents the estimated effect on the risk of homelessness using this approach. Despite the different methodological choices, the estimated effects are similar in magnitude and timing to those of the matched design, supporting the causal link between job displacement and housing instability.

Excluding Covid-19 years. Covid-19 affected both firms' economic performance and the policy response to layoffs. It also affected housing policies, with more individuals placed directly in hotels that had free beds due to the lockdowns, and it prolonged the winter eviction moratorium, making evictions more difficult. Figure B6 and Figure B7 show that the effects remain stable when excluding economic layoffs that occurred in 2020 and 2021.

The role of mental health. A key concern for the validity of my design is that mental health shocks might drive both job loss and homelessness. The focus on economic layoffs rather than dismissals due to individual performance alleviates this concern, but given the selection concerns discussed above, it cannot fully rule it out. Beauregard et al. (2026) show that mental health episodes often occur in the months immediately preceding homelessness. If mental health episodes were the primary driver, I would therefore expect most of the increase in homelessness to occur shortly after job loss. While some increase occurs within the first year, most of the effect materializes later, around the expiry of unemployment benefits, which makes it unlikely that the effects are driven by mental health events.

Moving as a response to homelessness risk. Individuals living in high-rent areas might respond to homelessness risk by moving to areas where rents are cheaper. To assess to what extent this might affect my results, I track moves through the employment records. I find that very few workers in my sample move across *départements*, with only roughly 1 percent doing so each year. Similarly, Bruneel-Zupanc et al. (2026), who study food and housing insecurity among the French poor, find very low migration rates, with only 1.2 percent of resident households moving to a different zip code and less than 0.1 percent to a different *département*. By contrast, roughly 40 percent of individuals who became homeless after an economic layoff are eventually observed in another *département*, underlining the importance of the nationwide coverage of my data (see Section 3).

6 The Effect of Shelter Access on Labor Market Trajectories

To study the reverse direction of the vicious cycle between homelessness and unemployment, I turn to the labor market consequences of homelessness itself. The ideal experiment would randomly assign housing among otherwise identical individuals who have just lost their home. I exploit the closest counterpart available in the shelter system: whether an individual's very first request for shelter is met. As the shelter system is highly saturated, this depends on the leniency of the call-center agent who happens to pick up the call. Because calls are quasi-randomly assigned to agents within a *département*, month, and shift, agent leniency isolates a positive housing shock, that is, rapid access to shelter rather than at least some period of street homelessness. I find that rapid shelter access reduces job separations among callers who are employed and raises job finding among callers who are unemployed.

6.1 Empirical Framework

This section presents the instrumental variables approach used to estimate the causal effect of rapid shelter access on the labor market outcomes of individuals experiencing homelessness. I first describe the research design and then the call-center agent leniency measure, which I use as an instrument for whether an individual's first request for shelter is met. Finally, I discuss the assumptions underlying the design and provide evidence supporting them.

6.1.1 Sample Construction

I focus on individuals' first request for shelter through the nationwide emergency shelter hotline and impose three further restrictions. First, because systematic call-level tracking only begins in 2020, I focus on calls made from then onwards. However, during the peak of the Covid-19 crisis, assignment rules differed from the regular regime as part of a large-scale effort to reduce street homelessness through the requisitioning of hotels. I therefore start the sample in May 2021, at the end of the extended eviction moratorium during Covid-19, after which the additional hotel capacity was progressively withdrawn, in particular from 2022 onwards. Second, to observe labor market outcomes for at least one year after the call, I restrict the sample to calls made before March 2024. Third, I restrict the sample to the working-age population, defined as individuals aged 18 to 60 at the time of the call.

The final sample consists of 97,385 individuals making their first request for shelter between May 2021 and March 2024. I consider two subpopulations separately: individuals who hold an employment contract at the time of the call (15,924 callers, the "employed" sample) and individuals without a contract at the time of the call but with a prior labor market history (81,461 callers, the "unemployed" sample).[18]

[18]Restricting the unemployed sample to individuals with a prior labor market history, in addition to the age restric-

A natural concern is selection into calling the hotline. Several features of the setting alleviate this concern. First, the hotline is the main entry channel into the shelter system. It receives over 14 million calls per year (Dihal, 2025).[19] As shown in Section 3, combining these calls with the shelter stay records covers the near-universe of the sheltered population and roughly 80 percent of the unsheltered population counted in the Paris street count. Second, because the system is saturated, the call records cover both individuals who are admitted to a shelter and those whose request is denied, so that the data also capture, at least in part, individuals who end up on the street. Nonetheless, interviews conducted during the annual street counts (*Nuit de la Solidarité*) indicate that many individuals sleeping rough have not called the hotline recently, consistent with discouragement after repeated denials (Fondation pour le logement (2025)). Restricting the analysis to the very first call an individual makes ensures that the sample is not selected on prior denials, as no discouragement can have occurred before the first call. It does not, however, capture individuals who never call the hotline.[20]

6.1.2 Instrumental Variables

The challenge for interpreting OLS in this setting is that whether a first request is met may be correlated with the caller's unobservable characteristics or with the timing of unobserved shocks that affect subsequent labor market outcomes. The descriptive analysis in Section 3 shows that many callers have experienced negative labor market shocks in the run-up to the call, which makes it especially difficult to disentangle the role of homelessness itself from the distress that precedes it. The aim here is therefore an instrument that is independent of these differences.

I leverage the quasi-random assignment of calls to call-center agents for identification. My instrument $Z_{j(i)}$ is the leave-one-out shelter-request acceptance rate of agent j who handles individual i's first call. I estimate the following population regression model using two-stage least squares:

$$S_i = \gamma Z_{j(i)} + X_i'\alpha + \psi_{c(i)} + \nu_i, \tag{3}$$

$$Y_i = \beta S_i + X_i'\delta + \psi_{c(i)} + \varepsilon_i, \tag{4}$$

where S_i is an indicator for whether individual i's first request for shelter is met, Y_i is the labor market outcome of interest, and X_i is a set of individual controls (age bins and nationality groups). The fixed effects $\psi_{c(i)}$ are for the cell within which assignment to agents is as good as

tion, brings the sample closer to the labor force. In combination with the nationality controls introduced below, this also addresses the concern that some callers remain out of the labor force because they lack a work permit.

[19] While there are a total of 14 million calls per year, the overall number of observed requests is much lower as only an estimated 27 percent of calls get picked up and not all of these calls lead to shelter requests (Dihal, 2025).

[20] If anything, first-time callers are likely to have better mental health and lower substance use than individuals who never call, since these conditions make it less likely that an individual is willing or able to call. In some cases, a third party, typically a social worker, places the request on the individual's behalf; these cases, however, represent less than 5 percent of all calls to the hotline requesting emergency shelter.

random: *département* × year-quarter × shift × household type.[21] Household type distinguishes single men, single women, single men with children, single women with children, couples with children, couples without children, and other household configurations. Under quasi-random assignment, the cell fixed effects are the only conditioning required for identification, and the baseline specification includes no further controls. X_i denotes a set of caller characteristics, age bins and nationality groups, that I add in a second specification to check the sensitivity of the estimates and to improve precision.[22] Standard errors are clustered at the agent level.[23] For this analysis I restrict the sample to calls handled by agents with at least 30 first calls in the data, so that the leniency measure can be constructed with reasonable precision.

If the IV assumptions hold and equations (3)–(4) are correctly specified (Blandhol, Bonney, Mogstad and Torgovitsky, 2026), the 2SLS estimand for β is a positively weighted average of the effect of rapid shelter access among compliers, that is, callers whose request would have been met had their call been picked up by a more lenient agent and denied had it been picked up by a less lenient one.

6.1.3 The Call-Center Agent Leniency Instrument

Although inspired by the judge stringency measures used in court settings to recover the causal effects of eviction (Collinson, Dutz, Humphries, Mader, Tannenbaum and van Dijk, 2026; Collinson et al., 2024), the leniency measure here differs in one important respect. If shelter places were freely available, agent leniency would not matter: every request would be met. The instrument therefore relies on the fact that agents face an overcrowded system in which they cannot admit everyone. Priority rules govern part of the allocation — families with small children and victims of domestic violence are prioritized — but within these rules agents handle requests differently. The margin of discretion is largest for single men, for whom no priority rule applies, and it is within this group that leniency varies most across agents in the same *département*, year-month, and shift.

To fix ideas, consider two single men calling the emergency shelter hotline of the same *département* during the same shift. The first reaches an agent who, by the leniency measure defined below, is comparatively strict. The agent checks the bed inventory, sees few places left and, knowing that a single man is not a priority case and that a young woman alone may call

[21] The *SIAO* is organized at the *département* level (96 units, see Section 2), and so is the allocation of shelter places; this is therefore the level at which the instrument and the fixed effects are constructed. The shift fixed effects ensure that agents are compared to other agents working the same shift in the same *département* and quarter. This rules out that agents who systematically work later shifts, when fewer places are left, appear more stringent purely because of capacity. Note that shift fixed effects do not, by themselves, address non-random sorting of calls to agents *within* a shift. This is what the balance tests below examine.

[22] Nationality is recorded in the *SI-SIAO* for every caller, whether or not the record is linked to the employment data. This is why, unlike the layoff design in Section 5, the analysis here does not need to be restricted to French nationals.

[23] Following Chyn, Frandsen and Leslie (2025), **??** in Appendix C shows that inference is robust to clustering at the agent-by-year and *cell* levels.

within the hour, tells the caller that no place is available for him. A common practice is to tell such callers that they may call back at the end of the day in case a bed is still free, which is unlikely. The request is recorded as unmet for lack of places. For the caller, calling again is costly, as waiting times to get through to the hotline are long, and he has just been told that his chances are low. In fact, most callers refused for lack of places do not call back within the day, the week or the month, and three in four are not placed within the year. The second man reaches an agent who is comparatively lenient. She sees the same inventory and makes the same judgement about the remaining places, but before refusing she calls structures that seldom take referrals from the platform or, as is common in Paris, a hotel in a neighboring *département* where capacity is less tight, and secures a bed for several nights. Once in a structure, a caller tends to remain attached to the system, as renewals of the stay are often arranged through the structure rather than through a new call. The two agents therefore applied the same eligibility rules, but the second spent more time on the call looking for a solution. Such agents are, on average, more experienced and better at navigating a capacity-constrained system. This is an important margin in a setting where turnover is high: a third of the agents active in a given year have left by the next, and four in ten calls are answered by an agent in her first year on the job. Which agent answers is determined by the order of calls in the queue, not by any characteristic of the caller, and this is the variation I use to estimate the effect of shelter access against street exposure. The leniency measure therefore captures an agent's propensity to secure a bed for a given caller: in part leniency, in part experience in knowing where to look when capacity binds.

I construct agent leniency as the leave-out mean placement rate of the agent who answered the call, computed on single-men calls and using all emergency shelter hotline call records rather than only the linked sample.[24] I proceed in four steps. First, I use calls from single men, who are typically not a priority group and for whom the agent's margin of discretion is therefore largest. Second, to avoid classifying as lenient an agent who simply faces lower capacity constraints, I residualise each placement decision on the call's *département*, year-month and shift; residualising on the shift matters because agents working shifts in which more places are available would otherwise look systematically more lenient.[25] Third, I average these residuals over all other single-men calls the agent handled between 2018 and 2026, excluding every call made by the focal caller, and set the measure to missing for agents with fewer than 30 remaining calls.[26] The average is taken over all callers, whether or not they can be matched to the administrative data. Fourth, I assign to every caller, not only single men, the placement propensity of the agent who answered his or her first request.

[24] Over 2018–2026, 6,965 distinct agents appear in the call records; 3,587 handle calls between May 2021 and March 2024, and 3,167 of them answer a first request in that window.

[25] Section 6.5 shows that the instrument is robust to finer constructions, including residualising within the day and hour of the call.

[26] 738 of the 3,167 agents answering a first request in the window fall below this cutoff. They together answer only 5,169 first requests, 1.3 percent of the total. Section 6.5 shows that raising the cutoff to 50, 100 or 200 calls leaves the first stage unchanged.

Assignment is random over the agents on duty in the caller's *département* on the day and shift of the call: which of them answers is determined by the order of calls in the queue. Because rotas are set in advance and do not depend on the callers expected on a given day, and because callers cannot choose or time their call to reach a particular agent, assignment can be taken as quasi-random within the coarser *département* × year-month × shift cell used to construct the instrument, and within the *département* × year-quarter × shift cell used as fixed effects in estimation. The shift is the essential element of this conditioning, since teams and bed inventories differ across shifts; household type is added in the regression as fixed effects not for random assignment but because priority rules make the placement margin differ across types.

Figure C1 shows the distribution of agent leniency across first requests, residualized within *département* × year-month × shift × household-type cells, for the full sample, the subset matched to the employment and unemployment records, and the two analysis samples of callers employed and unemployed at the call. The distribution is the same in the four panels; what differs is only the number of requests. The variation is substantial: the 10th-to-90th percentile range of agent leniency is 15 percentage points, so that, with a first stage of about one, moving from an agent at the 10th percentile to one at the 90th raises the probability that a first request is met from 19 to 34 percent.[27]

6.1.4 Validity of the IV Design

I next discuss tests of the assumptions under which agent leniency is a valid instrument and the IV estimand reflects a positively weighted average of local treatment effects on compliers.

Relevance. To assess the relevance of the instrument, Table C1 reports first-stage estimates from equation (3) for the full (matched and non-matched), the matched sample and the two main samples of the analysis: those that were employed the moment of their first call and those that were unemployed but with a prior work history. Agent leniency has a large and statistically significant effect on the probability that the first request is met. A 1 percentage point increase in leniency raises the probability of being placed by around 1 percentage point. The Kleibergen–Paap F-statistics are 702 in the employed sample and 1352 in the unemployed sample, providing evidence against weak instruments. As Table C2 shows, the first stage is essentially unchanged when adding controls for age and nationality.

The first request is however only the relevant treatment if its outcome is not undone by later calls, and it leads to a broader use of homelessness services beyond staying in a shelter for one night. Table C3 shows that both hold. Callers whose first request is denied do call back, however

[27] The spread is larger than in other leniency designs: in Collinson et al. (2026) the 10th-to-90th percentile range of judge stringency is 6 to 7 percentage points. My leniency measure is furthermore almost free of estimation error, a central concern that Goldsmith-Pinkham, Hull and Kolesár (2026) raise about hand-built leniency measures: the median agent's leniency is computed on about 2,800 calls, and splitting each agent's calls at random into two halves and correlating the two resulting measures across agents gives 0.98.

only roughly half of them: 37 percent have called the hotline again within a week and 52 percent within three months (column 1). Yet the next agent is drawn from the same pool and mostly denies them as well, so that calling back is a poor substitute for having been placed the first time: within a week only 10 percent of refused callers have obtained a place, within a month 18 percent and within a year 29 percent (column 2). On the other side, a place granted is not a place for a single night. Among placed compliers, 44 percent are still in an emergency bed one week after the call, 38 percent one month and 31 percent three months later, against 4 to 5 percent of refused compliers (columns 3 and 4). The levels in columns (3) and (4) understate shelter use, since the stay records are filled by the structures rather than by the hotline and cover about half of the places allocated at the call: hotel nights and stays in structures that do not use the homeless shelter management system are often not recorded in the earlier period of my sample, and some callers do not take up the place allocated to them.[28] The decline across rows is the informative part: relative to the night of the call, 80 percent of placed compliers recorded in a bed are still there one week later, 69 percent one month and 56 percent three months later. The first call is thus decisive for the subsequent engagement with homelessness services.

Exogeneity. Figure C2 presents evidence that socio-demographic characteristics and the caller's current and past labor market situation are not predictive of agent leniency, which provides empirical support to the quasi-random assignment of calls to agents. Panel (a) reports, for the employed sample, the coefficients from regressions of leniency on age, gender, nationality, education and the characteristics of the current contract (earnings, contract length, contract type, full-time status, and occupation group) and their past labor market history (number of days worked in the last 1 (2) years), all expressed in standard deviation units. Panel (b) repeats the exercise for the unemployed sample using past labor market history (days worked in the previous one and two years, time since the last job, and the earnings and contract type of the last job). The coefficients are uniformly small, with the exception of modest imbalances in nationality and age, which I include as controls in all specifications. An F-test that all coefficients are jointly equal to zero cannot reject the null in either sample ($p = 0.59$ for the employed sample and $p = 0.11$ for the unemployed sample).[29]

Monotonicity. The monotonicity assumption of Imbens and Angrist (1994) requires that callers whose request is met would also have had it met by a more lenient agent, and that callers whose

[28] The treatment the instrument shifts is therefore a place allocated at the call, including the cases in which the caller does not show up, which is the margin the agent controls. Furthermore, the overall first stage remains very high: 61 percent of placed compliers spend at least one night in accommodation during the month of the call, 47 percent in the following month, 39 percent three months and 26 percent twelve months later, against 9 to 11 percent of refused compliers.

[29] For the unemployed sample the p-value gets close to significance. However, given the latter sample size, the test can detect imbalances far too small to matter. The relevant check in this case is whether adding the characteristics reporting small imbalances as controls leaves the first stage unchanged, which Table C2 shows is the case. Furthermore, as the unemployed sample is larger we can also include more narrower fixed effects, such as year-month instead of year-quarter, which reports a p-value from the joint F-test of 0.2104, while leaving the second stage results unchanged.

request is denied would also have had it denied by a less lenient agent. Frandsen, Lefgren and Leslie (2023) show that a weaker, average monotonicity condition is sufficient for IV to have a causal interpretation in randomized examiner designs. It would violate average monotonicity, for example, if agents who are lenient with single men were stricter than average with other callers. This concern is more acute here than in court settings, because shelter places are rival within a shift: a place given to one caller is not available to the next. Two features of the setting nonetheless support the assumption. First, because the hotline is heavily used, agents collect only a minimum of information to assess priority — household composition and the stated reason for the call — so that the dimension along which they can rank callers differently is narrow. Statistics highlight that the median call lasts only three minutes (Dihal, 2025). Second, I follow Bhuller, Dahl, Løken and Mogstad (2020) and Norris, Pecenco and Weaver (2021) and construct leniency from one subpopulation — single men, the group for which no priority rule applies — and use it as the instrument for each other household type. I then test the average monotonicity assumption, namely whether an agent who is lenient with single men is also lenient with single women, single parents, and couples. Table C4 reports the first stage results: the first stage is positive and of similar size in every group. Agents thus move all callers in the same direction, which is consistent with average monotonicity holding in this setting.

Exclusion. A key assumption that cannot be tested directly is that agent leniency affects callers' labor market outcomes only through whether the first request for shelter is met. The setup of the hotline limits the scope for violations: the agent's role is to allocate a place according to priority rules, and the agent has no further contact with the caller once the place is assigned or denied. Due to the high saturation of the call center, agents spend only minimal time (a median of 3 minutes (Dihal, 2025)) on the phone with unhoused individuals, making it very unlikely that they impact their trajectory other than through their decision to grant the shelter bed. The one channel through which the agent could matter beyond the night's bed is the caller's subsequent calls, and Table C3 shows that it is small: later calls do not close the gap the first agent opens, as seven in ten refused callers have not obtained a place one year later.

Displacement. Finally, a concern that is usually left implicit in judge designs, where capacity rarely binds at the moment of the decision, is first-order here: beds are allocated against a fixed nightly inventory, so the bed a lenient agent finds might otherwise have gone to another caller. Table C5 tests this on all calls in the estimation window, exploiting that several agents answer callers of the same household type in the same *département* and shift. First, the placement rate of an agent's colleagues in the same shift does not fall when the agent is more lenient (-0.010, SE 0.011). Second, an agent's own placement rate falls only slightly when her colleagues are more lenient: by 0.055 per unit of leniency, about five percent of the effect of her own leniency. Third, the placement rate of a shift is unrelated to the leniency of the agents on duty the day before (-0.001, SE 0.017): beds found on one day are not missing the next. Within the *département*, the

beds lenient agents find are therefore additional. Displacement could however still occur across *départements*, yet its scope is bounded by the 12 to 19 percent of lenient agents' extra placements that are in structures outside the caller's *département*.

6.2 Main Results

In this section, I present the main estimates of the effect of rapid shelter access on labor market outcomes. Variation in whether the first request is met allows me to isolate a positive housing shock at the moment of entry into homelessness: the comparison is between individuals who are directly sheltered, a precarious situation that nonetheless provides a roof over their head, and individuals who spend at least some time without shelter because their request is denied.[30] I find that direct shelter access reduces job separations within three months of the call by 15 percent among those employed at the time of the call, and raises employment one year later by 58 percent among the unemployed.

Job separation. Table 5 reports the effects of rapid shelter access on job separation for the employed sample. Column (1) shows that workers whose first request is met are 6.4 percentage points less likely to see their contract end within three months of the call, a 15 percent reduction relative to the control mean of 41.6 percent. Column (2) shows the mirror image over a longer horizon: these workers are 8.2 percentage points, or 26 percent, more likely to still be employed in the same contract one year after the call.

Columns (3) to (7) decompose contract endings by the reason recorded by the employer in the employment registers. Individuals whose first request for shelter is met are 7.2 percentage points less likely to have their contract terminated within three months of the call for reasons linked to their own behavior, as opposed to the scheduled end of a fixed-term contract. This corresponds to an 80 percent reduction relative to the control mean. Two channels account for this reduction: sheltered workers are less likely to be dismissed for misconduct and less likely to be dismissed at the end of their probationary period. Dismissals for health-related reasons also decline, although the estimate is imprecise. There is no effect on quits, suggesting that the reduction in separations is driven by employer-initiated terminations rather than by workers leaving voluntarily. The pattern is consistent with shelter allowing workers to keep meeting the day-to-day demands of a job, such as showing up and showing up on time, during the most disruptive weeks of housing loss. A cautious interpretation of the decline in health-related dismissals also suggests that individuals in shelters are less exposed to health shocks than those on the street.

Figure C3 traces the effect of a met first request on the probability that the contract held at the time of the call has ended by month t. There is no effect in the first month, after which the

[30] This interpretation assumes that the caller has no outside option other than the street. Based on the housing situation reported at the time of the call, this holds for at least 50 percent of callers, who report sleeping rough, and for a further 15 percent, who report being in an emergency shelter that they must leave.

gap between placed and refused compliers opens quickly: a met request lowers the probability of having separated by 3.5 percentage points at two months, 5.6 at three and 8 at four, reaching about 10 percentage points between months five and eight, when it is most precisely estimated. From month nine the gap narrows, to 6.5 percentage points at twelve months, where it remains significant at the 10 percent level.

Job finding. Table 6 reports the effects on job finding for the unemployed sample. Rapid shelter access raises the probability of being employed one year after the call by 4.9 percentage points, a 58 percent increase over a control mean of 8.4 percent. The low control mean is itself informative: most unemployed individuals entering homelessness remain out of formal employment a year later. The effect is not confined to marginal jobs. The probability of holding an open-ended contract one year after the call rises by 1.4 percentage points (39 percent of the control mean), the number of days worked during the first year rises by 14 days (56 percent of the control mean), and unconditional monthly earnings one year after the call rise by €55, or 57 percent.

To examine the type of jobs individuals find, Table C7 turns from employment at a given point in time to the job finding hazard, that is, the probability of finding a job at any point within the first year after the call. While only 8.4 percent of those whose request was denied are employed one year after the call, 18 percent of them find a job at some point during the year, compared with 26 percent of those placed on their first call. In particular, placed individuals are more likely to find a job that pays more than 1,500 euros and is stable, that is, a permanent contract or a fixed-term contract lasting at least six months. Importantly, there is no difference in the likelihood of finding fixed-term contracts lasting only one day or one week, highlighting that shelter opens access to higher-quality jobs.

Figure C4 plots the effect of a met first request on monthly employment outcomes in the twelve months before and after the call, for the unemployed sample. In the year before the call, the estimates for employment, stable employment and days worked are flat and indistinguishable from zero: callers who reach a more lenient agent were not on a different trajectory before they called. After the call, the effects emerge within the first months and build over the year. The probability of being employed rises by one percentage point in the month of the call, two points at two to three months, three at four to five, and five to six points from month six onwards. Stable employment, defined as having an open-ended or fixed-term contract lasting at least six months, follows with a lag, becoming significant at three months and reaching four percentage points at the end of the year. Days worked increase by about one day per month from month four and by 1.7 days at twelve months, and monthly earnings by 10 euros at six months and about 20 euros at nine to twelve months. The effects are thus not a one-off at the time of placement but accumulate as placed callers find and keep jobs.

Taken together, the separation and finding results establish the reverse direction of the vicious cycle. Being denied shelter right away both destroys existing employment relationships and impedes the formation of new ones, in particular those that would allow individuals to

pass the landlord check. As emergency shelter provides only a lower bound for the effect of stable housing, these effects are likely to understate the role of rapid rehousing at the onset of homelessness.

6.3 Interpreting the IV estimates

The above reported estimates recover a weighted average of treatment effects for compliers, that is unhoused individuals whose placement outcomes depend on call center assignment, under the assumptions as highlighted in subsubsection 6.1.4. To better understand these estimates, I first characterize the complier population. I then characterize the treatment itself, that is, what a met first request implies in terms of subsequent shelter use.

Characterizing the compliers. Table C6 describes the characteristics of compliers, the callers whose first request is met only because they reached a lenient agent, following Bhuller et al. (2020).[31] Panel A describes socio-demographic characteristics, while panels B and C focus on the individual's labor market history. Columns (1) to (3) focus on the employed sample, and columns (4) to (6) on the unemployed sample. Three patterns emerge. First, compliers look like the average caller, not like the average placed caller. Placed callers are more often women (56 against 44 percent in the employed sample, 47 against 39 percent in the unemployed sample), which reflects the priority rules: these callers are placed by lenient and strict agents alike. Compliers, by contrast, are of the same age, as often women, as often without a high-school degree, and have the same work history, earnings and contract types as the sample as a whole. The agent's discretion thus operates on the unprioritized margin that the instrument was built to capture, and the LATE is informative about the typical caller rather than about those the rules already protect. Second, compliers are less often French nationals than the average caller (32 against 39 percent in the employed sample, 42 against 53 percent in the unemployed sample), so that non-nationals are over-represented among those for whom the first agent's decision is pivotal. Third, compliers are concentrated where capacity binds: 23 percent of compliers call from Paris, against 7 percent of all callers. Where beds are scarce, more callers sit at the margin on which the agent's decision matters, so that leniency has the most bite precisely in the *départements* with the tightest capacity.

Characterizing the treatment. Table C8 reports the effect of a met first request on the cumulative number of nights spent in accommodation within 1, 3, 6 and 12 months of the call (columns (1) to (4)), and on subsequent entry into transitional housing, the more stable form of accommodation within the shelter system (columns (5) to (8)). Compliers whose request is denied spend fewer than four nights in accommodation over the following year; those whose request is met spend 67, of which 8 in the first month. Placed compliers also enter transitional housing earlier:

[31] The complier means are estimated with the approach of Abadie (2003), described in the notes to the table, and are shown next to the means in the full sample and among placed callers.

4 percent have done so within three months against 1 percent of refused compliers, and 9 against 3 percent within a year. Panels B and C report the estimates separately for the employed and the unemployed sample. The dose is larger for callers employed at the time of the call, with 95 additional nights in accommodation over the year, than for the unemployed, with 57. The IV estimates that follow are therefore the effect of this bundle — roughly two months of accommodation over the year and, for some, the first step into transitional housing — relative to a counterfactual of a few nights sheltered.

6.4 Mechanisms

The size of the effects of a single met request may appear surprising at first: a shelter provides a roof, but the individual remains in a precarious housing situation. The previous section has shown that a met request is in fact a sustained period in accommodation and, for some, the first step into transitional housing. This section therefore aims to shed more light on the mechanisms underlying the estimates. First, I ask to what extent the estimated effects might run through entry into transitional housing, which is often considered a first step back into stable housing. Second, I conduct heterogeneity analysis to understand for whom shelter access matters.

Entry into transitional housing. The faster entry into transitional housing raises the question of how much of the labor market effect runs through this channel. The timing of the two effects indicates that it is a minor one. Entry into transitional housing is slow: a met request raises it by 2.3 percentage points within one month and by 2.7 within three (Table C8, panel C), with most of the additional entry occurring later in the year. The employment effect, by contrast, is already present three months after the call (Figure C4), before most of the additional entry has taken place. Transitional housing is nonetheless of interest in its own right, and Appendix D examines what it does for those who reach it, exploiting variation in waiting times among requesters for transitional housing to instrument for quick entry. Entering transitional housing within 3 months after the request also raises employment and days worked one year later, but the effects are smaller relative to the baseline than those of a met emergency shelter request, as requesters for transitional housing are more attached to the labor market (see Table D4). One potential reason for this is that they are less likely to be on the street, as many are already in emergency shelters or staying with friends and family.

Labor market attachment. Finally, the heterogeneity analysis shows that the positive effects of shelter access on the employed do not extend to the unemployed as a whole. Among the unemployed, the gains in employment are concentrated among callers who held a job within the three years before the request; for those detached from the labor market for longer, shelter access does not change their attachment. This difference is not one of treatment: the long-term detached receive the same housing response as other unemployed callers, with 57 additional nights in accommodation over the year in the unemployed sample (Table C8, panel C), and

compliers do not differ from the sample in their work history (Table C6). Together with the lower job separation among the employed documented in Section 6.2, the pattern suggests that a bed protects an existing job and creates employment only for those that are at least still marginally attached to the labor market. This is consistent with the model of Imrohoroglu and Zhao (2026), in which two populations experience homelessness: a transitional one, pushed into homelessness by labor market shocks, and a chronic one, whose homelessness stems from health shocks and who have typically been out of the labor market for longer. The callers for whom shelter access improves employment correspond to the first group; those detached from work for several years resemble the second.

6.5 Robustness

In this subsection, I assess the robustness of the results through a range of exercises: alternative constructions of the instrument, unbiased jackknife IV estimation with agent indicators, and sensitivity to the choice of controls, fixed effects and inference.

Instrument construction. I examine the robustness of the results to each of the three choices made in constructing the instrument: the minimum number of calls an agent must have handled, the level at which placement decisions are residualized, and the group of callers on whose decisions leniency is measured.

First, raising the minimum number of calls per agent from 30 to 50 or 100 leaves the first stage unchanged (Table C9). The second-stage estimates are largely robust (Table C10 and Table C11). For the employed sample, the effects on contract terminations are virtually identical to the baseline, while the effects on contract end within three months and, at the 50-call cutoff, on contracts lasting at least one year are smaller and less precisely estimated. For the unemployed sample, the effects on employment, days worked and earnings are somewhat smaller, within about one standard error of the baseline, and remain highly significant.

Second, instead of residualizing placement decisions at the year-month level, I residualize within *département* $\times$ day $\times$ hour cells, comparing an agent only with the colleagues working the same day and hour. The baseline uses the month because for some smaller *départements* there is little variation at the day $\times$ hour level; the two measures nonetheless correlate at 0.90 across first requests in the estimation sample. I keep the fixed effects at the year-quarter level, as finer fixed effects would drop too many singleton observations. The estimates (Table C12 and Table C13) are close to the baseline in both samples, with the exception of the effect on contract end within three months, which roughly halves.

Third, the baseline measures leniency on an agent's decisions on single men, the group to which no priority rule applies, so that the measure captures the agent's discretion. As an alternative, I measure leniency on the agent's decisions on callers of the same household type as the caller, which uses more of each agent's decisions. For single men the two instruments coincide;

for other callers, the alternative also reflects how strictly the agent applies the priority rules, and it shifts the weight across compliers of different household types. Under heterogeneous effects, the two estimates therefore need not coincide. For the unemployed sample, they are virtually unchanged (Table C15). For the employed sample (Table C14), the effect on contract terminations remains significant but is somewhat smaller.

Unbiased jackknife IV (UJIVE). The leniency instrument is constructed by hand as the leave-one-out mean placement rate of the agent who answered the first request. This construction allows me to measure leniency on all calls handled by an agent, including the calls that cannot be matched to the administrative employment data. Goldsmith-Pinkham et al. (2026) raise two concerns with this practice. First, while the leave-one-out mean removes the caller's own decision from the instrument, residualizing on the fixed effects that ensure as-good-as-random assignment is done on the full sample and thereby reintroduces part of the own-observation bias that the leave-one-out construction is meant to remove. Second, conventional second-stage standard errors treat the constructed leniency as a fixed regressor and ignore that it is itself estimated from roughly 2,000 agent effects. They therefore propose to estimate leniency designs with the unbiased jackknife IV estimator of Kolesár et al. (2013).

I implement their approach using the agent indicators directly as instruments, with *département* $\times$ year-quarter $\times$ shift cells absorbed as fixed effects. The limitation of this approach in my setting, and the reason the hand-constructed instrument remains the preferred measure, is that within the matched sample each agent's effect can only be estimated from her decisions on matched callers, a fraction of her caseload. This is also why the cells are at the year-quarter level and omit household type: with it, too few matched callers per agent and cell remain for the jackknife to be computed. For the same reason I conduct the exercise on the unemployed sample only, as the employed sample is too small to estimate agent effects from matched callers, and I further drop agents with fewer than (i) 10 and (ii) 30 first requests in the estimation sample. Table C16 compares the estimates to the baseline results of Table 6. With agents of at least 10 first requests, the UJIVE estimates are somewhat smaller than the baseline, but each within about one standard error of it and with the same pattern across outcomes; the effect on open-ended contracts is positive but imprecise in both. Restricting the sample further to agents with at least 30 first requests, which removes a third of the observations, moves the UJIVE estimates by less than a fifth of a standard error. The comparison speaks to both concerns raised by Goldsmith-Pinkham et al. (2026). The closeness of the point estimates indicates that the leave-one-out construction does not induce a bias of consequence, and the UJIVE standard errors are, if anything, smaller than the agent-clustered standard errors of the baseline, so that the latter do not understate the uncertainty in the estimates.

Robustness to the choice of controls and fixed effects. I next examine the sensitivity of the results to the choice of controls and absorbed fixed effects. The baseline includes no caller controls,

as the balance tests of Section 6.1.4 cannot reject that caller characteristics are jointly unrelated to leniency. Adding age and nationality controls barely moves the estimates in either sample (Table C17 and Table C18). Replacing the shift by the hour of the call in the fixed effects, i.e. absorbing *département* × year-quarter × hour × household-type cells, addresses the concern that callers sort across hours within a shift in a way correlated with which agents are on duty. The finer cells drop many singleton observations, so that the sample shrinks by about a fifth for the unemployed and by half for the employed (Table C19 and Table C20). For the unemployed sample, the estimates are unchanged. For the employed sample, the effects on contract terminations and on contracts lasting at least one year are of similar size and remain significant, while the estimates on the reasons for termination are noisier.

Robustness to the level of clustering. As is common practice in leniency designs, standard errors are clustered at the agent level. Goldsmith-Pinkham et al. (2026) note that this need not be the appropriate level in every setting. Clustering instead at the level of the *département* × year-quarter × shift × household-type cell, within which agents are as good as randomly assigned, yields standard errors that are smaller than the baseline for the unemployed sample and within 0.003 of it for the employed sample (Table C21 and Table C22).

7 Quantifying the Interaction Between Unemployment and Homelessness

The two quasi-experiments have shown that job loss increases homelessness, while homelessness in turn raises job separations and lowers job finding. This raises the question of how much the two effects reinforce each other, and how much longer they keep individuals in both unemployment and homelessness. I embed the two sets of estimates in a four-state Markov chain, a simple accounting framework that quantifies the identified channels, such as the higher risk of losing housing when unemployed and the lower job finding rate when homeless, and shows how each shapes steady-state homelessness and the time spent in both states at once. I conduct the exercise for low-skilled workers, defined as those who have not completed high school, since education is among the strongest predictors of homelessness risk (see e.g. Beauregard et al. (2026)).

7.1 A simple accounting framework

This subsection shows how the transition rates identified by the two quasi-experiments can be illustrated in a simple four-state Markov chain, in which workers move between employment and unemployment and between housing and homelessness. Appendix E shows the exact derivations and identifications of the transition rates for this population.

Four states. A low-skilled worker occupies one of four states, defined by employment status

$e \in \{E, U\}$ and housing status $h \in \{H, NH\}$. For the purpose of this exercise an individual counts as employed if she works at least fourteen days in a given month, which may be accumulated across contracts. Contracts shorter than a month are very common in this population, so the day threshold separates those more attached to the labor market from those working only a single day (Appendix E shows robustness checks under looser and narrower definitions). Housed means stable housing, and unhoused covers both sheltered and unsheltered individuals. I assume that an individual has exited homelessness when neither a shelter stay nor a request is observed for six consecutive months. The labor force is defined by having been observed in employment at some point in the sample period, since unhoused individuals are often removed from the public employment register for reasons unrelated to job search.[32]

Transition rates. Figure 7 shows the eight monthly transitions. Workers move between employment states by finding and losing a job, at rates f and s that depend on housing status, and between housing states at rates g and δ that depend on employment status. Each unhoused rate is written as the corresponding housed rate times a constant: $f_{NH} = \kappa_f f_H$, $s_{NH} = \kappa_s s_H$ and $\delta_E = \delta_U / \kappa_\delta$, so that κ_f and κ_s are the labor-market penalties of homelessness and κ_δ the insurance a job provides against losing one's home. I abstract from moves across the diagonal, since neither design identifies a simultaneous change in both statuses and they are also rare at monthly frequency.

Mapping the casual estimates to the transition rates. To quantify the interaction between homelessness and unemployment, I need to take into account selection into unemployment and into homelessness. The two quasi-experimental design allow me to address these selection issues: First, the shelter experiment identifies $\kappa_f = 0.68$ and $\kappa_s = 1.47$. Rather than taking the job finding and separation rates from the unhoused population, I compute them for the housed and impose these ratios. Because the experiment compares an immediate shelter spot with unsheltered homelessness, both of which count as unhoused in the Markov chain, they should be interpreted as lower bounds on what stable housing would do. Second, the layoff design identifies $\kappa_\delta = 2.92$. As it uses economic layoffs, it handles selection into unemployment directly. The main concern of selection is therefore selection into homelessness in particular for those who are employed as they may be facing other shocks pushing them into homelessness. I therefore take the entry rate of the unemployed as measured and recover the employed rate from the ratio. Six of the eight transitions are pinned down through these two quasi-experiments. To keep the two designs comparable I re-estimate both on the same population and impose only their ratios, not their control means, which are specific to each experimental sample.

Exit from homelessness by employment status. The two experiments do not speak to the transitions out of homelessness by employment status, and there is no direct causal evidence in

[32] Removal typically follows a failure to confirm monthly availability or having no address listed in their records, both of which are common without stable housing.

the literature that can be used instead. I therefore take this ratio from the raw transitions, which give $\kappa_g = 1.16$ (s.e. 0.01), and compare it with the estimates in Bruneel-Zupanc et al. (2026), who study transitions in and out of homelessness in a large French food distribution panel. Treating attrition from the panel as a return to housing, so that the whole sample is used, gives a gradient close to the one measured here. That calculation does not observe housing status after exit, however. Restricting to individuals present in two consecutive years, for whom it is recorded, and comparing housing status in the second year by employment status in the first, gives a ratio of roughly 3. Since remaining in the panel is itself correlated with both employment and rehousing, I use this as an upper bound.[33]

Validation against the raw transitions. Table E1 reports every rate used in the chain and its source. The unhoused rates are never an input, so the raw transitions provide a check (Table E2). Two of the three ratios are reproduced almost exactly. The separation penalty is 1.472 in the raw data against 1.474 from the instrument, and the entry gradient 2.94 against 2.92, even though in each case the two are built from different samples in different ways. Only the job finding penalty differs: the raw comparison gives 0.85 against a causal 0.68, understating the penalty by a fifth. Table C7 provides a possible explanation highlighting on which type of jobs the penalty exists. Among those individuals that just made the threshold of working 14 days per months, there might be many with one day or one week contracts for which the shelter experiment finds no penalty. Tightening the number of days worked restriction to 30 days, the pattern moves as expected and the job finding penalty gets larger.

7.2 Steady State Flows and Counterfactuals

Using the transition matrix constructed above, I can ask how the steady-state homelessness and unemployment rates would change if one particular channel were shut down. Writing each stock as a balance between inflows and outflows, and changing one rate at a time, identifies the transitions that matter most for each.

Steady State Homelessness In steady state the inflow into homelessness equals the outflow:

$$\underbrace{\delta_E \mu_{EH} + \delta_U \mu_{UH}}_{\text{entry}} = \underbrace{g_E \mu_{ENH} + g_U \mu_{UNH}}_{\text{exit}},$$

where μ_i is the share of the population in state i. The labor rates enter indirectly, by changing the composition of the unhoused: if the unhoused find work slowly and lose it quickly, most of them are unemployed, and the exit rate of the unemployed then governs the stock. The chain implies a stationary unhoused share of 1.4 percent, in line with the homelessness rate in this population group.

[33] I thank Bruneel-Zupanc et al. (2026) for computing these transition rates on their sample and sharing them with me.

Counterfactual Exercises. Table E3 reports the results. Each row sets one rate to the level faced by the other employment or housing group, re-solves for the stationary distribution, and reports the resulting change in the share unhoused; the last row switches all four off at once. Setting the housing-loss rate of the unemployed to that of the employed reduces steady-state homelessness by 39 percent using the raw transition exit rate, and by 46 percent at the higher exit gradient of 3 (Table E3). This is the counterfactual exercise where the qualitative results do not depend on the exit rate by employment status. How far unemployment prolongs a spell once it has begun depends entirely on that exit ratio: at the measured gradient of 1.16, the slower rehousing of the unemployed accounts for 7 percent of the steady state homelessness, and at the uppder bound of 3 implied by Bruneel-Zupanc et al. (2026) for 43 percent. The labor-market penalties, that is the lower job finding rate and higher job separation rate, follow the same pattern, accounting for 1 percent at the measured gradient and 9 percent at the higher one, where job finding contributes 7 points and separations 3 individually. Taken together, unemployment accounts for between 42 and 65 percent of steady-state homelessness.

Steady State Unemployment The mirror balance is

$$\underbrace{s_H \mu_{EH} + s_{NH} \mu_{ENH}}_{\text{entry}} = \underbrace{f_H \mu_{UH} + f_{NH} \mu_{UNH}}_{\text{exit}},$$

so that inflows into unemployment are governed by separations and outflows by job finding, both varying with housing status. How much homelessness can contribute to unemployment is bounded by the size of the unhoused population, 1.4 percent, and is correspondingly small. Table E4 reports the results for the same counterfactual exercise on steady state unemployment. Removing the job finding penalty changes non-employment by 0.48 percent and removing the separation penalty changes it by 0.36, together the two penalties account for 0.83 percent of non-employment. Restricting attention to the person-months accruing in spells past a year, which is 15.7 percent of all non-employment spells, the job finding penalty accounts for 2.0 percent.

7.3 Individual Trajectories and Counterfactuals

While homelessness does not affect the aggregate employment rate much, unemployment and homelessness may still reinforce each other for those in both states. To understand these feedback mechanisms better, I therefore turn from steady-state stocks to the trajectories of individuals.

Unemployment duration. I compare two workers who lose their job in the same month, one housed and one unhoused. A spell ends at the first job, so only the two non-employment states matter. Writing $t_{(U,H)}$ and $t_{(U,NH)}$ for the expected number of months until the first job from each

state,

$$t_{UH} = 1 + (1 - f_H - \delta_U)\, t_{UH} + \delta_U\, t_{UNH},$$
$$t_{UNH} = 1 + (1 - f_{NH} - g_U)\, t_{UNH} + g_U\, t_{UH}.$$

The housed worker finds work at f_H and loses her housing at δ_U; the unhoused worker faces the lower rate f_{NH} but may be rehoused at g_U, which returns her to the better one. Table E5 presents the results for this counterfactual exercise. Solving the system gives 7.0 months for the housed worker and 8.5 for the unhoused, 23 percent longer (Panel A).

Homelessness duration. The same exercise in addition gives the expected number of months a worker spends unhoused before reaching either housed state. At the measured exit gradient of 1.16, a worker who is unemployed when the spell begins spends 9.3 months unhoused against 8.8 for one in work, so being unemployed prolongs the spell by five percent. At $\kappa_g = 3$, the upper bound taken from Bruneel-Zupanc et al. (2026), the same two figures are 7.0 and 4.2 months and the gap is 68 percent (see Table E5, Panel B).

The interaction between homelessness and unemployment. The labor-market penalties therefore feed back onto housing: a worker who cannot find or keep work stays unemployed, and the unemployed are rehoused more slowly. Setting the job finding and separation rates of the unhoused to those of the housed, that is κ_f and κ_s to one, shortens the spell of a worker who is unemployed when she becomes homeless, without changing any housing rate. At the measured exit gradient of 1.16 the months spent homeless fall from 9.3 to 9.1, a reduction of 2 percent; at the upper bound of $\kappa_g = 3$ they fall from 7.0 to 6.3, a reduction of 11 percent (see Table E5, Panel B). How much the two directions reinforce each other therefore depends on how much faster the employed are rehoused.

Housing vs Job First. A current policy debate is *Job First* vs *Housing First*, that is whether individuals exit homelessness faster and more sustainably through employment or rehousing. The framework allows to visualize this debate: housing first takes an unemployed unhoused worker to (U, H), job first to (E, NH). The promise of job first depends on two rates. First, if the separation penalty is high, the worker risks losing the job before finding a home. Second, if the exit rate from homelessness is no higher for the employed, employment buys nothing on the housing margin. The chain gives the expected months to being both employed and housed from each state, and the two routes are equally long at $\kappa_g = 2.07$. Below that, employment is the slower route: rehousing while employed is not fast enough to outweigh the risk of losing the job first and falling back to unemployment while still unhoused, which leaves the worker further from recovery than when she started. At $\kappa_g = 1.16$ housing is worth 6.9 months and employment 3.0; at $\kappa_g = 3$ the ranking reverses. Job first is therefore promising where employment provides a clear pathway into housing, and weak where it does not, either because of discrimination in the

housing market or because the jobs on offer do not last long enough to secure a tenancy.

7.4 Interpreting the counterfactuals

Three limitations of this simple framework have to be taken into account when interpreting the counterfactual results: (i) the Markov chain is memoryless, (ii) the framework does not account for changes in behavior, and (iii) there are no causal estimates for the exit rates by employment status.

The chain is memoryless. The exercise allows to identify which channels are large and gives a sense of their magnitude, but it cannot compare policies acting at different points of a spell, because duration dependence in neither unemployment nor homelessness enters the transition rates.

Behavior is held fixed. Different policies change search effort and reservation wages in different ways, none of which the chain represents. The comparisons above are therefore between states and not between the direct policies. To get a sense of how sensitive the findings are to behavioral changes, the appendix reports how large a behavioral response would have to be to overturn the conclusions. A next step is to include this Markov chain framework inside a search model to directly observe the behavioral changes.

The type of employment matters. There are no causal estimates for the exit rate by employment status, and the raw data show little difference. Splitting employment by whether the job pays at least the monthly full-time equivalent minimum wage shows why. Housed and unhoused workers find low-paid work at similar rates, and the penalty appears only for the better jobs, precisely those that would allow a worker to secure a tenancy in the private market. The same pattern holds for exits: those in low-paid work leave homelessness more slowly each month, and the gap widens again for permanent contracts at that wage.

7.5 Back of the envelope calculation: The value of shelters

Lastly, the Markov chain provides a useful tool for interpreting the findings of the shelter access experiment, and the trajectory calculations give the quantities a cost-benefit comparison needs. A shelter place does not change housing status, since both sheltered and unsheltered individuals count as unhoused in the Markov chain, but it removes part of the job finding and separation penalty. The central question for a back of the envelope calculation is therefore how much employment it restores or protects and how that weighs against the nights it costs. I compare the months spent in employment over the twelve months after the first request with the nights spent in the shelter system, both estimated by the same instrument. For the unemployed, I restrict to callers with a labor market record in the preceding two years, for whom the shelter effect on employment are also the largest.

Cost and return. A placement adds 57.5 nights and 0.63 months of work for an unemployed caller, and 94.9 nights and 1.48 months for one already in work (Table E6). The jobs found pay around €1,500 a month, and at that wage a month of work returns €1,009 to the public finances once contributions, VAT and the transfers no longer paid are counted. The fiscal recovery is 44 percent for the unemployed and 63 percent for the employed, and the marginal value of public funds (MVPF, see Hendren and Sprung-Keyser (2020)) is 0.50 and 1.07 on earnings alone. The difference is not that the employed are cheaper to accommodate, since they are stay on average longer and are thus more expensive, but that the work a placement protects is worth more than the work it helps someone find.

Yet, a main limitation of this exercise is that these figures rest on transfers I do not observe. The matched records contain gross earnings but I cannot observe any transfers, which is why I impute welfare benefits, income subsidies and housing benefits from the statutory schedule for a single person with no dependants, assuming full take-up before and after the job. The three together account for a little over half the fiscal value of a month of work. With no transfers received at all that value falls from €1,009 to €538 and the private value rises from €634 to €1,170, so the fiscal recovery drops to 24 and 34 percent while the MVPF rises to 0.67 and 1.10: the MVPF is robust to the take-up assumption and the fiscal recovery is not.

Interpretation. These calculations only provide lower bounds for the cost-benefit of shelters. They count only twelve months, and Figure C4 shows that the employment effect has not plateaued. They treat the nights as a pure cost, without valuing the protection that comes from having a roof over one's head. Valuing them raises the MVPF to 1.35 and 2.35. These calculations further omit benefits beyond the labor market, for example Christopher et al. (2025) documents that providing emergency shelters reduce crime and emergency-room use.

8 Conclusion

This paper provides the first causal evidence on both directions of the relationship between homelessness and labor markets: job loss increases the risk of becoming homeless, and homelessness in turn increases job separation and reduces job finding. Combining both causal estimates in a Markov chain, I estimate that at least half of homelessness can be attributed to unemployment.

These results suggest that homelessness constitutes an uninsured risk of unemployment. Entries into homelessness are concentrated after unemployment benefits expire and among single, low-income workers, while low-income workers living in social housing in high-rent areas are less exposed. The risk of homelessness should therefore be taken into account when designing policies for job seekers, and labor market and housing policies for this population need to be combined.

Even temporary emergency shelters partly mitigate the negative labor market consequences of homelessness. As individuals in emergency shelters remain in an unstable and only temporary

housing situation, these effects are likely a lower bound for the effect of stable housing, pointing to the potential of Housing First over Job First policies. The effects of rapid shelter access are largest for individuals who worked within the three years before their shelter request, highlighting the importance of intervening quickly. Beyond that point, other factors, such as mental health, may become more important. More broadly, my findings highlight the importance of disentangling the role of labor markets as both a cause and a consequence of homelessness when designing policies to fight it.

Assessing these policies fully, however, requires understanding individuals' behavioral responses to them, which the Markov chain exercise does not capture. I plan to address this in future research by combining a field experiment on the effects of an intensive job search assistance program on homelessness with a job search model in which housing matters.

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Tables

Table 1: Snapshot: Homelessness Population in 2025

	# Individuals	% of individuals	% children	% women	% of bed-nights
	(1)	(2)	(3)	(4)	(5)
Household type					
Single adult, alone	214,927	40.0	0	30.1	26.7
Unaccompanied minor	4,037	0.8	100	22.5	0.3
Single parent with children	119,913	22.3	63	65.4	25.6
2+ adults, no children	39,621	7.4	0	48.5	6.4
2+ adults with children	159,357	29.6	50	50.2	41.0
Age					
Under 18	159,392	29.6	100	47.6	36.8
18–24	66,508	12.4	0	46.4	9.2
25–34	104,376	19.4	0	48.3	16.8
35–44	99,432	18.5	0	46.4	18.1
45–54	60,209	11.2	0	39.3	11.0
55–64	32,040	6.0	0	33.1	5.4
65+	15,898	3.0	0	36.8	2.7
All	**537,855**	100.0	30	45.2	100.0

Notes: This table describes the 537,855 individuals recorded as unhoused in France at any point in 2025, by household type (top panel) and age (bottom panel). Column (1) reports the number of distinct individuals in each group and column (2) the group's share of the total. Columns (3) and (4) report the share of children under 18 and of women within the group. Column (5) reports the share of all bed-nights in emergency and transitional accommodation in 2025 accounted for by the group. An individual is counted as unhoused if observed in a shelter stay or with a shelter request, met or unmet, in any month of 2025. Household type is recorded at the request. Source: SI-SIAO shelter stay and request records.

Table 2: Labor Market History for Workers Employed At Request

	All employed	Wage $>$ €1,500	Stable workers
	(1)	(2)	(3)
Panel A: Sociodemographics			
Female	0.44	0.40	0.41
Age (median)	37	38	36
With children	0.22	0.20	0.18
Panel B: Current job			
Monthly wage, € (median)	1,446	1,750	1,566
Permanent contract (CDI)	0.47	0.54	0.43
Full-time	0.59	0.78	0.65
Blue-collar	0.43	0.42	0.47
Days worked, last 2 yrs (median)	22	50	378
Observations	42,667	24,294	12,710

Notes: This table describes the labor market situation of individuals who hold an employment contract at the time of their first request to the emergency shelter hotline, for first requests between January 2018 and December 2024 by callers aged 18 to 60 linked to the employment records. Column (1) covers all employed callers; column (2) restricts to those whose current contract pays more than €1,500 per month; column (3) restricts to those who worked at least 185 days in the two years before the request. Panel A reports the share of women, the median age and the share with children at the time of the request. Panel B describes the contract held at the request: median gross monthly earnings, the share on an open-ended contract (CDI), the share working full time, the share in blue-collar occupations, and the median number of days worked in the two years before the request. Source: SI-SIAO linked to DSN.

Table 3: Labor Market History For Those Unemployed At First Request

	All	Job before	Last job < 2 yrs	Stable workers
	(1)	(2)	(3)	(4)
Panel A: Sociodemographics				
Female	0.42	0.37	0.39	0.42
Age (median)	35	36	34	35
With children	0.24	0.17	0.17	0.26
Panel B: Employment history				
Days since last job (median)		659	253	149
Days worked, last 2 yrs (median)		0	60	346
Last monthly wage, € (median)		247	222	877
Last job: permanent contract (CDI)		0.21	0.21	0.36
Last job: temp agency		0.31	0.30	0.19
Last job: full-time		0.60	0.59	0.62
Last job: blue-collar		0.54	0.54	0.48
Observations	1,026,993	220,322	116,138	32,968

Notes: This table describes the labor market history of individuals without an employment contract at the time of their first request to the emergency shelter hotline, for first requests between January 2018 and December 2024 by callers aged 18 to 60 linked to the employment or unemployment records. Column (1) covers all such callers, including those with no observed employment contract since 2017, for whom Panel B is undefined; column (2) restricts to callers with at least one contract since 2017; column (3) to those whose last contract ended less than two years before the request; column (4) to those who worked at least 185 days in the two years before the request. Panel A reports the share of women, median age and share with children. Panel B describes the last contract before the request: median days since it ended, median days worked in the two years before the request, median gross monthly earnings in the last contract, and the shares on an open-ended contract, in temporary agency work, working full time and in blue-collar occupations. Source: SI-SIAO linked to DSN and France Travail registers.

Table 4: The Effect of Job Loss on Labor Market Outcomes and Homelessness Risk

	Employment	Employment (Open-Ended)	Earnings	Probability of Homelessness
	(1)	(2)	(3)	(4)
Economic Layoff	-0.32771***	-0.41779***	-661.46718***	0.00025**
	(0.002)	(0.002)	(8.890)	(0.000)
Mean outcome, treated at t= 0	1	1	2,109	0.0005
Effect relative to mean	**-32%**	**-41%**	**-31%**	**50%**
Number of obs.	2,681,175	2,681,175	2,681,175	2,681,175

Notes: This table reports the estimated impact of an economic layoff event on labor market outcomes and the risk of homelessness, utilizing the matched difference-in-differences framework specified in Equation 2. The treatment group comprises workers displaced in economic layoffs of two or more workers in a given firm-month between 2017 and 2023. The control group consists of matched workers who were not employed at firms engaged in economic layoffs during the analysis period and remained employed during the displacement year. The row "Mean outcome, treated at t= 0" reports the mean of each outcome in the quarter of the economic layoff event; "Effect relative to mean" divides the coefficient by it. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Rapid Shelter Access Reduces Job Separation

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Same Contract at 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.064*	0.082**	-0.072***	-0.029***	-0.005	-0.033**	-0.005*
	(0.037)	(0.041)	(0.020)	(0.010)	(0.013)	(0.013)	(0.003)
Mean, refused	0.416	0.316	0.092	0.023	0.031	0.037	0.002
Complier mean, refused E[Y(0)|C]	0.413	0.298	0.100	0.032	0.021	0.044	0.003
Kleibergen–Paap F	701.61	701.61	701.61	701.61	701.61	701.61	701.61
Observations	12,094	12,094	12,094	12,094	12,094	12,094	12,094

Notes: This table reports 2SLS estimates of the effect of a met first shelter request on job separation among the 12,094 callers who held an employment contract at the time of their first request to the emergency shelter hotline between May 2021 and March 2024. The endogenous variable is an indicator for the first request being met, instrumented by the leave-out placement rate of the agent who answered the call. Column (1): the contract held at the request ends within three months of the call. Column (2): the caller is still employed in the same contract twelve months after the call. Column (3): the contract is terminated within three months by the employer or the employee, as opposed to the scheduled expiry of a fixed-term contract. Columns (4) to (7) decompose column (3) by the reason recorded by the employer: dismissal for misconduct, quit, dismissal at the end of the probationary period, and health-related termination. The row "Mean, refused compliers" reports E[Y(0) | C], the complier mean under refusal, estimated following (Abadie, 2003). All specifications include *département* × year-quarter × shift × household-type fixed effects. The Kleibergen–Paap F-statistic of the first stage is 702. Standard errors in parentheses are clustered at the agent level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

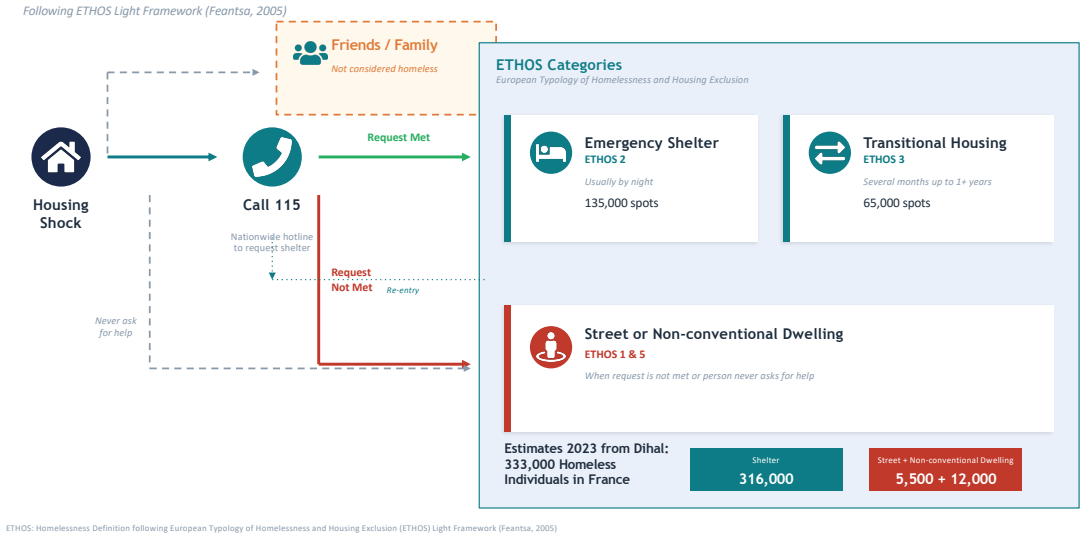
Table 6: Rapid Shelter Access Raises Job Finding

	Unemployed at Request			
	Employed at 1 Year	Open-ended contract at 1 Year	Days Worked Year 1	Monthly Earnings at 1 Year
	(1)	(2)	(3)	(4)
First Request Met	0.049***	0.014*	14.142***	54.781***
	(0.013)	(0.008)	(3.688)	(15.692)
Mean, refused	0.084	0.036	25.102	97.625
Complier mean, refused E[Y(0)|C]	0.09	0.04	24.91	96.98
Kleibergen–Paap F	1,351.80	1,351.80	1,351.80	1,351.80
Observations	71,236	71,236	71,236	71,236

Notes: This table reports 2SLS estimates of the effect of a met first shelter request on job finding among the 71,236 callers without an employment contract at the time of their first request to the emergency shelter hotline between May 2021 and March 2024, but with at least one employment contract or unemployment registration since 2017. The endogenous variable is an indicator for the first request being met, instrumented by the leave-out placement rate of the agent who answered the call. Column (1): the caller is employed twelve months after the call. Column (2): the caller holds an open-ended contract twelve months after the call. Column (3): number of days worked in the twelve months following the call. Column (4): gross monthly earnings twelve months after the call, in euros, set to zero for callers not employed. The row "Mean, refused compliers" reports E[Y(0) | C], the complier mean under refusal. All specifications include *département* × year-quarter × shift × household-type fixed effects. The Kleibergen–Paap F-statistic of the first stage is 1,352. Standard errors in parentheses are clustered at the agent level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

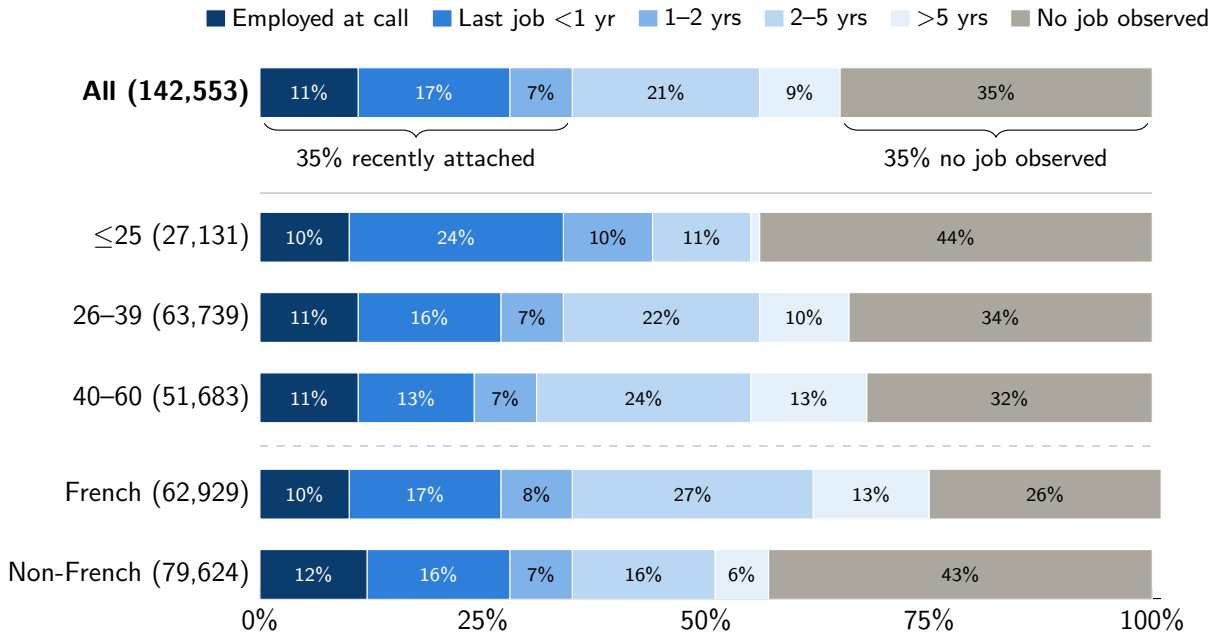
Figures

Figure 1: Pathways After a Housing Shock in France



Notes: This figure shows the pathways open to an individual after a housing shock in France. Individuals may stay with family or friends, remain on the street, or request a place through the emergency shelter hotline (115), which is the main entry into the shelter system. A met request leads to emergency shelter or, through a social worker, to transitional housing; an unmet request leaves the individual on the street or in a non-conventional dwelling. Emergency shelter and transitional housing correspond to ETHOS categories 2 and 3, the street to category 1 (Feantsa, 2005) In 2025, according to point-in-time estimates France counted about 350,000 unhoused individuals, of whom 334,000 were in shelters (Dihal, 2025).

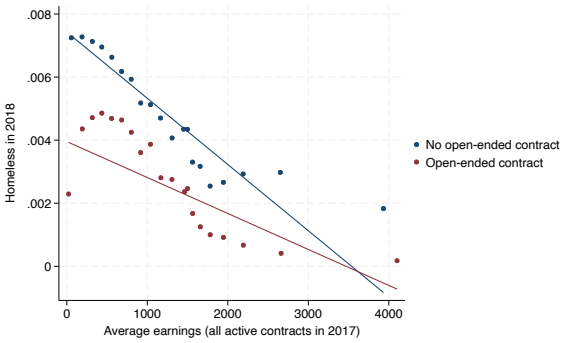
Figure 2: Labor market attachment at the time of the call



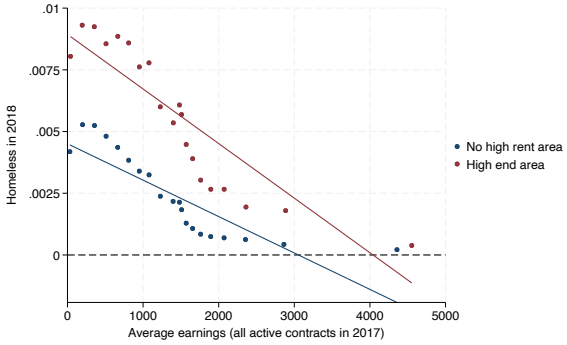
Notes: This figure shows the labor market attachment of individuals making their first request to the emergency shelter hotline between May 2021 and March 2024, for callers aged 18 to 60 linked to the employment records (N = 142,553). Each bar divides a group into callers employed at the call and callers whose last employment contract ended less than one year, one to two years, two to five years, or more than five years before the call; the last segment is callers with no employment contract observed since 2017. The top bar is the full sample; the bars below split it by age at the call and by nationality. Numbers in parentheses are group sizes. Source: SI-SIAO linked to DSN.

Figure 3: Homelessness Risk and Precarious Labor Market Conditions

(a) Earnings and open-ended contract

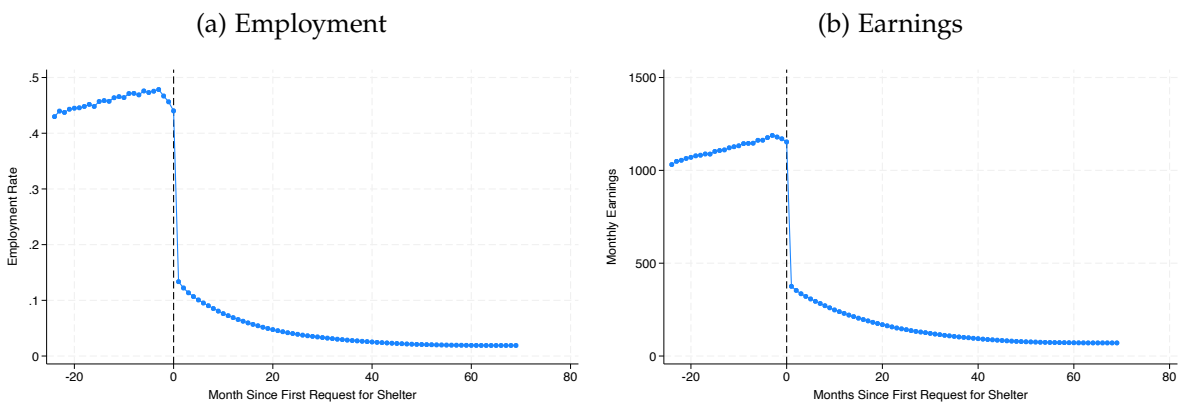


(b) Earnings and rent burden



Notes: This figure shows the relationship between earnings and the risk of becoming homeless in the following year. The sample consists of all individuals registered with France Travail at some point between 2017 and 2025 who held at least one employment contract in 2017. The horizontal axis is average gross monthly earnings over all contracts held in 2017, winsorized at the 99th percentile; the vertical axis is the share with a shelter stay or a shelter request in 2018. Each point is the mean within one of 20 equal-sized bins of earnings; lines are linear fits on the underlying individual data. Panel (a) splits the sample by whether the individual held an open-ended contract in 2017. Panel (b) splits it by whether the individual lived in a high-rent area, defined as a département with median rent above €17 per square meter in 2018.

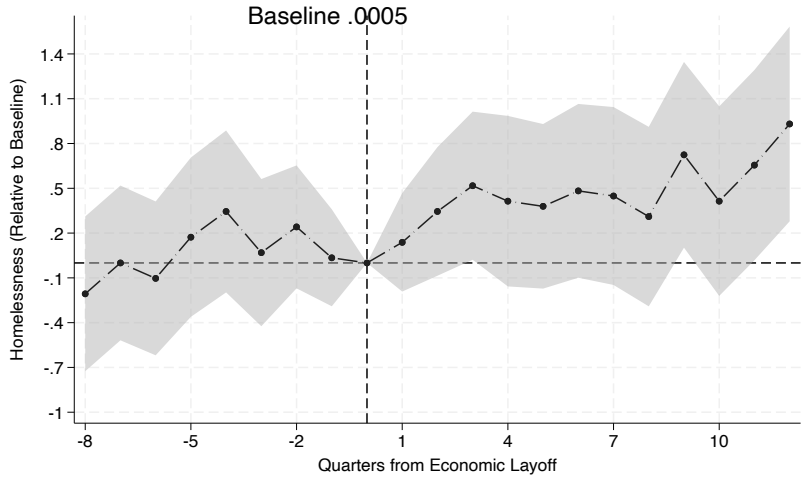
Figure 4: Earnings and Employment Trajectories Around Entry Into Homelessness



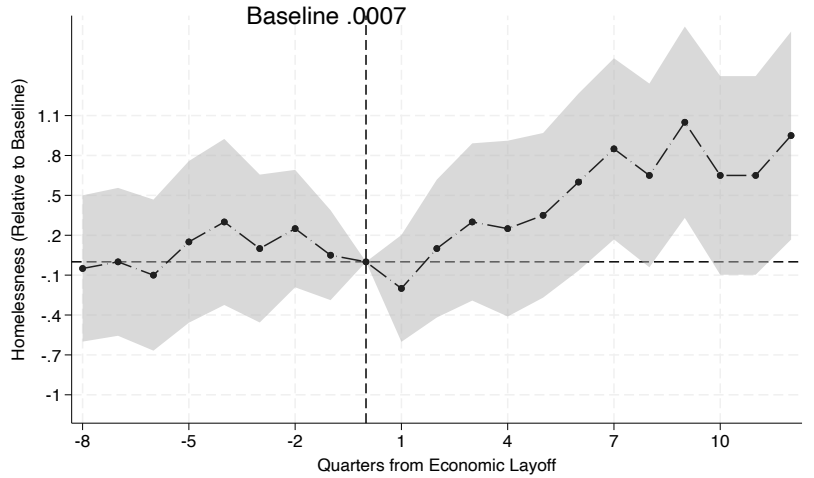
Notes: This figure shows the employment rate (panel a) and average gross monthly earnings (panel b) of individuals in the months around their first request for shelter through the emergency shelter hotline, for first requests between 2018 and 2025 by callers aged 18 to 60 linked to the employment records. The sample is restricted to individuals who worked at least 180 days in the five years before the request Because this restriction conditions on employment before the request but not after it, the level of employment before month 0 is raised mechanically; the figure is therefore informative about the shape of the trajectories, in particular the persistence of the decline, rather than about the size of the drop at the request. Earnings are set to zero in months without employment. The vertical dashed line marks the month of the first request.

Figure 5: The Effect of Job Loss on Homelessness Risk — Relative to Baseline

(a) Dynamic Event Study — Full Sample



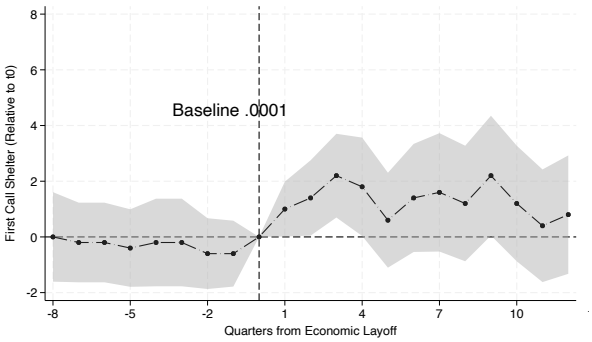
(b) Dynamic Event Study — Men



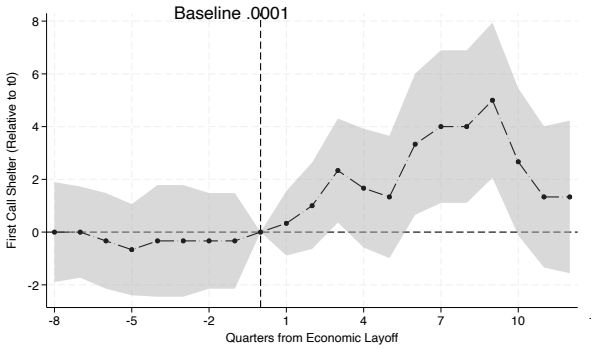
Notes: This figure shows the dynamic effect of job loss in an economic layoff on the probability of being homeless, from the matched difference-in-differences specification in Equation 1. Each point is the coefficient on the interaction of treatment and quarter relative to the layoff, divided by the mean of the outcome at period 0 (0.05 percent), so that a value of 1 corresponds to a doubling of the baseline risk. The quarter before the layoff is the omitted category. Panel (a) uses the full matched sample (53,619 treated and 53,619 control workers); panel (b) restricts to men. Shaded areas are 90 percent confidence intervals based on standard errors clustered at the individual level.

Figure 6: Timing of Entry into Homelessness: Effect of Job Loss on the First Call for Shelter

(a) Dynamic Event Study — Full Sample

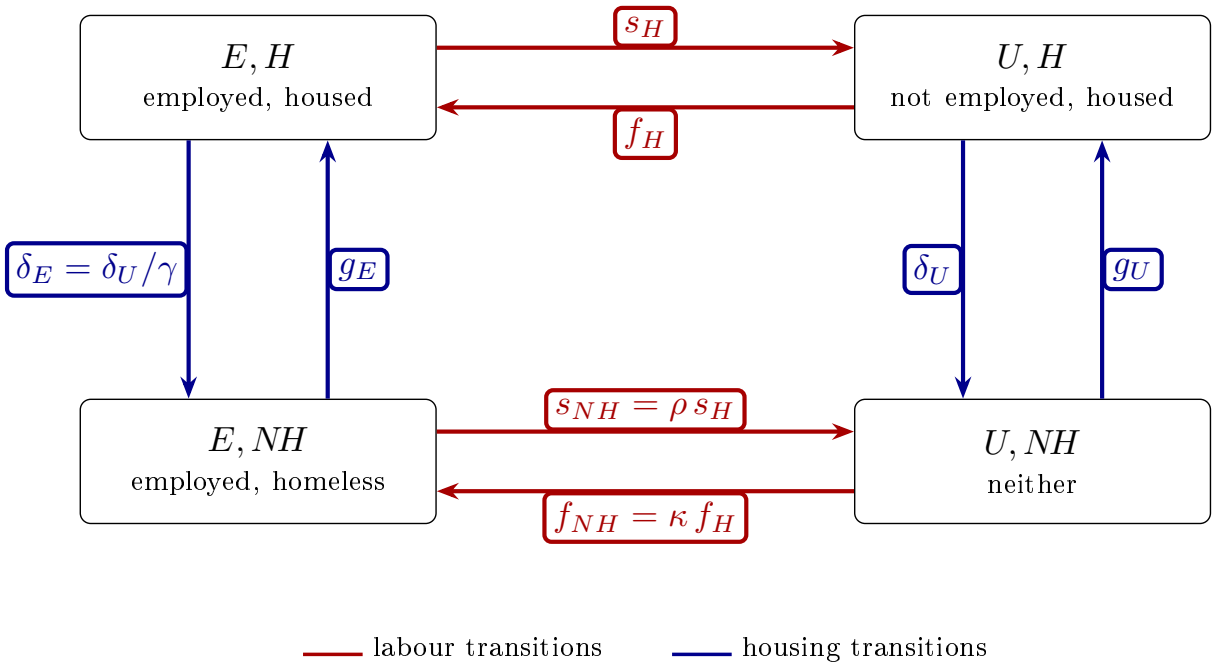


(b) Dynamic Event Study — Men



Notes: This figure shows the dynamic effect of job loss in an economic layoff on the probability of making a first call to the emergency shelter hotline in the quarter, from the matched difference-in-differences specification in equation 1. Each point is the coefficient on the interaction of treatment and quarter relative to the layoff, divided by the mean of the outcome at period 0 (0.01 percent per quarter), so that a value of 1 corresponds to a doubling of the baseline risk.. The quarter before the layoff is the omitted category. Panel (a) uses the full matched sample; panel (b) restricts to men. Shaded areas are 90 percent confidence intervals based on standard errors clustered at the individual level.

Figure 7: Transition rates in four-state Markov process



Notes: This figure shows the four states of the Markov chain of Section 7 and the monthly transition rates between them. Rows of boxes distinguish housing status (housed, H; unhoused, NH) and columns employment status (employed, E; unemployed, U). Red arrows are labor market transitions: job separation at rate s and job finding at rate f. Blue arrows are housing transitions: housing loss at rate δ and rehousing at rate g. The rates of the housed, f_H, s_H and δ_U, and the rehousing rates g_U and g_E are measured directly in the data; the rates of the unhoused are obtained from the causal ratios identified Section 5 and Section 6: $\delta_E =_U /\gamma$ with γ from the layoff design, and $f_{NH} = \kappa f_H$ and $s_{NH} = \rho s_H$ with κ and ρ from the shelter design. **??** lists the values and sources of all rates.

A Data and Sample Construction

A.1 Matching Procedure and Results

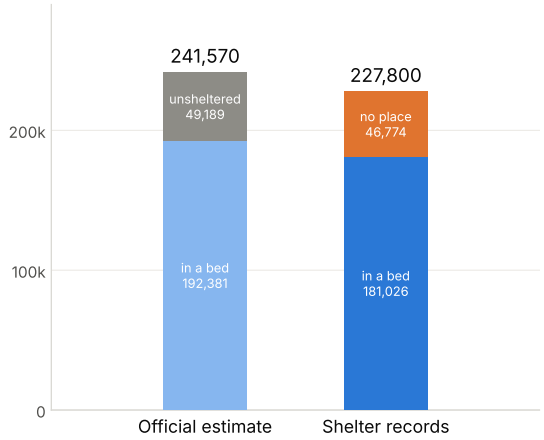
As the *SI-SIAO* does not collect social security identifiers compatible with administrative systems, the linkage proceeds in three steps, ordered from most to least restrictive. Step 1 performs an exact match on first name, last name, and date of birth, recovering approximately 94% of all eventual matches. Step 2 applies the Jaro–Winkler similarity metric to first and last name to identify strongly similar (but not identical) names, combined with an exact match on date of birth; this step recovers approximately 5% of matches. Step 3 uses the Jaro–Winkler similarity for names combined with an exact match on either month or year of birth and an exact match on phone number; this step recovers approximately 0.08% of matches. Of the 3,262,677 unique individuals appearing in the SI-SIAO since 2014, 1,111,354 are successfully linked to administrative records.[34]

[34] While the system was established in 2014, it was not universally adopted until 2020. Since then, shelter allocation has been strictly required to occur through this centralized platform.

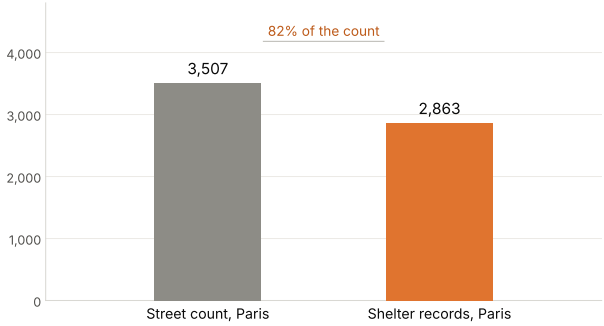
A.2 Data Coverage and Descriptives

Figure A1: Estimates of data coverage: sheltered and unsheltered population in 2025

(a) Comparison to point-in-time estimates

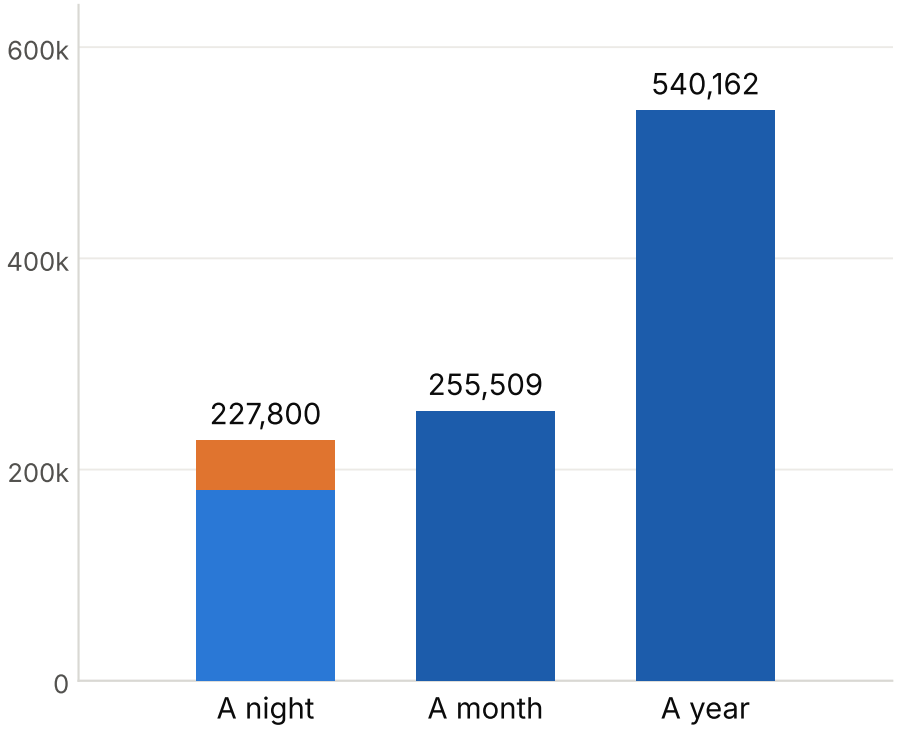


(b) Comparison to street counts



Notes: Panel A compares two independently constructed totals for France in 2025. *Official estimate* is the published national figure once the asylum reception system is removed, 241,570. It is a capacity count rather than a count of individuals: 192,381 is the generalist accommodation park multiplied by its occupancy rate, and 49,189 is an estimate for rough sleeping and informal settlements that rests on a 2016 census base. *My data* is the SI-SIAO shelter database, stays and calls together, counting individuals observed on an average night: 181,026 hold an accommodation place and 46,774 are recorded as homeless with none. The 181,026 is 94 percent of the occupied generalist park, and 90 to 96 percent across plausible occupancy rates. Panel B focuses on the coverage of the unsheltered population. It compares estimates from street counts to estimates from the shelter records. *Street count, Paris* is the eighth Nuit de la Solidarité, conducted on the night of 23–24 January 2025 by roughly 4,000 volunteers, which counted 3,507 unsheltered individuals across the city. *Shelter records, Paris* counts individuals with no accommodation place that night who had either a shelter request refused that same day or an earlier request declaring *Rue* still unresolved within their current spell: 2,863 individuals, 82 percent of the count. The residual can be interpreted as the individual who never contacts the 115 and for whom no social worker or outreach team files a request.

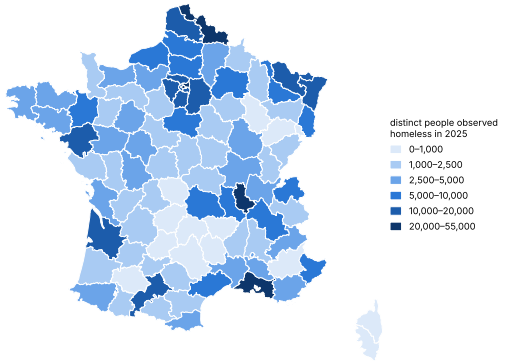
Figure A2: Homelessness Flow Measure



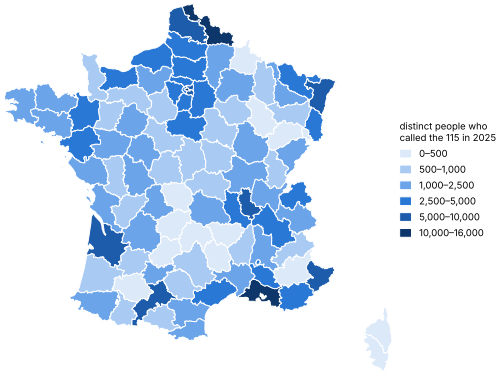
Notes: This figure shows the different estimates for the number of unhoused individuals, on a single night, within a month and within the year in 2025.

Figure A3: Geography of Homelessness Population in France, 2025

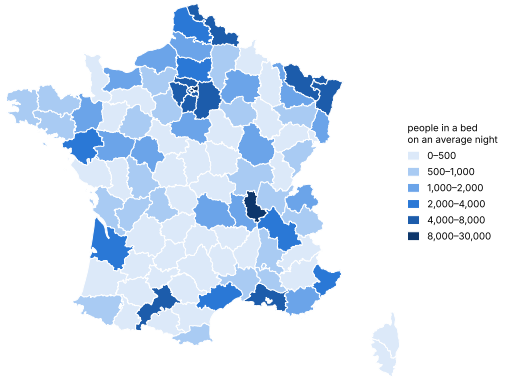
(a) People observed homeless over the year



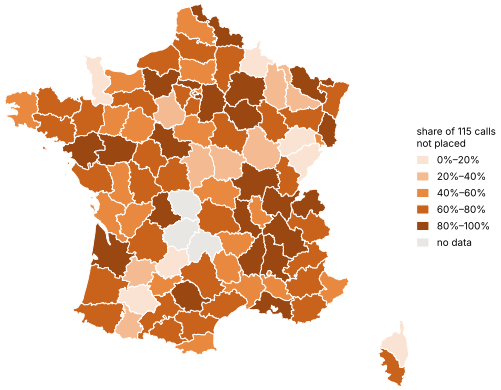
(b) People who called the 115 at least once



(c) People in a shelter bed on an average night



(d) Share of daily 115 calls not placed



Notes: SI-SIAO administrative records, 2025; 96 metropolitan départements, overseas omitted. Panel A counts individuals recorded as sheltered or on the street in any month of 2025, bridging gaps of up to six months (540,162 nationally, 99.6 percent assigned to a département). Panel B counts individuals placing at least one call to the 115 (230,573). Panel C divides emergency and transitional-housing bed-days by 365 (181,068 beds). Panel D is the mean across days of the département-day share of 115 calls ending without a placement (1,435,790 calls, 69.9 percent unplaced); départements with fewer than 300 calls or calls on fewer than 200 days are shown as missing. Individuals are assigned to the département of SI-SIAO file creation in Panels A–C and of the call in Panel D. Class limits differ across panels.

Table A1: Sample Characteristics of Matched vs Non-Matched Homeless Individuals

	Not Matched		Matched		Δ		
	$\bar{x}$	sd	$\bar{x}$	sd	$\Delta\bar{x}$	Δ sd	p-val
Panel A: Information about Request							
Year of Request	2021	2.506	2020	2.615	-0.519	0.003	0.000
Household Type:							
Couple with Children	0.085	0.279	0.066	0.248	-0.019	0.000	0.000
Couple without Children	0.027	0.163	0.025	0.155	-0.003	0.000	0.000
Single Women	0.139	0.346	0.165	0.371	0.026	0.000	0.000
Single Women with Kids	0.120	0.325	0.140	0.347	0.020	0.000	0.000
Group with Kids	0.135	0.342	0.101	0.302	-0.034	0.000	0.000
Adults without Kids	0.071	0.257	0.062	0.241	-0.009	0.000	0.000
Isolated Man	0.390	0.488	0.421	0.494	0.031	0.001	0.000
Man with Kids	0.012	0.107	0.015	0.121	0.003	0.000	0.000
Panel B: Socio-demographic Characteristics							
Age	35.839	19.838	36.515	12.470	0.676	0.017	0.000
Female	0.413	0.492	0.425	0.494	0.012	0.001	0.000
French Nationality	0.298	0.458	0.535	0.499	0.236	0.001	0.000
Observations	1,118,749		1,044,769		2,163,518		

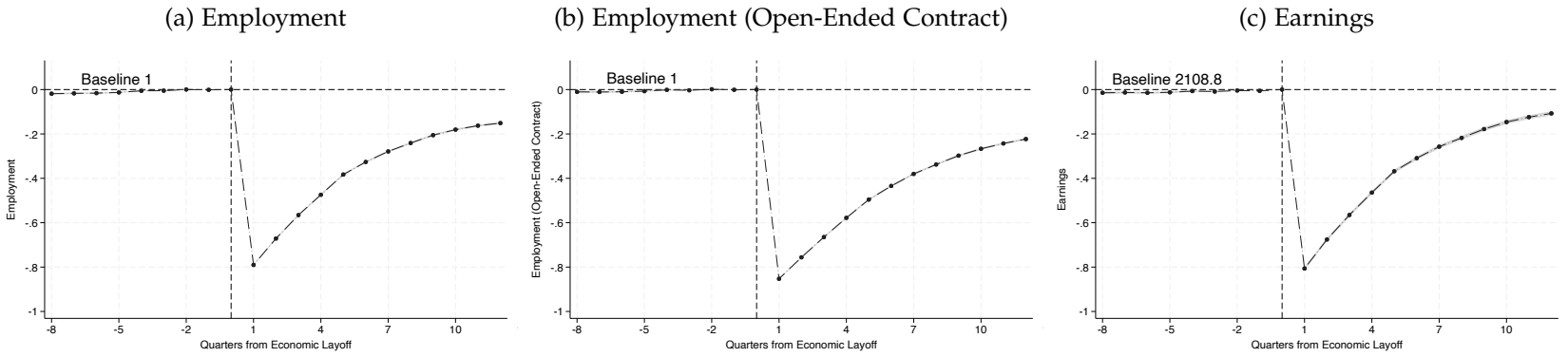
Notes: This table compares the baseline characteristics of individuals who submitted a shelter request and were successfully linked to administrative labor market records against those who remained unmatched. The sample is restricted to adults without a recent migratory history; this exclusion is necessitated by the fact that both minors and recent arrivals to France have a low probability of appearing in national employment and unemployment registries, which would lead to mechanical attrition in the matching process.

B The Effect of Job Loss on Homelessness: Additional Results and Robustness

This appendix reports robustness checks and additional heterogeneity results for the matched difference-in-differences design of Section 5.

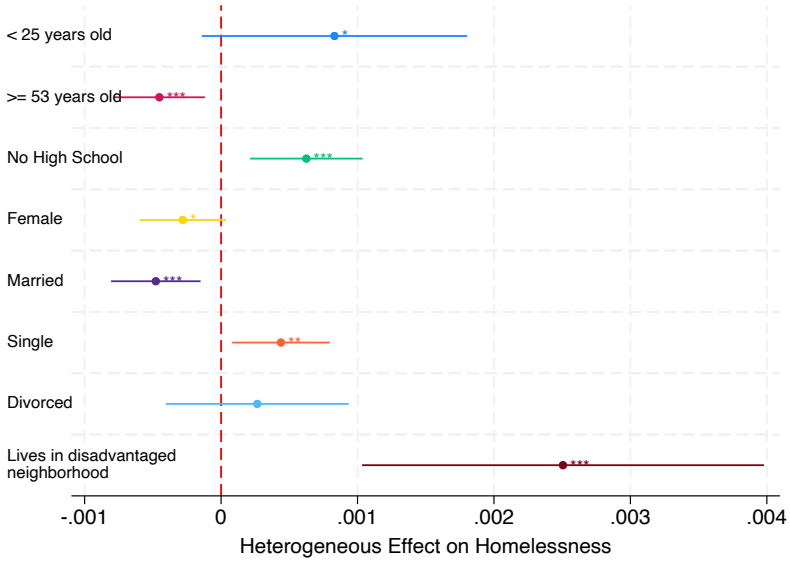
B.1 Additional Results

Figure B1: The Effect of Job Loss on Labor Market Outcomes — Relative to Baseline



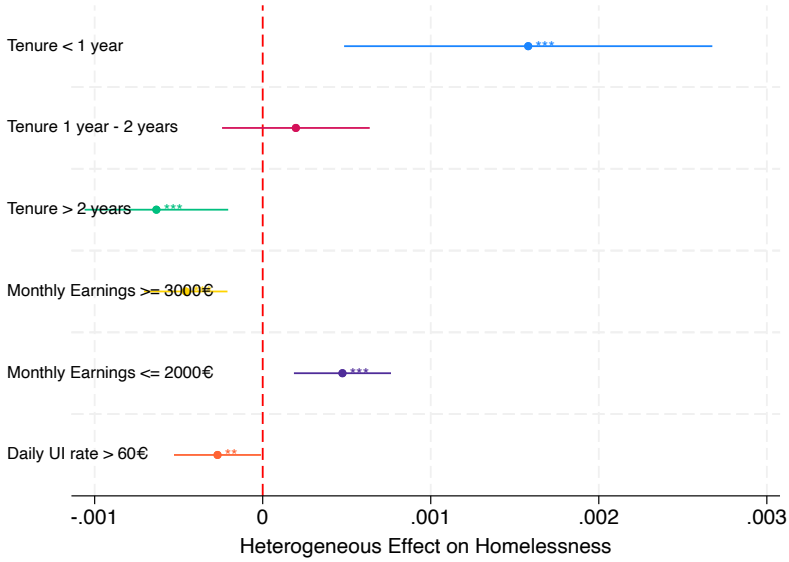
Notes: This figure presents the estimated impact of job loss on labor-market outcomes using the matched difference-in-differences specification in Equation 1. Estimates are normalized relative to the baseline mean of the outcome at the time of displacement. The horizontal axis represents quarters relative to the economic layoff event. The treatment group comprises workers displaced in an economic layoff (at least two workers in a given firm-month for economic reasons) between 2017 and 2023. The control group consists of matched workers who were not employed at firms engaged in economic layoffs during the analysis period and remained employed during the displacement year. Shaded areas denote 90% confidence intervals.

Figure B2: Heterogeneity Analysis: Sociodemographic Characteristics



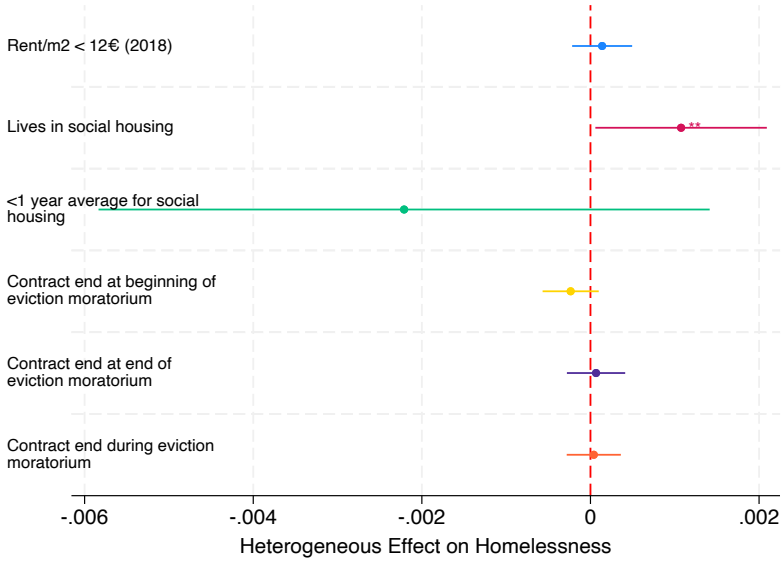
Notes: This figure plots the interaction coefficients from a specification that augments Equation 2 with interactions between the treatmentxpost indicator and worker characteristics. Outcome variable: probability of homelessness. Confidence intervals are 95% and based on standard errors clustered at the individual level. Household composition, age, and education are all significantly related to the homelessness response to job loss.

Figure B3: Heterogeneity Analysis: Job Characteristics



Notes: This figure plots the interaction coefficients from a specification that augments Equation 2 with interactions between the treatmentxpost indicator and worker characteristics. Outcome variable: probability of homelessness. Confidence intervals are 95% and based on standard errors clustered at the individual level. Short tenure and low earnings sharply elevate homelessness risk after job loss; a high UI replacement rate provides partial protection.

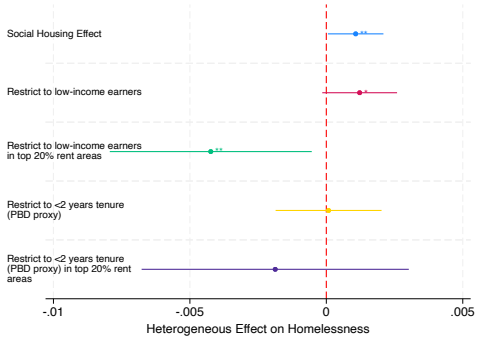
Figure B4: Heterogeneity Analysis: Housing Policies



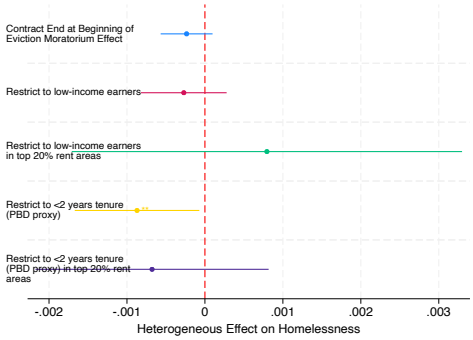
Notes: This figure plots the interaction coefficients from a specification that augments Equation 2 with interactions between the treatment indicator and pre-displacement housing-policy characteristics. Outcome variable: probability of homelessness. Variables include indicators for rent per square meter below €12 in 2018, residence in social housing, short average wait time for social housing, and whether the worker's contract ended at the beginning, end, or during an eviction moratorium period. 95% confidence intervals based on standard errors clustered at the plant level.

Figure B5: Heterogeneity Analysis: Combination of Housing and Income

(a) Social Housing



(b) Eviction Moratorium

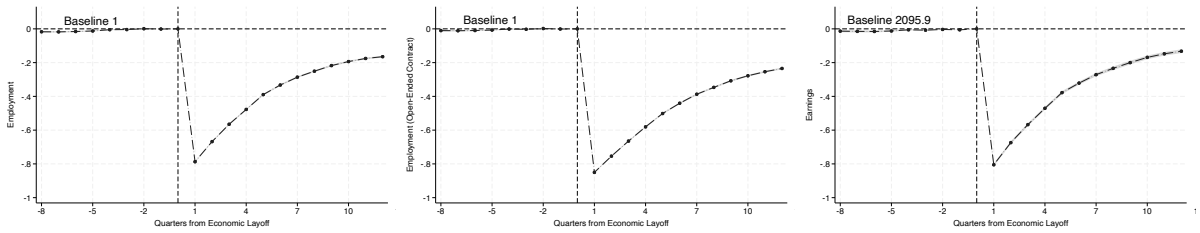


Notes: This figure plots heterogeneous treatment effects on homelessness combining housing policy variables with income/tenure restrictions. Left panel: social housing interacted with low-income, high-rent area, and short-tenure (PBD proxy) subsamples. Right panel: contract-end timing relative to the eviction moratorium interacted with the same subsamples. 95% confidence intervals based on standard errors clustered at the plant level.

B.2 Robustness

Figure B6: Robustness — Excluding 2020 and 2021: Effect on Labor Market Outcomes (Relative to Baseline)

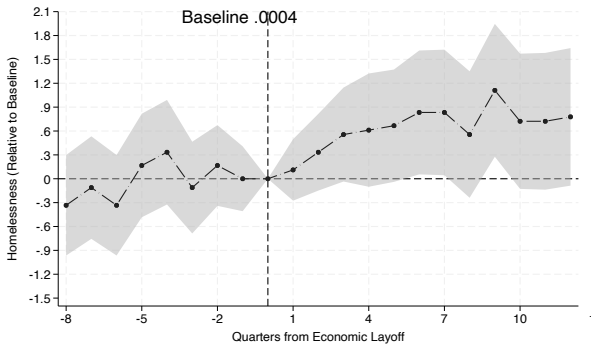
(a) Employment (b) Employment (Open-Ended) (c) Earnings



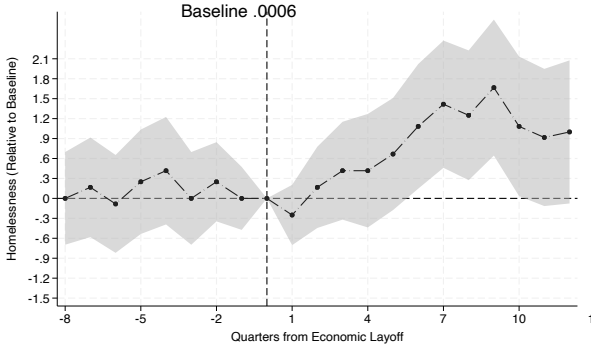
Notes: Robustness check excluding economic layoffs occurring in 2020 and 2021 to ensure that the main findings are not driven by the COVID-19 pandemic and associated labor-market disruptions. All other specification choices follow the main analysis.

Figure B7: Robustness — Excluding 2020 and 2021: Effect on Homelessness (Relative to Baseline)

(a) Dynamic Event Study — Full Sample

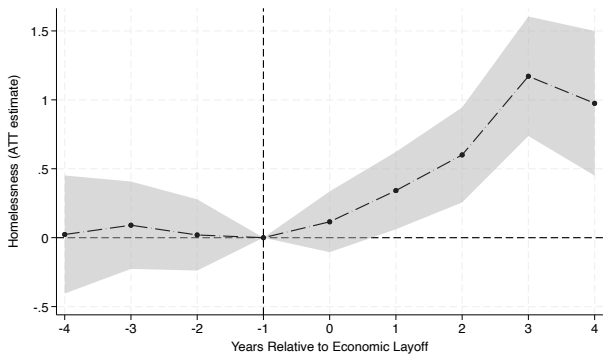


(b) Dynamic Event Study — Men



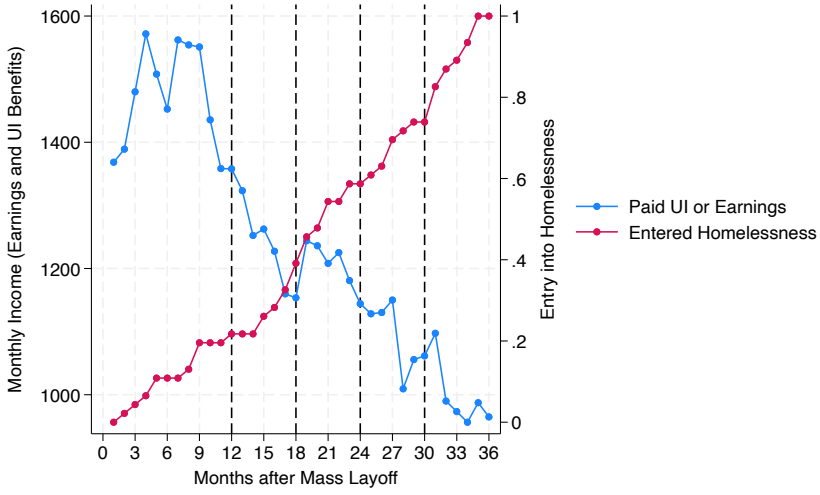
Notes: Robustness check excluding economic layoffs occurring in 2020 and 2021. All other specification choices follow the main analysis. Left panel: full matched sample. Right panel: men only.

Figure B8: Robustness — Only Treated: Effect on Homelessness (Relative Effects)



Notes: Robustness check restricting the sample to treated workers only and using staggered-timing variation to identify dynamic effects on homelessness. All other specification choices follow the main analysis.

Figure B9: Income and Cumulative Entry into Homelessness After an Economic Layoff



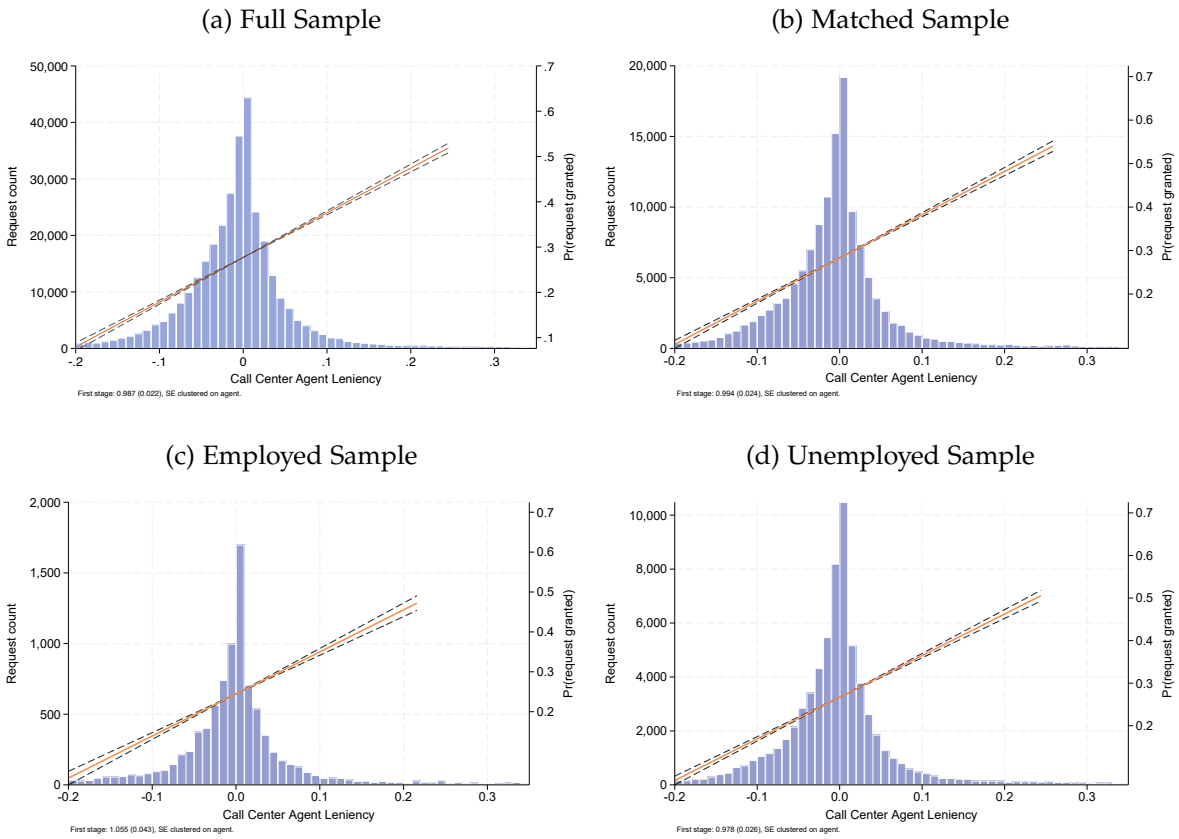
Notes: Sample restricted to workers who experience an economic layoff and subsequently enter homelessness. The left axis (blue) plots average monthly income, combining earnings and unemployment insurance benefits. The right axis (red) plots the cumulative share who have entered homelessness. The horizontal axis is months after the layoff event. Income peaks 4–6 months after displacement (severance plus UI onset), declines as UI benefits exhaust, and homelessness entry accelerates exactly as income falls.

C The Effect of Shelter Access: Validity Checks and Additional Results

This appendix reports the validity checks and robustness results for the leniency design of Section 6.

C.1 Sample and Instrument

Figure C1: Distribution and First Stage of Call-Center Agent Leniency



Notes: This figure plots the distribution of the leave-out agent leniency instrument together with the first stage it generates. Blue bars show the number of requests in each of 55 equally spaced bins of width 0.010, measured against the left vertical axis; each request is placed at the leniency of the agent who handled it, so an agent with a large caseload contributes many requests at a single point. The solid orange line, plotted against the right vertical axis, gives fitted values from the first-stage regression of placement on leniency with cell fixed effects, and dashed lines give the corresponding 95% confidence interval. Panel (a) shows the distribution of the full sample, while panel (b) to (d) zoom in on the main analysis samples, that is those that could be matched to the administrative records. Panel (c) focuses on those workers that were employed at their first request and panel (d) focuses on those that were unemployed but have a prior work history at the moment of their request.

C.2 Validity of the IV Design

Table C1: First Stage: First Request Met on Agent Leniency, by Sample

	Full sample	All matched	Employed at call	Unemployed, prior job
	(1)	(2)	(3)	(4)
Agent leniency	0.987***	0.998***	1.055***	0.978***
	(0.022)	(0.021)	(0.043)	(0.026)
Kleibergen–Paap F	2,000	2,189	702	1,352
Agents	2,406	2,323	1,606	2,238
Observations	341,545	136,734	12,094	71,236

Notes: *** $p < 0.01$. First-stage regressions of an indicator for the first request being met on agent leniency, with *département* × year-month × shift × household-type fixed effects, the cell within which the instrument is constructed (singleton cells dropped). Columns (1)–(2): all first requests to the 115, May 2021 to March 2024, callers aged 18–60, and the subset matched to the employment records, both estimated on the SI-SIAO data; the matched sample estimated inside the CASD gives 0.994 (0.024). Columns (3)–(4): matched callers employed at the call, and unemployed at the call with a prior job, estimated inside the CASD. Agent leniency is the leave-out placement rate computed on single-men calls. Standard errors in parentheses, clustered by agent; with one instrument the Kleibergen–Paap rk Wald F equals the squared t-statistic of the first-stage coefficient.

Table C2: First Stage With and Without Caller Controls

	Full sample		All matched	
	(1)	(2)	(3)	(4)
Agent leniency	0.987***	0.988***	0.998***	0.998***
	(0.022)	(0.022)	(0.021)	(0.021)
Controls	No	Yes	No	Yes
Kleibergen–Paap F	2,000	1,989	2,189	2,191
Agents	2,406	2,406	2,323	2,323
Observations	341,545	341,545	136,734	136,734

Notes: *** $p < 0.01$. First-stage regressions of an indicator for the first request being met on agent leniency, with *département* × year-month × shift × household-type fixed effects, the cell within which the instrument is constructed (singleton cells dropped). Columns (2) and (4) add five-year age-bin dummies and nationality-group dummies (French, EU, non-EU, missing). All columns use the estimation sample: first requests to the 115 made between May 2021 and March 2024 by callers aged 18 to 60 (SI-SIAO data); columns (3)–(4) restrict to the subset matched to the employment and unemployment records. Agent leniency is the leave-out placement rate computed on single-men calls over 2018–2026. Standard errors in parentheses, clustered by agent; with one instrument the Kleibergen–Paap rk Wald F equals the squared *t*-statistic of the first-stage coefficient.

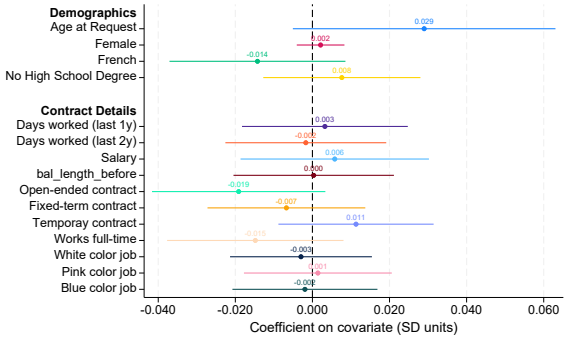
Table C3: The First Call Is Decisive

Horizon t	Refused at first request: share who by day t …		In an emergency bed on night t, compliers		
	called again	were placed on a later call	refused E[Y(0)|C]	placed E[Y(1)|C]	2SLS
	(1)	(2)	(3)	(4)	(5)
Night of the call	–	–	0.00	0.55	0.550 (0.020)
1 day	0.21	0.04	0.02	0.50	0.480 (0.020)
1 week	0.37	0.10	0.04	0.44	0.406 (0.020)
1 month	0.47	0.18	0.04	0.38	0.335 (0.020)
3 months	0.52	0.23	0.05	0.31	0.258 (0.020)
1 year	0.59	0.29	0.04	0.17	0.129 (0.019)
First requests	136,734	136,734	136,734	136,734	136,734
Agents	2,323	2,323	2,323	2,323	2,323

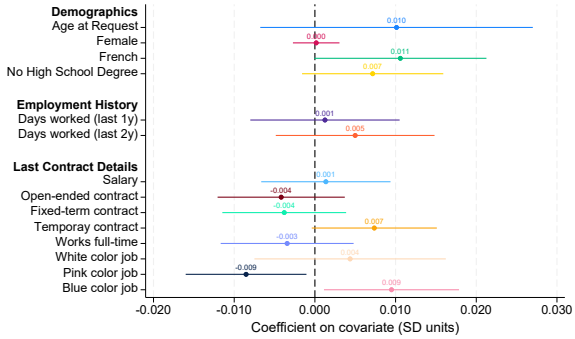
Notes: Matched sample of first requests to the 115, May 2021 to March 2024, callers aged 18–60. Columns (1)–(2): among callers refused at the first request, the share who called the 115 again by day t and the share who obtained a place on a later call by day t. Columns (3)–(5): the outcome is being in an emergency bed (CHU or hotel) on night t after the call, from the stay records; (3) and (4) are the complier means under refusal and placement, from 2SLS of $Y(1-D)$ on $(1-D)$ and of YD on D, (5) their difference (2SLS), with département $\times$ year-month $\times$ shift $\times$ household-type fixed effects and standard errors clustered by agent. Stay records cover about half of the placements on the night of the call, so the levels in columns (3)–(4) understate shelter use; the decline across rows is the informative part. Full sample in **??**.

Figure C2: Balance Check: Caller Characteristics and Agent Leniency

(a) Employed at request



(b) Unemployed at request



Notes: Each point is the coefficient from a regression of residualized agent leniency on the characteristic listed, expressed in standard deviation units, with 95% confidence intervals based on standard errors clustered at the agent level. Panel (a): demographics and characteristics of the current contract at the time of the first request. Panel (b): past labor market history. Joint F-test p-values: 0.59 for the employed sample and 0.11 for the unemployed sample.

Table C4: Single-Sample Monotonicity Checks by Household Type

	Full Sample		Matched Sample	
	(1)	(2)	(3)	(4)
Main	0.987***	0.988***	0.998***	0.998***
	(0.022)	(0.022)	(0.021)	(0.021)
Man alone	1.051***	1.050***	1.044***	1.043***
	(0.025)	(0.025)	(0.026)	(0.026)
Woman alone	0.994***	0.997***	1.002***	1.006***
	(0.029)	(0.029)	(0.030)	(0.030)
Woman with children	0.887***	0.888***	0.882***	0.883***
	(0.036)	(0.036)	(0.040)	(0.040)
Couple with children	0.941***	0.943***	0.969***	0.978***
	(0.042)	(0.042)	(0.061)	(0.059)
Couple without children	0.913***	0.907***	0.997***	0.993***
	(0.044)	(0.044)	(0.057)	(0.057)
Group of adults	0.876***	0.876***	0.988***	0.978***
	(0.055)	(0.055)	(0.075)	(0.074)
Age and nationality controls	No	Yes	No	Yes
Observations (main)	341,545	341,545	136,734	136,734

Notes: *** $p < 0.01$. First-stage regressions of an indicator for the first request being met on agent leniency, with *département* × year-month × shift × household-type fixed effects; each row runs the first stage on the subsample of first requests of the stated household type, with the same leniency measure (built on single-men calls) in every row. Columns (2) and (4) add five-year age-bin dummies and nationality-group dummies (French, EU, non-EU, missing). All columns use the estimation sample: first requests May 2021 to March 2024, callers aged 18 to 60; columns (3)–(4) restrict to the subset matched to the employment records. Standard errors in parentheses, clustered by agent.

Table C5: Do Lenient Agents Displace Other Callers?

Dependent variable, regressor	Coefficient
Colleagues' placement rate in the same shift, on the agent's own leniency (holding colleagues' mean leniency fixed)	−0.010 (0.011)
Agent's own placement rate, on colleagues' mean leniency (holding the agent's own leniency fixed)	−0.055 (0.012)
Shift placement rate, on the previous day's mean on-duty leniency (holding the same day's mean on-duty leniency fixed)	−0.001 (0.017)
For reference: agent's own placement rate, on own leniency	0.827 (0.037)
Shift-cells (dép × day × shift × household type)	289,724
Calls	4,150,777

Notes: All 115 calls, May 2021 to March 2024, in shift-cells with at least two agents on duty. Rows 1, 2 and 4: unit is the agent × shift-cell, weighted by the agent's calls in the cell; colleagues' placement rate and colleagues' mean leniency exclude the agent's own calls. Row 3: unit is the shift-cell, weighted by calls. All regressions include *département* × year-month × shift × household-type fixed effects. Standard errors in parentheses, clustered by agent (rows 1, 2, 4) or by *département* (row 3).

Table C6: Characterizing Compliers

	Employed sample			Unemployed sample		
	All	Placed	Complier	All	Placed	Complier
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Characteristics						
Age	35.9	36.8	34.7	36.4	36.7	35.9
			(0.6)			(0.4)
Female	0.440	0.561	0.427	0.389	0.473	0.377
			(0.034)			(0.020)
French	0.388	0.386	0.319	0.530	0.538	0.421
			(0.028)			(0.019)
No high-school degree	0.559	0.555	0.535	0.627	0.614	0.592
			(0.027)			(0.013)
Panel B. Labor market history						
Days worked in previous year	232.5	240.1	226.4	29.1	28.6	32.8
			(7.6)			(1.8)
Days worked in previous two years	407.9	420.8	391.6	71.3	70.2	80.4
			(16.2)			(4.0)
Panel C. Current contract (employed) / last job (unemployed)						
Monthly earnings (euros)	1,401	1,393	1,395	564	570	534
			(46)			(20)
Contract length (days)	491	547	408			
			(57)			
Open-ended contract	0.459	0.459	0.490	0.220	0.223	0.228
			(0.029)			(0.011)
Fixed-term contract	0.343	0.348	0.373	0.467	0.484	0.486
			(0.027)			(0.013)
Temporary agency work	0.116	0.096	0.103	0.294	0.273	0.269
			(0.019)			(0.012)
Full-time	0.622	0.597	0.630	0.597	0.589	0.572
			(0.026)			(0.012)
White-collar occupation	0.033	0.033	0.036	0.029	0.029	0.028
			(0.010)			(0.005)
Pink-collar occupation	0.403	0.446	0.379	0.356	0.384	0.380
			(0.029)			(0.014)
Blue-collar occupation	0.424	0.386	0.438	0.535	0.509	0.502
			(0.028)			(0.013)
Observations	12,094			71,236		

Notes: This table reports sample means, means among callers whose first request was met, and complier means in the employed and unemployed samples. Sample and placed means for a characteristic Y_i are estimated from sample averages. The complier mean is estimated following Abadie (2003), from a 2SLS regression with outcome $Y_i(1-D_i)$, endogenous variable $(1-D_i)$, agent leniency as the instrument and the same fixed effects as in the main IV specification, where D_i indicates that the first request was met; for a characteristic fixed at the time of the call, this identifies the mean of that characteristic among compliers. Standard errors in parentheses are clustered by agent. Panel A variables are measured at the time of the first call. Panel B counts days worked in the one and two years before the call. Panel C refers to the current contract for the employed sample and to the last job before the call for the unemployed sample; contract length is only observed for the employed sample.

C.3 Additional Results

Table C7: Rapid Shelter Access Raises Job Quality

	Job Within 1 Year	Job (> €1500) Within 1 Year	Job (> €1800) Within 1 Year	Stable Job Within 1 Year	Job (1 Day) Within 1 Year	Job (1 Week) Within 1 Year
First Request Met	0.080***	0.041***	0.019**	0.045***	0.004	0.021
	(0.020)	(0.013)	(0.009)	(0.014)	(0.010)	(0.013)
Mean, refused	0.183	0.082	0.045	0.100	0.059	0.100
Complier mean, refused E[Y(0) | C]	0.18	0.08	0.05	0.10	0.06	0.10
Kleibergen—Paap *F*	1,352	1,352	1,352	1,352	1,352	1,352
Observations	71,236	71,236	71,236	71,236	71,236	71,236

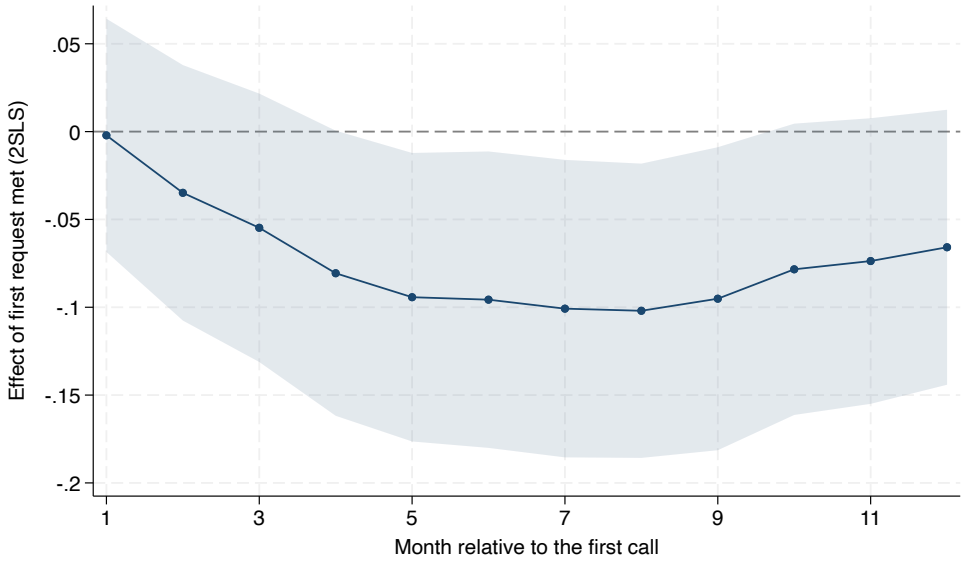
Notes: Stable job refers to a permanent contract or a fixed term contract lasting at least six months.

Table C8: First Request Met and the Housing Trajectory

Within	Nights in accommodation, cumulative				Entered transitional housing (CHRS)			
	1 month	3 months	6 months	12 months	1 month	3 months	6 months	12 months
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Labor force								
First request met	7.9***	20.9***	37.3***	63.3***	0.023***	0.031***	0.042***	0.056***
	(0.6)	(1.6)	(3.1)	(5.5)	(0.006)	(0.009)	(0.011)	(0.012)
Mean, refused	0.5	1.8	3.6	7.0	0.010	0.017	0.026	0.037
Refused compliers, E[Y(0)|C]	0.2	0.8	1.8	3.7	0.01	0.01	0.02	0.03
Kleibergen–Paap F	1,430	1,430	1,430	1,430	1,430	1,430	1,430	1,430
Observations	80,257	80,257	80,257	80,257	80,257	80,257	80,257	80,257
Panel B. Employed at first call								
First request met	9.9***	28.8***	53.5***	94.9***	0.039***	0.060***	0.071***	0.076***
	(1.1)	(3.2)	(5.8)	(11.0)	(0.013)	(0.018)	(0.021)	(0.020)
Mean, refused	0.5	1.5	3.1	5.8	0.012	0.017	0.022	0.027
Refused compliers, E[Y(0)|C]	0.2	0.8	1.5	3.0	0.01	0.01	0.01	0.02
Kleibergen–Paap F	660	660	660	660	660	660	660	660
Observations	11,537	11,537	11,537	11,537	11,537	11,537	11,537	11,537
Panel C. Unemployed at first call								
First request met	7.5***	19.4***	34.3***	57.5***	0.021***	0.027***	0.040***	0.057***
	(0.6)	(1.6)	(3.0)	(5.1)	(0.006)	(0.009)	(0.011)	(0.014)
Mean, refused	0.6	1.8	3.7	7.3	0.010	0.017	0.026	0.039
Refused compliers, E[Y(0)|C]	0.1	0.7	1.6	3.2	0.00	0.01	0.02	0.03
Kleibergen–Paap F	1,294	1,294	1,294	1,294	1,294	1,294	1,294	1,294
Observations	65,586	65,586	65,586	65,586	65,586	65,586	65,586	65,586

Notes: Each cell reports the 2SLS coefficient on the first request being met, instrumented by agent leniency, with département × quarter × shift × household-type fixed effects and controls for age and nationality; standard errors in parentheses are clustered by agent. Columns (1)–(4): cumulative number of nights spent in emergency or transitional accommodation within the given number of months after the first call. Columns (5)–(8): indicator for having entered transitional housing (CHRS) within the given number of months. Mean, refused is the sample mean among callers whose first request was denied; the complier mean under refusal is estimated following **?**, from a 2SLS regression of $Y_i(1 - D_i)$ on $(1 - D_i)$. Panel A pools the employed and unemployed samples; panels B and C report the two samples separately.

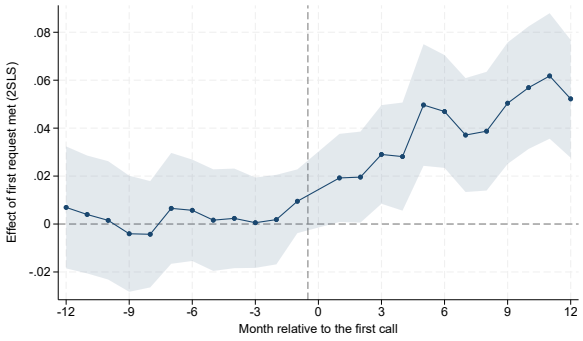
Figure C3: The Effect of Shelter Access on Job Separations



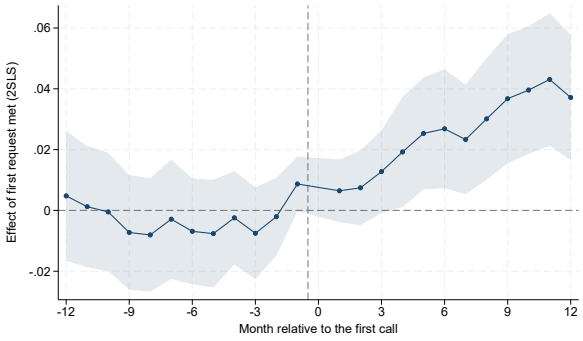
Notes: This figure plots the 2SLS estimates of the effect of rapid shelter access on separation hazard within 12 months after the shelter request. 95% confidence bands are plotted.

Figure C4: The Effect of Shelter Access on Employment (Unemployed Sample)

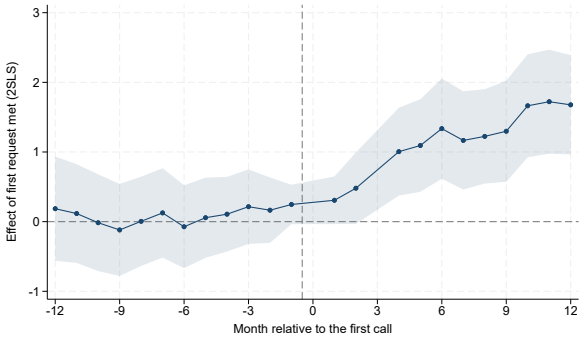
(a) Monthly employment



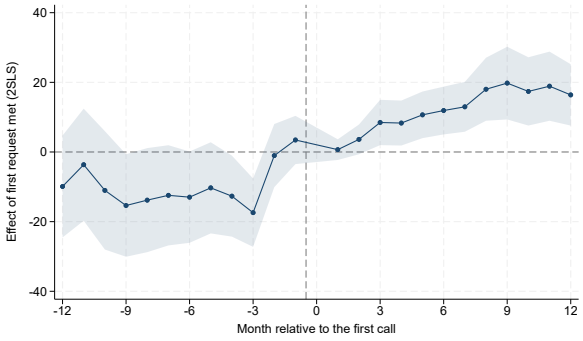
(b) Monthly stable employment



(c) Monthly Days Worked (Unconditional)



(d) Monthly earnings



Notes: This figure plots the 2SLS estimates of the effect of rapid shelter access on employment outcomes in the 12 months before and after the callers first call. 95% confidence bands are plotted.

C.4 Robustness

C.4.1 Instrument Construction

Table C9: First Stage by Minimum Number of Calls Used to Measure Leniency

	Minimum caseload		
	30 (baseline)	50	100
	(1)	(2)	(3)
Panel A. Matched sample			
Agent leniency	0.998***	1.000***	0.998***
	(0.021)	(0.023)	(0.027)
SD of leniency (within cell)	0.102	0.099	0.090
Kleibergen–Paap F	2,189	1,931	1,401
Agents	2,325	2,202	1,987
Observations	136,737	135,931	133,704
Panel B. Full sample			
Agent leniency	0.987***	0.990***	0.983***
	(0.022)	(0.023)	(0.027)
SD of leniency (within cell)	0.103	0.100	0.093
Kleibergen–Paap F	2,000	1,818	1,323
Agents	2,408	2,270	2,040
Observations	341,549	339,987	335,195

Notes: *** $p < 0.01$. Each column reports the first-stage coefficient of an indicator for the first request being met on agent leniency, where leniency is the agent's leave-out mean of residualized placement decisions over all her other calls from single men in the data (2018–2026, across all cells), and the agent is retained only if the total number of such calls is at least the number given in the column header; column (1) is the baseline measure. First requests to the 115, May 2021 to March 2024, callers aged 18–60, estimated on the SI-SIAO data with département × year-month × shift × household-type fixed effects (singleton cells dropped); Panel A is the subset matched to the employment records, Panel B all first requests. SD of leniency is the standard deviation of the instrument within cells. Standard errors in parentheses, clustered by agent; with one instrument the Kleibergen–Paap rk Wald F equals the squared t-statistic. Observations and agents fall across columns as agents below the cutoff, and their callers, are dropped.

Table C10: Robustness to the Minimum Number of Calls per Agent – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Agents with at least 50 calls							
First Request Met	-0.036	0.066	-0.068***	-0.029***	-0.000	-0.034**	-0.005*
	(0.040)	(0.043)	(0.022)	(0.011)	(0.014)	(0.015)	(0.003)
Mean, refused	0.421	0.312	0.091	0.023	0.030	0.036	0.002
Kleibergen–Paap F	597	597	605	597	597	597	597
Observations	11,444	11,444	11,503	11,444	11,444	11,444	11,444
Panel B: Agents with at least 100 calls							
First Request Met	-0.049	0.085*	-0.069***	-0.029**	-0.002	-0.035**	-0.004
	(0.045)	(0.049)	(0.024)	(0.012)	(0.015)	(0.017)	(0.003)
Mean, refused	0.421	0.311	0.091	0.023	0.030	0.036	0.002
Kleibergen–Paap F	516	516	529	516	516	516	516
Observations	11,209	11,209	11,268	11,209	11,209	11,209	11,209

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ shift $\times$ household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Leniency is measured on agents who handled at least 50 (Panel A) or 100 (Panel B) calls, instead of 30 in the baseline. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C11: Robustness to the Minimum Number of Calls per Agent – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
Panel A: Agents with at least 50 calls				
First Request Met	0.036***	0.008	12.352***	38.978**
	(0.013)	(0.008)	(3.833)	(15.767)
Control mean	0.085	0.036	25.272	98.158
Kleibergen–Paap F	1,247	1,247	1,247	1,247
Observations	67,834	67,834	67,834	67,834
Panel B: Agents with at least 100 calls				
First Request Met	0.037***	0.007	10.602***	46.968***
	(0.013)	(0.008)	(3.694)	(16.607)
Mean, refused	0.084	0.036	25.239	97.853
Kleibergen–Paap F	875	875	875	875
Observations	66,710	66,710	66,710	66,710

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Leniency is measured on agents who handled at least 50 (Panel A) or 100 (Panel B) calls, instead of 30 in the baseline. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C12: Robustness to Residualizing Leniency at the Day × Hour Level – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.033	0.076*	-0.069***	-0.034***	-0.000	-0.030**	-0.006**
	(0.042)	(0.043)	(0.023)	(0.012)	(0.015)	(0.015)	(0.003)
Mean, refused	0.421	0.312	0.091	0.023	0.031	0.036	0.002
Kleibergen–Paap F	335	335	335	335	335	335	335
Observations	11,552	11,552	11,611	11,552	11,552	11,552	11,552

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Placement decisions are residualized within *département* × day × hour cells instead of at the year-month level. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C13: Robustness to Residualizing Leniency at the Day × Hour Level – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
First Request Met	0.055***	0.009	14.791***	58.123***
	(0.014)	(0.009)	(3.952)	(17.480)
Mean, refused	0.085	0.036	25.279	98.236
Kleibergen–Paap F	695	695	695	695
Observations	68,226	68,226	68,226	68,226

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Placement decisions are residualized within *département* × day × hour cells instead of at the year-month level. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C14: Robustness to Measuring Leniency on Callers of the Same Household Type – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.039	0.029	-0.044**	-0.024**	0.016	-0.033***	-0.004*
	(0.034)	(0.036)	(0.022)	(0.009)	(0.014)	(0.012)	(0.002)
Mean, refused	0.416	0.316	0.092	0.023	0.031	0.037	0.002
Kleibergen–Paap F	785	785	779	785	785	785	785
Observations	12,287	12,287	12,347	12,287	12,287	12,287	12,287

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ shift $\times$ household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Leniency is measured on the agent's decisions on callers of the same household type as the caller, instead of on single men. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C15: Robustness to Measuring Leniency on Callers of the Same Household Type – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
First Request Met	0.045***	0.013*	13.629***	50.360***
	(0.011)	(0.007)	(3.311)	(13.492)
Mean, refused	0.084	0.036	25.126	97.783
Kleibergen–Paap F	1,968	1,968	1,968	1,968
Observations	72,068	72,068	72,068	72,068

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Leniency is measured on the agent's decisions on callers of the same household type as the caller, instead of on single men. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Unbiased Jackknife IV (UJIVE)

Table C16: Baseline 2SLS and Unbiased Jackknife IV Estimates, Unemployed Sample

	Employed at 1 Year	Open-ended contract at 1 Year	Days Worked Year 1	Monthly Earnings at 1 Year
	(1)	(2)	(3)	(4)
Baseline 2SLS, leniency instrument	0.049***	0.014*	14.1***	54.8***
	(0.013)	(0.008)	(3.7)	(15.7)
UJIVE, agents with ≥ 10 requests	0.038***	0.007	11.8***	36.5***
	(0.008)	(0.005)	(2.0)	(9.9)
UJIVE, agents with ≥ 30 requests	0.041***	0.006	10.8***	37.3***
	(0.009)	(0.006)	(2.5)	(11.7)
Mean, refused	0.084	0.036	25.1	97.6
Kleibergen–Paap F, baseline	1,352	1,352	1,352	1,352
Observations, baseline	71,236	71,236	71,236	71,236
Observations, UJIVE ≥ 10	66,471	66,471	66,471	66,471
Observations, UJIVE ≥ 30	48,500	48,500	48,500	48,500

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Unemployed sample: callers without an employment contract at the time of the first request and with at least one employment contract or unemployment registration since 2017. First row: baseline 2SLS of the outcome on the first request being met, instrumented by the leave-out agent leniency, as in **??**; standard errors clustered by agent. Second and third rows: UJIVE estimator of **?** with the full set of agent indicators as instruments and *département* $\times$ quarter $\times$ shift $\times$ household-type fixed effects absorbed; heteroskedasticity-robust standard errors. Because the jackknife requires each agent's effect to be estimable from her other observations, the second row excludes agents with fewer than 10 first requests in the estimation sample and the third row agents with fewer than 30. All specifications control for age and nationality. Mean, refused is the mean of the outcome among callers whose first request was denied, baseline sample.

C.6 Controls and Fixed Effects

Table C17: Robustness to Controlling for Age and Nationality – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.051	0.069*	-0.066***	-0.027***	-0.004	-0.031**	-0.005*
	(0.037)	(0.038)	(0.021)	(0.010)	(0.013)	(0.014)	(0.003)
Mean, refused	0.416	0.316	0.092	0.023	0.031	0.037	0.002
Kleibergen–Paap F	682	682	687	682	682	682	682
Observations	12,094	12,094	12,154	12,094	12,094	12,094	12,094

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ shift $\times$ household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. All regressions additionally control for the caller's age and nationality. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C18: Robustness to Controlling for Age and Nationality – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
First Request Met	0.051***	0.014*	14.735***	56.863***
	(0.012)	(0.008)	(3.639)	(15.481)
Mean, refused	0.084	0.036	25.102	97.625
Kleibergen–Paap F	1,345	1,345	1,345	1,345
Observations	71,236	71,236	71,236	71,236

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ shift $\times$ household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. All regressions additionally control for the caller's age and nationality. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C19: Robustness to Hour-of-Call Fixed Effects – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.058	0.108*	-0.066**	-0.010	-0.021	-0.027	-0.008
	(0.057)	(0.062)	(0.030)	(0.012)	(0.016)	(0.021)	(0.005)
Mean, refused	0.414	0.316	0.089	0.023	0.029	0.036	0.002
Kleibergen–Paap F	422	422	422	422	422	422	422
Observations	5,811	5,811	5,811	5,811	5,811	5,811	5,811

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ hour-of-call $\times$ household-type fixed effects, instead of shift in the baseline. Control mean is the mean outcome among callers whose first request was refused. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C20: Robustness to Hour-of-Call Fixed Effects – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
First Request Met	0.055***	0.015	16.395***	60.241***
	(0.016)	(0.009)	(4.727)	(19.222)
Mean, refused	0.085	0.035	24.184	97.538
Kleibergen–Paap F	1,040	1,040	1,040	1,040
Observations	55,482	55,482	55,482	55,482

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* $\times$ year-quarter $\times$ hour-of-call $\times$ household-type fixed effects, instead of shift in the baseline. Control mean is the mean outcome among callers whose first request was refused. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.7 Level of Clustering

Table C21: Robustness to the Level of Clustering – Employed at Request

	Employed at Request						
				Reasons Contract End (Within 3 Months)			
	Contract End $\leq$ 3 Months	Contract End $\geq$ 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Clustered at the agent × year-quarter level							
First Request Met	-0.064	0.082**	-0.072***	-0.029***	-0.005	-0.033**	-0.005*
	(0.040)	(0.039)	(0.021)	(0.009)	(0.013)	(0.014)	(0.003)
Mean, refused	0.416	0.316	0.092	0.023	0.031	0.037	0.002
Kleibergen–Paap *F*	869	869	879	869	869	869	869
Observations	12,094	12,094	12,154	12,094	12,094	12,094	12,094
Panel B: Clustered at the fixed-effect cell level							
First Request Met	-0.064*	0.082**	-0.072***	-0.029***	-0.005	-0.033**	-0.005*
	(0.039)	(0.036)	(0.021)	(0.010)	(0.013)	(0.013)	(0.003)
Mean, refused	0.416	0.316	0.092	0.023	0.031	0.037	0.002
Kleibergen–Paap *F*	995	995	994	995	995	995	995
Observations	12,094	12,094	12,154	12,094	12,094	12,094	12,094

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Standard errors in parentheses are clustered at the agent × year-quarter level (Panel A) and at the level of the fixed-effect cell (Panel B). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C22: Robustness to the Level of Clustering – Unemployed at Request

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
Panel A: Clustered at the agent × year-quarter level				
First Request Met	0.049***	0.014*	14.142***	54.781***
	(0.011)	(0.007)	(3.129)	(14.132)
Mean, refused	0.084	0.036	25.102	97.625
Kleibergen–Paap F	3,089	3,089	3,089	3,089
Observations	71,236	71,236	71,236	71,236
Panel B: Clustered at the fixed-effect cell level				
First Request Met	0.049***	0.014*	14.142***	54.781***
	(0.011)	(0.007)	(3.175)	(14.515)
Mean, refused	0.084	0.036	25.102	97.625
Kleibergen–Paap F	3,367	3,367	3,367	3,367
Observations	71,236	71,236	71,236	71,236

Notes: 2SLS estimates. *First Request Met* is instrumented with the leniency of the agent who answered the caller's first request. All regressions absorb *département* × year-quarter × shift × household-type fixed effects. Control mean is the mean outcome among callers whose first request was refused. Standard errors in parentheses are clustered at the agent × year-quarter level (Panel A) and at the level of the fixed-effect cell (Panel B). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Alternative Design: Entry into Transitional Housing Shelters

D.1 Descriptive Statistics

Table D1: Labor market history for workers employed at transitional housing request

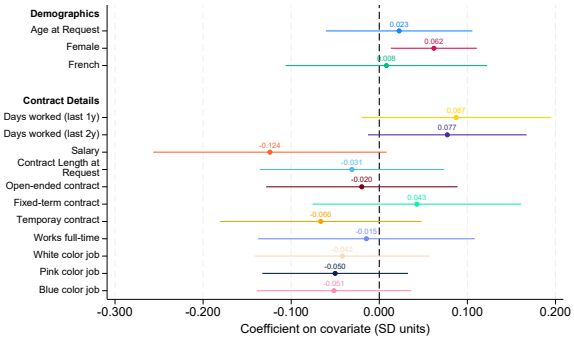
	All employed	Wage $>$ 1,500	Stable workers (185+ days / 2 yrs)
	(1)	(2)	(3)
Panel A: Sociodemographics			
Female	0.39	0.35	0.39
Age (median)	34	35	34
With children	0.24	0.24	0.23
Panel B: Current job			
Monthly wage (median)	1,236	1,733	1,408
Permanent contract (CDI)	0.38	0.46	0.37
Full-time	0.57	0.74	0.61
Blue-collar	0.48	0.47	0.50
Observations	43,727	20,703	12,288

Table D2: Labor market history for workers unemployed at transitional housing request

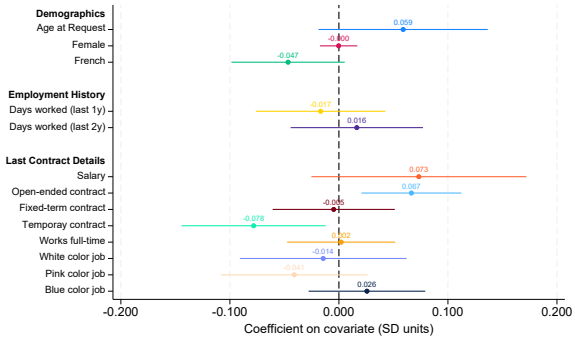
	All	Had a job before	Last job $<$ 2 yrs ago	Stable workers (185+ days / 2 yrs)
	(1)	(2)	(3)	(4)
Panel A: Sociodemographics				
Female	0.41	0.39	0.39	0.39
Age (median)	35	36	35	37
With children	0.27	0.22	0.22	0.24
Panel B: Employment history				
Days since last job (median)		221	150	115
Days worked, last 2 yrs (median)		60	105	339
Last monthly wage (median)		182	161	678
Last job: permanent contract (CDI)		0.20	0.19	0.29
Last job: temp agency		0.32	0.33	0.26
Last job: full-time		0.59	0.58	0.60
Last job: blue-collar		0.56	0.56	0.51
Observations	149,753	76,022	61,136	21,305

Figure D1: Balance Check: Unhoused Individuals' Characteristics and Shelter Capacity Instrument

(a) Employed at request



(b) Unemployed at request



Notes: Each point is the coefficient from a regression of the indicator variable of having a short average wait time ($<$ 3 months, calculated by leave-one out wait-time of all other individuals requesting transitional housing shelters in the same year-month and *département*) on the characteristic listed, expressed in standard deviation units, with 95% confidence intervals based on standard errors clustered at the agent level. Panel (a): demographics and characteristics of the current contract at the time of the first request. Panel (b): past labor market history. Joint F-test p-values: 0.1703 for the employed sample and 0.1911 for the unemployed sample.

D.2 Balance

D.3 Main Results

Table D3: Rapid Shelter Access Reduces Job Separation

	Employed at Request					
				Reasons Contract End (Within 3 Months)		
	Contract End $\leq$ 3 Months	Same Contract at 1 Year	Contract Termination (Within 3 Months)	Misconduct	Quit	End Trial Period
	(1)	(2)	(3)	(4)	(5)	(6)
Transitional Housing Within 3M	-0.061*	0.001	-0.024	-0.021**	0.009	-0.009
	(0.035)	(0.037)	(0.022)	(0.010)	(0.017)	(0.016)
Mean, refused	0.406	0.329	0.059	0.014	0.023	0.021
Complier mean, refused E[Y(0) | C]	0.50	0.29	0.08	0.02	0.02	0.04
Kleibergen—Paap F	99	99	96	99	99	99
Observations	10,968	10,968	11,031	10,968	10,968	10,968

Notes: This table reports 2SLS estimates of the effect of quick entry into transitional housing request on job separation among the 12,094 callers who held an employment contract at the time of their first request for transitional housing between May 2021 and March 2024. The endogenous variable is an indicator for entering a transitional housing shelter three months after the request, instrumented by having a short leave-out average wait-time at the moment of the request, in the area of the request. Column (1): the contract held at the request ends within three months of the call. Column (2): the caller is still employed in the same contract twelve months after the call. Column (3): the contract is terminated within three months by the employer or the employee, as opposed to the scheduled expiry of a fixed-term contract. Columns (4) to (7) decompose column (3) by the reason recorded by the employer: dismissal for misconduct, quit and dismissal at the end of the probationary period. In this population, there are too few health-related terminations to estimate the effect on this layoff reason in comparison to the caller sample. The row "Mean, refused compliers" reports E[Y(0)|C], the complier mean under refusal, estimated following (Abadie, 2003). All specifications include *département* × year-month × household-type fixed effects. The Kleibergen–Paap F-statistic of the first stage is 99. Standard errors in parentheses are clustered at the *département* level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D4: Rapid Shelter Access Raises Job Finding

	Unemployed at Request			
	Employed at 1 Year	Open-ended contract at 1 Year	Days Worked Year 1	Monthly Earnings at 1 Year
	(1)	(2)	(3)	(4)
Transitional Housing Within 3M	0.044**	0.019	21.488***	48.532*
	(0.017)	(0.014)	(5.292)	(27.135)
Mean, refused	0.393	0.168	143.035	480.377
Complier mean, refused E[Y(0) | C]	0.34	0.13	118.41	419.88
Kleibergen—Paap F	146	146	146	146
Observations	36,452	36,452	36,452	36,452

Notes: This table reports 2SLS estimates of the effect of quick entry into transitional housing on job finding among the 34,452 individuals requesting transitional housing shelers without an employment contract at the time of their first request between May 2021 and March 2024, but with at least one employment contract or unemployment registration since 2017. TThe endogenous variable is an indicator for entering a transitional housing shelter three months after the request, instrumented by having a short leave-out average wait-time at the moment of the request, in the area of the request. Column (1): the caller is employed twelve months after the call. Column (2): the caller holds an open-ended contract twelve months after the call. Column (3): number of days worked in the twelve months following the call. Column (4): gross monthly earnings twelve months after the call, in euros, set to zero for callers not employed. The row "Mean, refused compliers" reports E[Y(0)|C], the complier mean under refusal. All specifications include *département* × year-month × household-type fixed effects. The Kleibergen–Paap F-statistic of the first stage is 146. Standard errors in parentheses are clustered at the *département* level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Quantifying the Interaction Between Unemployment and Homelessness

E.1 Main tables

Table E1: Monthly transition rates and their sources

		value	s.e.	source
Levels, measured where selection into homelessness cannot operate				
f_H	job finding, housed	0.1444	(0.0003)	$(U,H) \to (E,H)$
s_H	separation, housed	0.0685	(0.0002)	$(E,H) \to (U,H)$
δ_U	housing loss, not employed	0.00290	(0.00001)	$(U,H) \to (U,NH)$
g_U	exit without a job	0.1030	(0.0004)	$(U,NH) \to (U,H)$
Ratios, from the two quasi-experiments				
$\kappa_f = f_{NH}/f_H$	job finding penalty	0.68	(0.05)	shelter instrument
$\kappa_s = s_{NH}/s_H$	separation penalty	1.47	(0.08)	shelter instrument
$\kappa_\delta = \delta_U/\delta_E$	entry gradient	2.92	(1.16)	mass layoffs
Implied				
f_{NH}	job finding, unhoused	0.0982		$\kappa_f f_H$
s_{NH}	separation, unhoused	0.1009		$\kappa_s s_H$
δ_E	housing loss, employed	0.00099		δ_U/κ_δ
Exit while employed: no design				
$\kappa_g = g_E/g_U$	exit gradient	1.16	(0.01)	measured
		3.00		Bruneel-Zupanc et al. (2026)

Notes: Standard errors on the measured rates are clustered on the individual. g_E sums the flows into employment and housing from both unhoused states. The standard error on κ_δ is from 500 bootstrap replications of the two layoff regressions, resampled jointly; κ_s is from the twelve-month retention outcome, 0.270 against 0.418.

Table E2: The causal ratios against the raw transitions

	raw transitions	quasi-experiment	gap
job finding penalty κ_f	0.845	0.68	-20%
separation penalty κ_s	1.472	1.474	$+0\%$
entry gradient κ_δ	2.939	2.923	-1%

Notes: The unhoused rates are never an input: they are the measured housed rates scaled by the causal ratios, so the raw transitions provide an over-identification check. The separation penalty and the entry gradient are reproduced almost exactly, and since each pair is built from a different sample in a different way the agreement is not mechanical. Only the job finding penalty differs; Table E9 shows that the gap narrows as the employment threshold is tightened.

Table E3: Contribution of each channel to steady-state homelessness

	set equal to	exit gradient κ_g	
		1.16	3.00
share unhoused	—	1.43%	0.92%
entry when unemployed	$\delta_U = \delta_E$	39%	46%
slower exit when unemployed	$g_U = g_E$	7%	43%
job finding penalty	$\kappa_f = 1$	1%	7%
separation penalty	$\kappa_s = 1$	1%	3%
all four		42%	65%

Notes: Each row sets one rate to its counterpart for the other group, resolves for the stationary distribution μ^c, and reports $(n - n^c)/n$. The first column uses the exit gradient measured in the transition data, the second the upper bound from Bruneel-Zupanc et al. (2026). The entry channel is a counterfactual on a primitive and barely moves with the exit gradient; the other three do.

Table E4: Contribution of homelessness to non-employment

	set equal to	all non-employment	spells past 12 months
job finding penalty	$f_{NH} = f_H$	0.48%	2.0%
separation penalty	$s_{NH} = s_H$	0.36%	0.1%
both		0.83%	2.0%

Notes: At $\kappa_g = 1.16$; at 3 the figures are 0.56% and 1.8%. The second column follows one month's inflow into non-employment forward and reports the share of person-months accruing after the twelfth, so it is a cohort measure rather than a stock share. 15.7 percent of all non-employment accrues in spells past a year.

Table E5: Individual trajectories

	exit gradient κ_g	
	1.16	3.00
Panel A. Months to the first job, from unemployment		
housed	7.0	
unhoused	8.5	
longer by	23%	
Panel B. Months unhoused before reaching a housed state		
starting employed	8.8	4.2
starting unemployed	9.3	7.0
longer by	5%	68%
starting unemployed, $\kappa_f = \kappa_s = 1$	9.1	6.3
shorter by	2%	11%
Panel C. Months to employment and housing		
from (U, H)	7.1	7.0
from (E, NH)	10.9	5.2
from (U, NH)	13.9	11.1
housing is worth	6.9	4.1
employment is worth	3.0	5.9

Notes: Panel A solves the two-state system in which a spell of non-employment ends at the first job. It does not depend on the exit gradient: both employment states are absorbing, so the rate at which the employed leave homelessness never enters, and the figures are reported once. Panels B and C use the fundamental matrix of the chain with the relevant states made absorbing; in Panel B both housed states absorb, in Panel C only (E, H) does. The two unhoused states are equidistant from recovery at $\kappa_g = 2.07$.

Table E6: The cost and return of one shelter placement

	unemployed at first call	employed at first call
Cost		
additional nights in accommodation, 12 months	57.5	94.9
	(5.1)	(11.0)
at €25 a night	€1,438	€2,372
Return		
additional months worked, 12 months	0.629	1.480
	(0.256)	(0.505)
complier mean under refusal	1.59	7.79
fiscal value at €1,009 a month	€635	€1,494
net government cost	€803	€879
fiscal recovery	**44%**	**63%**
95% interval	[9, 79]	[21, 105]
MVPF, earnings only	0.50	1.07
$\Pr(\text{MVPF} > 1)$	17%	53%
$\Pr(\text{net cost} < 0)$	0.1%	4.2%
MVPF including the value of the nights	1.35	2.35
MVPF assuming no benefit take-up	0.67	1.10
fiscal recovery in that case	24%	34%

Notes: Sample: callers with less than a high school education; the unemployed arm is further restricted to those with a labour market record in the preceding two years. Nights and months worked are 2SLS estimates instrumented by agent leniency, standard errors clustered by agent. Mean gross monthly earnings among the employed are €1,500; the fiscal value of a month of work is built in Table E8. Intervals are from 20,000 draws of the months-worked coefficient. The MVPF is convex in months worked and unbounded where the net cost reaches zero, so the probability mass is reported rather than an upper bound.

E.2 Derivations

E.2.1 The transition matrix

Sample and states. The chain is estimated on person-months in the matched SIAO–DSN–Fichier Historique sample over 2021 to 2024. I draw a 5 percent random sample of individuals never observed unhoused and retain all individuals ever observed unhoused, keeping every month for each; all transitions are weighted accordingly, so the housed rows carry weight 20 and the unhoused rows weight 1. I restrict to individuals with less than a high school education who have an observed labour market record in the sample period, following the shelter experiment, which identifies an employment penalty only among those who worked in the three years before entering homelessness. The four states are $\{(E,H),(U,H),(E,NH),(U,NH)\}$, with E denoting at least fourteen days of covered work in the month, summed across contracts, and NH an observed shelter stay or request; an exit from homelessness is recorded when neither is observed for six consecutive months. Person-months with no labour market record form a fifth state, which is excluded and over which the four rows are rescaled.

Collect the one-month transition probabilities in the matrix P, where P_{ij} is the probability of moving from state i to state j. Ordering rows and columns $(E,H),(U,H),(E,NH),(U,NH)$,

$$P = \begin{pmatrix} 1-s_H-\delta_E & s_H & \delta_E & 0 \\ f_H & 1-f_H-\delta_U & 0 & \delta_U \\ g_E & 0 & 1-g_E-s_{NH} & s_{NH} \\ 0 & g_U & f_{NH} & 1-g_U-f_{NH} \end{pmatrix}. \tag{E5}$$

The four zeros are the joint transitions discussed next.

Joint transitions. A worker can change both statuses in the same month, and neither design identifies such a move. Three of the four joint flows are small and fold into the diagonal. The exception is $(U,NH)\to(E,H)$, a job and a tenancy secured together, at 0.021 a month against an exit rate of 0.103; I count it as an exit achieved through employment and include it in g_E.

E.2.2 The causal ratios

Three ratios enter the chain, and they run in the two directions the paper identifies. The labour-market penalties of homelessness are $\kappa_f = f_{NH}/f_H$ for job finding and $\kappa_s = s_{NH}/s_H$ for separation, each comparing the rate when unhoused with the rate when housed. The entry gradient runs the other way: $\kappa_\delta = \delta_U/\delta_E$ compares the rate of losing housing when unemployed with the rate when employed.

The shelter instrument identifies the first two by comparing compliers whose first request was met with those whose request was refused, so the offered arm stands in for the housed rate

and the refused arm for the unhoused one; since both groups remain unhoused in the sense used here, these are lower bounds on what stable housing would do. The instrument reports outcomes over twelve months, so for both I write P for the probability of an event within that window and convert it to the monthly hazard the chain requires. The layoff design identifies κ_δ and needs a different treatment, set out below.

κ_f. The chain needs a single monthly job finding rate, but the shelter instrument does not find a constant monthly effect. As a simplification I therefore take the twelve-month hazard and convert it, $f = 1 - (1 - P)^{1/12}$. Among compliers refused a place the probability of finding a job within twelve months is 0.1766, and the 2SLS coefficient of 0.0830 puts the offered arm at 0.2596, so

$$\kappa_f = \frac{0.1766}{0.2596} = 0.68, \qquad \text{s.e. } 0.05. \tag{E6}$$

Applied to the housed rate this gives $f_{NH} = 0.68 \times 0.1444 = 0.0982$. The ratio is invariant to the shape of the hazard as long as it declines at the same rate in both arms: letting it decay geometrically moves the implied levels by a factor of two across the range in Table E7 and the ratio in the fourth decimal.

Table E7: κ_f under a declining hazard

decay rate d	f_0, refused	f_0, offered	κ_f
1.00 (constant)	0.0162	0.0247	0.657
0.95	0.0212	0.0322	0.658
0.90	0.0271	0.0412	0.658
0.85	0.0340	0.0517	0.658

Notes: The hazard is allowed to decline geometrically, $f_t = f_0 d^t$, and f_0 is chosen to reproduce the same twelve-month probability. The implied levels double across the range while the ratio moves in the fourth decimal.

κ_s. The outcome here is retention: among compliers refused a place 0.270 are still in the same job after one year, and the 2SLS coefficient of 0.148 (s.e. 0.032) puts the offered arm at 0.418. The probability of separating within twelve months is therefore 0.730 if refused and 0.582 if offered, and converting each as above gives monthly separation hazards of 0.103 and 0.070, so

$$\kappa_s = \frac{0.103}{0.070} = 1.47, \qquad \text{s.e. } 0.08. \tag{E7}$$

Applied to the housed rate this gives $s_{NH} = 1.47 \times 0.0685 = 0.1009$.

κ_δ. The layoff design gives the effect of displacement on homelessness, not a ratio of hazards by employment status, so an exposure correction is needed. With R the relative effect on homeless-

ness over the window and E the excess share of the window spent not employed,

$$R = (\kappa_\delta - 1)\, E \qquad \Longrightarrow \qquad \kappa_\delta = 1 + \frac{R}{E} = 1 + \frac{0.675}{0.351} = 2.92. \tag{E8}$$

Dividing by E converts an effect per displaced worker into an effect per month unemployed. The standard error, 1.16, is from 500 bootstrap replications of the two regressions resampled jointly, since R and E share a sample. Both are measured over the same three-year window, but the window truncates them differently: the employment gap has nearly closed by the twelfth quarter while the homelessness effect is still rising, so κ_δ is a lower bound. Applied to the measured rate this gives $\delta_E = 0.00290/2.92 = 0.00099$.

κ_g. The exit gradient $\kappa_g = g_E/g_U$ compares the rate at which the employed leave homelessness with the rate at which the unemployed do, and neither design identifies it. I therefore take it from the transition data. The denominator g_U is the flow from (U, NH) into (U, H), an exit without a job, at 0.1030. The numerator adds the two ways a worker can reach a housed and employed state: from (E, NH), keeping a job she already had, at 0.0978, and from (U, NH), the joint move discussed above, at 0.0213. Together $g_E = 0.1191$ and

$$\kappa_g = \frac{0.1191}{0.1030} = 1.16, \qquad \text{s.e. } 0.007. \tag{E9}$$

Bruneel-Zupanc et al. (2026) report two figures from a French food distribution panel. Treating attrition from the panel as a return to housing, so that the whole sample is used, gives $g_E = 0.090$ and $g_U = 0.079$ and a gradient of 1.14, close to the one measured here. Restricting to individuals present in two consecutive years, for whom housing status is recorded, 13.6 percent of the employed move into housing within the year against 4.6 percent of the non-employed, which converts to monthly rates of 0.0121 and 0.0039 and a gradient of 3.11. Since remaining in the panel is correlated with both employment and rehousing, the second is an upper bound, and every result that depends on this parameter is reported at both values.

E.2.3 Objects computed from the chain

Stationary distribution and attribution. Let μ be the stationary distribution over the four states, that is the long-run share of person-months spent in each. With P the transition probability matrix in (E5), μ solves

$$\mu' P = \mu', \qquad \mu' \mathbf{1} = 1, \tag{E10}$$

so that the share unhoused is $n = \mu_{EN} + \mu_{UN}$ and the share not employed is $u = \mu_{UH} + \mu_{UN}$. Replacing one rate in P by its counterpart for the other group and re-solving gives a counterfactual distribution μ^c. The contribution of that channel to homelessness is $(n - n^c)/n$, and its contribution to non-employment $(u - u^c)/u$; the first is used for the channels in Table E3 and the

second for those in Table E4. Setting $\delta_U = \delta_E$ is a counterfactual on a primitive, since δ is the rate at which a tenancy ends given income and is what the layoff design identifies; setting $f_{NH} = f_H$ is not, since in a search framework f adjusts with the reservation wage, so that exercise bounds how much of the aggregate the measured penalty can account for rather than describing a policy. Because homelessness has risen over the sample period the population is not in steady state, so μ is the distribution the measured rates would generate rather than the one observed.

Long-term non-employment. A chain with constant rates has no notion of elapsed duration in its stock, so the share of non-employment that is long-term cannot be read off μ. I therefore follow a cohort. Take the workers who enter non-employment in a single month, $s_H \mu_{EH}$ of them into (U, H) and $s_{NH} \mu_{EN}$ into (U, NH), and iterate the two-state non-employment block forward until they have all found work. Summing the person-months they accumulate gives

$$\text{long-term share} \;=\; \frac{\text{person-months in month 13 and after}}{\text{person-months in all months}}, \tag{E11}$$

which is 15.7 percent at the measured rates. This is a cohort measure, not a stock share: it is the fraction of all time spent not employed that is spent in spells already longer than a year.

The counterfactual works as before, with the change applied in both places the rate enters. Setting $f_{NH} = f_H$, say, alters the stationary distribution and so the composition of the entering cohort, and it alters the block the cohort is iterated through. Recomputing the share under the new rates gives L^c, and the contribution of that channel to long-term non-employment is $(L - L^c)/L$, reported in the second column of Table E4.

Expected durations. The trajectory results ask how long it takes to reach a given state. A spell of non-employment ends at the first job, so employment is absorbing and only $\{(U, H), (U, NH)\}$ remain. With t_i the expected months until the first job from state i,

$$t_{(U,H)} = 1 + (1 - f_H - \delta_U)\, t_{(U,H)} + \delta_U\, t_{(U,NH)}, \tag{E12}$$

$$t_{(U,NH)} = 1 + (1 - f_{NH} - g_U)\, t_{(U,NH)} + g_U\, t_{(U,H)}, \tag{E13}$$

or $t = \mathbf{1} + At$, where A is obtained from P by deleting the rows and columns of the absorbing states, here (E, H) and (E, NH), so that

$$t = (I - A)^{-1}\mathbf{1}. \tag{E14}$$

The entry $(I - A)^{-1}_{ij}$ is the expected months spent in j before absorption, starting from i. Panel B of Table E5 applies the same construction with both housed states absorbing, and Panel C with only (E, H) absorbing.

The housing-first threshold. Solving $t_{(E,NH)} = t_{(U,H)}$ on the chain gives $\kappa_g^* = 2.07$, against a measured 1.16.

The two terms behind it are visible in the decomposition of $t_{(E,NH)}$. A worker in (E, NH) leaves that state at $g_E + s_{NH}$, so she spends $1/(g_E + s_{NH}) = 4.55$ months there, and with probability $s_{NH}/(g_E + s_{NH}) = 0.46$ she leaves by losing the job rather than by being rehoused. That returns her to (U, NH), from which recovery takes $t_{(U,NH)} = 13.92$ months, so

$$t_{(E,NH)} \;=\; \frac{1}{g_E + s_{NH}} \;+\; \frac{s_{NH}}{g_E + s_{NH}}\, t_{(U,NH)} \;=\; 4.55 + 0.46 \times 13.92 \;=\; 10.93, \tag{E15}$$

against $t_{(U,H)} = 7.06$. Employment is the better route only when g_E is large enough that the first term falls and the second is reached less often. Because a separation returns her to (U, NH) rather than to (U, H), the penalty is severe: she is then further from recovery than when she started.

The shelter calculation. The policy priced is one additional placement for the marginal applicant; leniency compliers approximate who enters under a capacity expansion, since more places shift the same rationing margin. The cost is the measured effect on nights spent in the shelter system, with refused compliers as the comparison group. The return is the effect on months worked within the following twelve, $m = \sum_{t=1}^{12} \mathbf{1}\{\text{employed in month } t\}$, which requires no assumption about when within the window a job starts or ends. With C the cost and τ and w the fiscal and private value of a month of work, the MVPF is $mw/(C - m\tau)$. All sampling error enters through m; since the ratio is convex in m and unbounded where the net cost reaches zero, I draw m from its sampling distribution and report the median, the probability that the MVPF exceeds one, and the probability that the net cost is negative.

Table E8: The fiscal and private value of one month of work

	€	source
gross monthly earnings	1,500	mean among the employed, matched sample
Public finances		
employee contributions and CSG/CRDS	330	22% of gross, applied rate
employer contributions	68	4.5% of gross, applied rate
VAT on the additional consumption	76	12% effective rate
RSA no longer paid	636	single-person rate, imputed
prime d'activité now paid	−180	at this wage, imputed
APL reduction	80	imputed
fiscal value per month	**1,009**	
assuming no benefit take-up	538	
The worker		
net wage	1,170	gross less employee contributions
prime d'activité	180	
APL forgone	−80	
RSA forgone	−636	
private value per month	**634**	
assuming no benefit take-up	1,170	

Notes: Gross earnings are observed in the matched records. The contributions are applied rates at this wage rather than observed amounts; near the minimum wage the réduction générale removes almost all employer contributions, so the employer wedge is €68 rather than the several hundred a headline rate would imply. The three transfers are not observed either and are imputed from the statutory schedule for a single person with no dependants, assuming full take-up before and after the job; they account for a little over half the fiscal value, and the prime d'activité offsets part of the RSA saving. Health, emergency-room and justice costs are unobserved, so the fiscal return is a lower bound; entry into transitional housing, which a placement also raises, is omitted from both sides for want of a unit cost.

E.3 Robustness

The employment threshold. Table E9 reports the rates under a one-day rule, the fourteen-day rule used throughout, and a whole-month rule. The whole-month rule records a separation whenever a single month falls between two fixed-term contracts, which in a population where most hires are on contracts shorter than a month puts s_H an order of magnitude above the aggregate French rate. The one-day rule removes the measured job finding penalty, because the unhoused find short low-paid work at least as often as the housed. The job finding penalty widens monotonically as the threshold tightens and the separation penalty narrows, which is what a composition effect in job quality predicts and selection on workers does not.

Table E9: Transition rates under three employment definitions

	one day	14 days	30 days
Measured rates, housed			
f_H	0.1797	0.1500	0.1819
s_H	0.0634	0.0699	0.1518
Raw ratios			
κ_f	0.897	0.813	0.600
κ_s	1.509	1.442	1.060
κ_δ	2.40	3.25	3.53
κ_g	1.36	1.14	0.93
Implied			
mean months in a job, $1/s_H$	15.8	14.3	6.6
stationary share not employed	26.2%	31.9%	45.7%

Notes: Employment is at least the stated number of days of covered work in the month, summed across contracts, except under the thirty-day rule, which requires the whole month. Ratios are computed from the probability matrices with the joint transitions left in, so the fourteen-day figures differ slightly from the 0.845 and 1.472 in Table E2, which use the convention in Section E.2. The causal κ_f of 0.68 lies between the fourteen- and thirty-day values.

Job quality. Table E10 splits employment by whether the job pays at least the monthly full-time equivalent minimum wage. The unhoused find non-qualifying work at 89 percent of the rate of the housed and qualifying work at 44 percent, so the job finding penalty falls almost entirely on the better jobs. Only those carry a higher exit rate, 1.16 against 1.04, which is why the pooled exit gradient is close to one; and the entry gradient is 5.66 against qualifying employment but 2.74 against non-qualifying, so a good job insures against losing housing roughly twice as well as a bad one. The causal κ_f of 0.68 lies between the two quality-specific values, as it should if the instrument's compliers are a mix of the two.

Table E10: Rates by job quality, six-state chain

	qualifying	non-qualifying	pooled
Job finding			
housed	0.0266	0.1233	0.1444
unhoused	0.0118	0.1102	0.1220
κ_f	**0.44**	**0.89**	0.85
Separation			
housed	0.0395	0.0849	0.0685
unhoused	0.0586	0.1085	0.1008
κ_s	1.49	1.28	1.47
Exit from homelessness			
employed	0.1239	0.1120	0.1191
not employed		0.1072	0.1030
κ_g	**1.16**	**1.04**	1.16
Entry into homelessness			
employed	0.00047	0.00096	0.00099
not employed		0.00264	0.00290
κ_δ	5.66	2.74	2.94

Notes: Qualifying means a job paying at least the monthly full-time equivalent minimum wage. The six-state chain splits both employment states by job quality and is otherwise constructed as the four-state chain, with the same treatment of the joint transitions. The pooled column is reproduced from Table E2 and Table E1 and is not the average of the other two, since the rates are weighted by different stocks. The unhoused find non-qualifying work at 89 percent of the rate of the housed and qualifying work at 44 percent; only qualifying work carries a higher exit rate; and a qualifying job insures against losing housing roughly twice as well as a non-qualifying one.

Behavioral responses. The comparison between states holds behaviour fixed, so a worker moved into housing is assumed to search as a housed worker does. Let ψ scale the job finding rate she faces once rehoused,

$$f_H^{\text{rehoused}} = \psi\, f_H, \tag{E16}$$

so that $\psi = 1$ is the baseline and $\psi < 1$ captures scarring or reduced search effort. The two unhoused states are equidistant from recovery at $\psi = 0.56$: rehousing would have to cut job finding by more than two fifths to overturn the comparison, and the Housing First literature reports no deterioration of this kind. Waiting times for transitional housing vary across local systems for reasons unrelated to the applicant, so spell duration can be instrumented by the local average wait, which would estimate ψ rather than bound it.